Pythagorean Specializations of an Elliptic K3 Surface: Geometry, Conductors, Root Numbers and Rank Jumps

Status report, 6 November 2026. Each result carries one of five labels: ***Proved****,* ***Certified*** *(exact computation that was independently re-checked),* ***Observed*** *(finite computation),* ***Conditional*** *(depends on a stated hypothesis) and* ***Open****.*

Disclosure. The research and the write-up of this paper were done with a large language model (Claude, by Anthropic) working with the person who directed the project. The model proposed arguments and wrote proofs, code and text; the person chose the questions, supplied some outside computations and reviews, and decided what to keep. No independent human referee has checked the text. See "Limitations and provenance" below.

Abstract

The curves $E_t : y^2 = x^3 + 4A(t)x - 8B(t)$, where $(A, B) = ((1 - t^2)/(1 + t^2), 2t/(1 + t^2))$ is a rational point of the unit circle, are the fibres of an elliptic K3 surface S over $\mathbb{P}^1$ with Picard number 16 and generic Mordell–Weil rank 0. We determine the conductor and the global root number of E_t for every rational t, show that infinitely many t have rank $\geqslant 1$, exhibit fibres of rank 2 and 3, and study how the rank jumps. Along the orbit $t_n = 1 - x(nP)$ of the point $P = (1, 1)$ on the elliptic curve $y^2 = z^3 - 2z^2 + 2z$, the root number is the product of a periodic sign (period 6 along nP, period 3 along $nP + T$), local factors at 2 and 3 that are completely determined, and one remaining bit: the parity of the number of prime factors of the values of an irreducible binary sextic. That bit is the only obstacle to infinitely many members of rank $\geqslant 2$ (Problem H, open). The paper proves everything else, records the routes that cannot work (identities of functions, degree reduction, primitive divisors, a second section on a constant-x cover), and reports computations that exclude a second section for 18 such covers.

Results at a glance

Each row carries one of the five labels of the paper: Proved, Certified (exact computation, independently re-checked), Observed (finite computation), Conditional (depends on a stated hypothesis) or Open.

#	Result	Status	Where	Depends on
1	S has fibres $2\cdot III^* + 6\cdot I_1$, Picard number 16 and generic rank 0	Proved	§2	nothing
2	Kodaira types, conductor exponents and conductor of E_t for every rational t	Proved	§3	nothing
3	Global root number W(E_t) for every rational t, in closed form, with no squarefree assumption on F	Proved	§4, F.28	nothing; checked analytically at eight values of t
4	The sign ε_n is periodic: period 6 along nP, period 3 along nP + T	Proved	Appendix B, F.21	finitely many exact local evaluations
5	Valuation laws $v_2(D_n) = v_2(n)$, $v_3(D_n) = v_3(n)$; the 2- and 3-adic valuations of F_n and G_n	Proved	F.20, F.22	nothing
6	gcd(F_n, G_n) = 1 for every n	Proved	F.25	finite check at the primes 7, 103 and 519521
7	F_n is not squarefree for infinitely many n (23^2 divides F_{32}); the squarefree part of F_n tends to infinity	Proved	Appendix B, F.18	Faltings' theorem (ineffective)
8	No identity of rational functions relates the values F_n at multiples, translates or doubles of n	Proved	F.19, F.34, F.36, F.41	the points over the roots of h are non-torsion
9	W(E_{t_n}) = ε_n·(−1)^{ω(N_n)}·η_n, so Problem H is equivalent to one parity statement	Proved	F.24, F.40	rows 3 and 4
10	Infinitely many t of rank ⩾ 1 (the cover C_2)	Proved	§6	specialization theorem

#	Result	Status	Where	Depends on
11	Ranks 0, 1, 2 and 3 occur (four rank-3 fibres certified; three more reported, not recomputed)	Certified	§5	exact computation
12	The twist by $(1 - t)(1 + t^2)$ has geometric rank exactly 1	Conditional	§7	Tate conjecture for K3 surfaces over finite fields (known for $p \geq 5$)
13	The same for 18 values of c (2, 13/9, −11/5, 14/45, 1, 4 and 12 more): no second section on these covers	Conditional	F.44–F.46, Appendix G	Tate conjecture, Artin–Tate formula, a sign rule validated by direct counts
14	If W(E_t) = +1 and E_t has a non-torsion point: dim $Sel_2 \geq 2$, and rank ⩾ 2 or Sha[2^∞] infinite	Proved	§6	2-parity theorem
15	Conjectured patterns of the parity bit (period 4; δ_{2n} = δ_n) are false	Observed	F.31–F.38	complete factorizations up to 163 digits
16	Infinitely many t with rank ⩾ 2	Open	Problem H, F.24	the parity barrier

Reader's guide

Part	Content	Read it for
§1–4	the family, the K3 surface, conductors, root numbers	the definitions and the exact formula for W(E_t)
§5–6	ranks of fibres, sources of rank jumps, the conditional theorem	where rank ⩾ 2 would come from
§7	the quadratic twist and its Picard number	why the cover C_2 gives rank 1 only
§8–9	negative results and open problems	what has been tried
Appendix A	script checking the root-number predictions	reproducing §4

Part	Content	Read it for
Appendix B	Problem H, the periodicity theorem and its lemmas, data	the core of the parity problem
Appendices C–E	coprimality by finite orbits, reflection and doubling, isogeny and Galois groups	tools and side results
Appendix F	sieve, parity programme, searches, covers (index at the start)	everything added after the first version
Appendix G	code and validation values for the Picard-number computations	reproducibility

Notation, cross-references and corrections

Notation.

Symbol	Meaning
J, P, T	$J : y^2 = x^3 - 2x^2 + 2x$; $P = (1, 1)$ of infinite order; $T = (0, 0)$ of order 2; the cover C_2 is J itself
t_n, U_n, V_n	$t_n = 1 - x(nP) = U_n/V_n$ in lowest terms; $V_n = D_n^2$ is a square
f, h, F	$f(t) = t^6 - 30t^4 - 24t^2 - 1$; $h(x) = f(1 - x)$; $F(U, V) = V^6 f(U/V)$
F_n, G_n, N_n	$F_n = F(U_n, V_n)$; G_n the value along $nP + T$; N_n the part of the absolute value of F_n prime to 6
ε_n, a_n, W_n	ε_n the periodic sign; a_n = (−1)^{ω(N_n)}η_n; W_n = W(E_{t_n}) = ε_n·a_n
C_c, S^{d_c}	C_c the genus-one cover carrying the section $x = c$; S^{d_c} the K3 surface of the twist by $(1 + t^2)N_c(t)$; ρ its Picard number

Cross-references. The periodicity theorem of Appendix B (the period-6 law for ε_n and the exact definition of ε) was cited in earlier drafts of F.20–F.46 as "F.17"; those citations now read "Appendix B". The paragraph F.17 itself is the elliptic-net test. The group S_4 cited in F.26 is the one of Appendix E.

Corrections made in this revision.

#	What was wrong	Where fixed
1	Problem H was called equivalent to Chowla's conjecture for F(u, v); Helfgott's equivalence concerns averages over the whole family, not the thin orbit	F.24
2	dim $\mathrm{Sel}_2 \geqslant 2$ was equated with the alternative rank ⩾ 2 or Sha[2^∞] infinite; they are two statements with different proofs	§6, first item of the conditional theorem
3	The claim that all algebraic routes to a parity law are excluded was too strong	F.34
4	The discriminant class at p = 19 used the wrong sign of the functional equation (c = 2: 1 should be 11; c = 13/9: 73 should be 15 or 69); the conclusion ρ = 13 is unchanged	F.45
5	A point-counting routine failed for fibres whose points have small order (found at p = 37, repaired and revalidated)	F.44, F.45
6	Four paragraphs of F.8 and F.9 stood after F.46 and were moved back	Appendix F
7	A period-4 pattern and Conjecture D for the parity bit were proposed; complete factorizations refute both	F.31–F.38

Limitations and provenance

This section states what the paper rests on without independent verification. Where a paragraph of Appendix F reports an independent check, it is named in the last column.

Use of language models. The research and the write-up were done in conversation with a large language model (Claude, by Anthropic) over many sessions; the model versions are not recorded for all of them. The model proposed and checked arguments, wrote the proofs and most of the text, wrote and ran the Python and C code in a sandbox, and carried out the searches and the factorizations, except where a paragraph says that a result was supplied. The person directing the project chose the questions, supplied outside computations (Magma output and a PARI/GP factorization) and review comments, whose own provenance is not recorded here, and accepted or rejected the model's suggestions. Language models make characteristic errors: confident wrong

statements, references recalled from memory, and arguments that look complete but are not. Several occurred in this paper and are listed in the corrections log: an equivalence that was claimed and is false, a wrong sign in a computation, and citations of the wrong paragraph. Every result labelled Proved should therefore be checked by a human before it is relied on, and the code and validation values of Appendix G are included so that the computational claims can be rerun.

Item	What is not independently verified	How to check, or where a check exists
Sections 1–7, Appendices A–E	The labels Proved and Certified in these parts were written before this revision. This revision did not re-derive them; it re-ran only the checks named in F.20–F.46.	Read the proofs of §3.1, §4.1 and Appendix B. The root-number formula has an independent analytic check at eight values of t (F.28, F.40).
Ranks in §5	The rank certificates come from computer-algebra output (generators, regulators, Selmer bounds) that this revision did not recompute. Four rank-3 fibres ($t = -46/9, -12/11, -17/31, 15/8$) are certified there; three more ($t = -86/27, 55/37, 11/60$) are reported from an independent note and their ranks were not recomputed. This revision did not check how the upper bounds behind "rank exactly 3" were obtained.	Rerun the rank computation for these curves; the generators and regulators are listed in §5.

Item	What is not independently verified	How to check, or where a check exists
Point counts and L-functions (F.43–F.46, Appendix G)	The counts come from one C program written for this revision and repaired once (F.44). The sign rule for the functional equation is an empirically validated rule, not a theorem (two direct N_5 checks and four cases admitting only one sign). The conclusion $\rho = 13$ also uses the Tate conjecture for K3 surfaces over finite fields (a theorem for $p \geqslant 5$; its hypotheses were not re-checked here) and the square order of the Brauer group. No second implementation was run.	Compare with Magma or PARI on the validation table of Appendix G. Computing N_5 at $p = 19$ for $c = 13/9$ would decide between the classes 15 and 69.
Factorizations (F.31, F.32, F.38)	Prime factors were accepted on a strong probable-prime test, not proved prime. The 78-digit factorization of F.32 was supplied by the user with PARI/GP. The 94-digit cofactor of N(10P) and the 136-digit composite of N(11P) are not factored.	Prove primality (ECPP) of the large factors; factor the open cofactors.
Heuristic statements	The expectation that rational c with $\rho = 14$ are rare (F.46), the Noether–Lefschetz density on which it rests, the Poisson-parity models and the "expected" statements of F.16 and F.40 are heuristics or are stated from general knowledge, not checked against sources.	Treat them as conjectures.

Item	What is not independently verified	How to check, or where a check exists
Literature	The works in references 7–16 were added in this revision from memory and have not been checked against the sources. F.11 and F.12 are based on abstracts and an introductory chapter, and F.27 on two searches that are not exhaustive.	Verify every bibliographic detail and the hypotheses of every cited theorem before use.
This front matter	The abstract, the results table and the corrections log were written in this revision by the same assistant that wrote much of Appendix F; their labels have not been checked by an independent reader.	Check the table row by row against the cited paragraphs.

Introduction and main results

A primitive Pythagorean triple (a, b, c) gives a rational point (a/c, b/c) of the unit circle, and a point (A, B) of the circle gives the elliptic curve $y^2 = x^3 + 4Ax - 8B$. The sequence $A + iB = ((3 + 4i)/5)^n$ led us to this family. Writing the circle point as $t = \tan(\theta/2)$ makes all these curves fibres E_t of one elliptic surface over $\mathbb{P}^1$. We show that this surface is a K3 surface with generic rank 0, determine the conductor and the root number of E_t for every rational t, and study the ranks of the fibres and the mechanisms that produce rank jumps over the generic rank 0.

The paper has three parts. §§1–4 treat the surface and the arithmetic of the individual fibres: the geometry and $\rho(S) = 16$ (§2), the conductors and local data (§3) and the root numbers (§4). §§5–7 treat ranks: certified examples and a scan (§5), the sources of jumps (§6) and a quadratic twist (§7). §§8–9 record negative results and open problems. Each proof is given directly after the statement it proves (§2.1, §3.1, §4.1). Appendix A contains the script used to check the predictions, Appendices B–F the later results on the parity problem and on the covers, and Appendix G the code for the Picard-number computations. The main results are the following.

- **A (Proved).** The surface S has fibres $2\cdot III^* + 6\cdot I_1$, Picard number 16 and generic Mordell–Weil rank 0 (§2).

- **B (Proved).** The Kodaira types, conductor exponents and the conductor of E_t are determined for every $t \in \mathbb{Q}$, with the finite tables at 2 and 3 and the cases $p \geqslant 5$ (§3).
- **C (Proved; tested on all curves checked).** The global root number of E_t, including the collapse of all primes dividing c to the single Jacobi symbol $(2/c)$ (§4).
- **D (Certified).** Ranks 0, 1, 2 and 3 all occur, with bases and regulators for the curves of rank $\geqslant 1$ listed in §5.
- **E (Proved).** Infinitely many rational t have rank $\geqslant 1$, through the elliptic curve $w^2 = z^3 - 2z^2 + 2z$. The x-fibration has generic rank 0, so jumps are not produced by a section of either fibration. E_t has a rational 2-torsion point for only finitely many t (§6, §8).
- **F (Proved).** The quadratic twist E^d has rank exactly 1 over $\mathbb{Q}(t)$ and geometric rank exactly 1 (§7).
- **Open.** Infinitely many t with rank $\geqslant 2$ (Problem H, reduced to one parity statement; §9, F.24).

Results added after the first version, in Appendices B–F, extend this list:

- **G (Proved).** Along the orbit nP the sign of the root number is periodic with period 6 (period 3 along $nP + T$) up to one explicit parity bit; the 2- and 3-adic valuations and the exact formula for $W(E_t)$, valid for every rational t, are proved (Appendix B, F.20–F.28).
- **H (Proved).** Problem H, infinitely many n with $W(E_{t_n}) = +1$, is equivalent to a parity statement about the number of prime factors of F_n; every route examined leaves exactly that statement (F.13–F.42).
- **I (Conditional).** For 18 constant-x covers C_c the twist has geometric rank 1, so no second section exists over $\bar{\mathbb{Q}}$ (F.44–F.46).
- **J (Observed).** Complete factorizations up to 163 digits refute a period-4 pattern and Conjecture D for the parity bit, and confirm every proved law (F.31–F.38).

1. The family

For $t \in \mathbb{Q}$ let $A(t) = (1 - t^2)/(1 + t^2)$, $B(t) = 2t/(1 + t^2)$, so $A^2 + B^2 = 1$, and

$E_t : y^2 = x^3 + 4A(t)x - 8B(t)$.

Write $t = u/v$ in lowest terms, $(a, b, c) = ((v^2 - u^2)/d, 2uv/d, (v^2 + u^2)/d)$ with $d = \gcd(v^2 - u^2, 2uv, v^2 + u^2) \in \{1, 2\}$, so $a^2 + b^2 = c^2$, c is odd and $A = a/c$, $B = b/c$. Put $F = 4a^3 + 27b^2c$. The integral model $Y^2 = X^3 + 4ac^3X - 8bc^5$ has $c_4 = -192ac^3$, $c_6 = 6912bc^5$ and $\Delta = -1024c^9F$. The orbit $(3 + 4i)^n$ corresponds to $(a, b, c) = (a_n, b_n, 5^n)$ and $t_n = b_n/(c_n + a_n)$.

2. Geometry of the surface

Fibres (Proved). The integral model $Y^2 = X^3 + 4(1 - t^2)(1 + t^2)^3X - 16t(1 + t^2)^5$ is globally minimal, with $\Delta = 4096(1 + t^2)^9D(t)$ and $D(t) = t^6 - 30t^4 - 24t^2 - 1$ (disc $D = 2^6 \cdot 3^{18} \cdot 23^2$). The fibres are $2 \cdot III^*$ (at $t = \pm i$), $6 \cdot I_1$ and a smooth fibre at ∞. The Euler number is 24, so S is a K3 surface with trivial lattice $U \oplus E_7 \oplus E_7$ of rank 16.

ρ(S) = 16 and generic rank 0 (Proved). The good-reduction lemma of Kazalicki–Naskręcki (Lemma 3.11) applies to every prime $p > 5$, $p \neq 23$; I checked that the radicals of a_4, a_6, Δ, f and g have discriminants involving only 2, 3 and 23. At p = 13 the degree-6 L-polynomial is $1 - 12T + 208T^2 - 3042T^3 + 35152T^4 - 342732T^5 + 13^6T^6$. Independent counts of the points of all fibres over $\mathbb{F}_{13}$, $\mathbb{F}_{13^2}$ and $\mathbb{F}_{13^3}$ give exactly its first three coefficients −12, 208 and −3042; since the third is nonzero, the functional equation has sign +1 and determines the rest of the polynomial. All roots have modulus 13, and c(13X) is irreducible over $\mathbb{Q}$, so no zero has the form 13·(root of unity). Tate's inequality gives $\rho(S_{\bar{\mathbb{F}}_{13}}) \leqslant 16$, and injectivity of Néron–Severi specialization gives ρ(S) = 16 and rank $E(\bar{\mathbb{Q}}(t)) = 0$. The same counts at p = 17 give 12, 17 and −6936, which determine $1 + 12T + 17T^2 - 6936T^3 + 4913T^4 + 1002252T^5 + 17^6T^6$ and lead to the same conclusion.

Context. For the lattice-polarized family $H \oplus E_7 \oplus E_7$ a generic member has finite Mordell–Weil group in every Jacobian fibration (Clingher–Malmendier). What is new here is ρ = 16 for this explicit $\mathbb{Q}$-model.

2.1 Details of the proof of ρ(S) = 16

Lower bound. The two III* fibres contribute two root lattices E_7, and together with the zero section and the fibre class the trivial lattice has rank 2 + 7 + 7 = 16. By Shioda–Tate, $\rho(S) = 16 + \text{rank } E(\bar{\mathbb{Q}}(t))$, so $\rho(S) \geqslant 16$ and the rank is at most 4. It remains to show $\rho(S) \leqslant 16$.

Hypotheses of the good-reduction lemma. The nonzero a_i are a_4 and a_6, and $j = 1728(t^2 - 1)^3/D(t)$. The radicals and their discriminants are:

h	disc(h)
$\text{rad}(a_4) = t^4 - 1$	-2^8
$\text{rad}(a_6) = t^3 + t$	-2^2
$\text{rad}(f) = t^2 - 1$	2^2
$\text{rad}(g) = D$	$2^6 \cdot 3^{18} \cdot 23^2$
$\text{rad}(\Delta) = (t^2 + 1)D$	$2^{20} \cdot 3^{18} \cdot 23^2$

Only the primes 2, 3 and 23 occur, so the lemma applies to every prime $p > 5$ with $p \neq 23$. At $p = 13$ the reduction stays globally minimal: D remains squarefree of degree 6 and coprime to $1 + t^2$ ($\mathrm{Res}(1 + t^2, D) = 64$), and $s^8 a_4(1/s)$ equals -4 at $s = 0$.

Upper bound. H^2 of the special fibre has dimension 22. It splits as the 16-dimensional part spanned by the classes of the trivial lattice, on which Frobenius acts with eigenvalues $q\cdot$(roots of unity), plus a 6-dimensional part $W = H^1(\mathbb{P}^1, j_*R^1\pi_*\mathbb{Q}_\ell)$, where $\det(1 - FT \mid W) = L(E/\mathbb{F}_q(t), T)$ and $6 = \deg(\text{conductor}) - 4$ (the conductor has degree $2\cdot 2 + 6 = 10$). By Tate's inequality, $\rho(S_{\bar{\mathbb{F}}_q})$ is at most the number of Frobenius eigenvalues on H^2 of the form $q\cdot\zeta$ with ζ a root of unity. A root of unity of degree at most 6 has order m with $\varphi(m) \leqslant 6$, and if the irreducible polynomial $c(13X)$ had such a root it would be a cyclotomic polynomial with integer coefficients; the coefficient $-12/13$ shows it is not. Hence no zero of L has the form 13ζ, and $\rho(S_{\bar{\mathbb{F}}_{13}}) \leqslant 16$. Injectivity of Néron–Severi specialization gives $\rho(S) \leqslant 16$.

3. Conductors and local data for all $t \in \mathbb{Q}$

Theorem (Proved, using Dokchitser–Dokchitser, Ogg's formula and Rizzo's tables). $F \neq 0$ for every rational t. For primes $p \geqslant 5$: good reduction if $p \nmid cF$; type I_k with $k = v_p(F)$, $f_p = 1$ if $p \mid F$ (then $p \nmid abc$); and for $p \mid c$ (then $p \equiv 1 \bmod 4$) with $m = v_p(c)$: $f_p = 0$ if $4 \mid m$ and 2 otherwise, with types good, III^*, I_0^*, III for $m \equiv 0, 1, 2, 3 \pmod 4$. At 3, with $A, B \in \mathbb{Z}_3$: good if $B \equiv 0 \pmod 3$; otherwise $f_3 = 2, 3, 4$ for $A \equiv 0, 3, 6 \pmod 9$. At 2: $f_2 = 5$ if $A \equiv 1 \pmod 4$, and 6 if $A \equiv 3 \pmod 4$ or A is even. Hence

$$N(E_t) = 2^{f_2}\cdot 3^{f_3}\cdot \prod_{p \mid c,\ 4 \nmid v_p(c)} p^2 \cdot \prod_{p \mid F,\ p \geqslant 5} p.$$

The 2-adic exponents use Table III of Rizzo, whose rows were compared with the printed table (§4). **Observed:** the formula matched the conductor computed by Sage for all 33 parameters scanned and for the orbit $n \leqslant 5$.

3.1 Proof of the local-data theorem

Tools.

- **T1** (Dokchitser–Dokchitser, Theorem 1). For an integral Weierstrass equation over a complete discrete valuation ring with perfect residue field, put $d = \mathrm{Disc}(x^3 + a_2x^2 + a_4x + a_6)$. If $\min v(a_i)/i$ equals $1/6$ the equation is minimal of type II, if it equals $1/4$ type III, if it equals $1/2$ and $v(d) = 6$ type I_0^*, and if it equals $3/4$ type III*.
- **T2** (Dokchitser–Dokchitser, Proposition 2). With $b_6 = a_3^2 + 4a_6$ and $n > 0$: if $v(a_2) = 1$, $v(a_i) \geqslant i/2 + \lfloor (i-1)/2 \rfloor \cdot n/2$ for all i, and ($v(d) = n + 6$ and $v(b_6) \geqslant n + 3$ for n even, or $v(d) \geqslant n + 6$ and $v(b_6) = n + 3$ for n odd), then the equation is minimal of type I^*_n.
- **T3** (Ogg's formula). $f_p = v_p(\Delta_{\min}) + 1 - m$ with $m = 1$ for I_0 and II, 2 for III, 5 for I_0^*, 8 for III* and $n + 5$ for I^*_n.

- **T4.** An integral equation with $v_p(c_4) = 0$ is minimal (a change of variables with $v_p(u) > 0$ would give $c_4' = u^{-4}c_4$ of negative valuation). Its reduction is good if $v_p(\Delta) = 0$ and multiplicative otherwise, with $f_p = 1$ (Silverman, AEC VII.5.1).

Setup. Since $\gcd(u, v) = 1$, the odd prime factors of $v^2 + u^2$ are all $\equiv 1 \pmod 4$ and $3 \nmid v^2 + u^2$. Also $c \equiv 1 \pmod 4$: for $d = 1$, $v^2 + u^2 \equiv 1$, and for $d = 2$, $(v^2 + u^2)/2 \equiv 1$. The numbers a, b, c are pairwise coprime. The integral model $Y^2 = X^3 + 4ac^3X - 8bc^5$ comes from $x = X/c^2$, $y = Y/c^3$. Also $4A^3 + 27B^2 = F/c^3 = -4D(t)/(1 + t^2)^3$ with $D(t) = t^6 - 30t^4 - 24t^2 - 1$, a monic polynomial with $D(\pm 1) = -54$, so it has no rational root and $F \neq 0$. If a prime $p \geqslant 5$ divides F then $p \nmid abc$ (if $p \mid a$ then $p \mid 27b^2c$, so $p \mid b$ or $p \mid c$, and in either case p divides all of a, b, c); if $p \mid c$ then $F \equiv 4a^3 \not\equiv 0 \pmod p$.

Primes $p \geqslant 5$. If $p \nmid cF$ the integral model has $v_p(\Delta) = 0$ and the reduction is good. If $p \mid F$ then $v_p(c_4) = 0$, so by T4 the reduction is multiplicative of type I_k with $k = v_p(F)$, $f_p = 1$. If $p \mid c$ with $m = v_p(c) = 4q + r$, the integral model has $(v_p(c_4), v_p(c_6), v_p(\Delta)) = (3m, 5m, 9m)$ exactly, which reduces mod $(4, 6, 12)$ to $(0, 2q, 0)$, $(3, 2q + 5, 9)$, $(2, 2q + 4, 6)$ and $(1, 2q + 3, 3)$ for $r = 0, 1, 2, 3$. These give good reduction, III*, I_0^* and III (Rizzo's Table I, or T1 on the scaled model), and $f_p = 0, 2, 2, 2$ by T3.

The prime 3. Since $3 \nmid c$, $A, B \in \mathbb{Z}_3$ with $B^2 = 1 - A^2$. If $3 \mid B$ then $A \equiv \pm 1$ and $v_3(4A^3 + 27B^2) = 0$, so the reduction is good. Otherwise $3 \mid A$ and B is a unit. Write $A = 3\alpha$. Then $4A^3 + 27B^2 = 27(4\alpha^3 + B^2)$ and $4\alpha^3 + B^2 \equiv 4\alpha^3 + 1 \pmod 9$, which is a unit for $\alpha \equiv 0, 1 \pmod 3$ and $\equiv -3 \pmod 9$ for $\alpha \equiv -1 \pmod 3$. Hence $v_3(\Delta) = 3$ for $A \equiv 0, 3 \pmod 9$ and $v_3(\Delta) = 4$ for $A \equiv 6 \pmod 9$. The singular point of the reduction has $x \equiv -B$; translating $x = X - B$ gives $a_1 = a_3 = 0$, $a_2 = -3B$, $a_4 = 3(1 + 4\alpha - 9\alpha^2)$ and $a_6 = -B(9 + 12\alpha - 9\alpha^2) \equiv -3\alpha B \pmod 9$. So $v(a_2) = 1$, $v(a_4) \geqslant 1$, and $v(a_6) = 1$ exactly when $3 \nmid \alpha$. By T1: if $3 \nmid \alpha$ ($A \equiv 3, 6 \bmod 9$) then $\min v(a_i)/i = 1/6$ and the type is II; if $3 \mid \alpha$ ($A \equiv 0 \bmod 9$) then $v(a_6) \geqslant 2$ and $v(a_4) = 1$, so the minimum is $1/4$ and the type is III. By T3, $f_3 = 4, 3, 2$ for $A \equiv 6, 3, 0 \pmod 9$.

The prime 2. If A is odd then $A^2 \equiv 1 \pmod 8$, so $B^2 \equiv 0 \pmod 8$, $4 \mid B$ and $A \equiv 1$ or $7 \pmod 8$. If A is even then B is odd and $A^2 = 1 - B^2 \equiv 0 \pmod 8$, so $4 \mid A$. We use $c_4 = -192A$ and $c_6 = 6912B$, with $192 = 2^6 \cdot 3$ and $6912 = 2^8 \cdot 27$.

- **A odd.** Then $v_2(c_4) = 6$, $v_2(c_6) = 8 + v_2(B) \geqslant 10$, and $v_2(4A^3 + 27B^2) = 2$, so $v_2(\Delta) = 12$. The reduced triple is $(2, \geqslant 4, 0)$. Rizzo's Table III gives type I_2^* with $v(N) = 6$ when $A \equiv 3 \pmod 4$, and type I_3^* with $v(N) = 5$ when $A \equiv 1 \pmod 4$.
- **A even.** Then $v_2(c_6) = 8$, $v_2(c_4) \geqslant 8$ and $v_2(\Delta) = 10$, so the triple $(\geqslant 8, 8, 10)$ lies in the row $(\geqslant 7, 8, 10)$: type I_0^*, $v(N) = 6$.

These exponents agree with Ogg's formula: $12 + 1 - 7 = 6$, $12 + 1 - 8 = 5$, $10 + 1 - 5 = 6$.

Which classes occur. $A + iB = (v + iu)/(v - iu)$, and $(A, B) \bmod 16$ depends only on $(u, v) \bmod 16$ (for u, v of opposite parity the denominator is a unit and squares mod 16 depend on $u, v \bmod 8$; for both odd one divides numerator and denominator by 2 and squares mod

32 depend on u, v mod 16). Running over the 192 residue pairs with u, v not both even gives exactly the 16 classes (A, B) mod 16 listed in §4; the other 16 solutions of $A^2 + B^2 \equiv 1 \pmod{16}$ do not occur. At 3 the denominator is a unit, (A, B) mod 9 depends on (u, v) mod 9, and 12 classes occur.

The orbit $(3 + 4i)^n$. For (a, b, c) = $(a_n, b_n, 5^n)$, $F = D_n = 4a_n^3 + 27 \cdot 5^n b_n^2$ and, with ε = −1 (the sign of this model),

prime	condition	type	v	f
2	n even	I_3^*	12	5
2	n odd	I_2^*	12	6
3	n even	I_0	0	0
3	$n \equiv 1, 11 \pmod{12}$	II	4	4
3	$n \equiv 5, 7 \pmod{12}$	II	3	3
3	$n \equiv 3, 9 \pmod{12}$	III	3	2
5	$n \equiv 0, 1, 2, 3 \pmod{4}$	I_0, III*, I_0^*, III	0, 9, 6, 3	0, 2, 2, 2

The classes mod 9 follow from $(a_n, b_n) \equiv$ (1,0), (3,4), (2,6), (0,8), (4,6), (6,7), (8,0), (6,5), (7,3), (0,1), (5,3), (3,2) for n = 0, …, 11 (period 12). For this sequence the types at 2 can also be obtained without tables from T2: with $A_n \equiv 1$, $B_n \equiv 0 \pmod 8$ for n even and (7, 4) for n odd, the substitution x = X + 2 (and y = Y + 4 for n even) gives coefficient valuations ($v(a_2)$, $v(a_3)$, $v(a_4)$, $v(a_6)$) = (1, ∞, 3, 5) for n odd and (1, 3, 4, ⩾ 6) for n even, with v(d) = 8, $v(b_6) = 7$ and v(d) = 10, $v(b_6) = 6$, so T2 gives I_2^* and I_3^*. At 5 the scaling x = 5^{−2k}X with k = ⌈n/4⌉ gives the valuations of the scaled model ($v(a_4')$, $v(a_6')$, $v(\Delta')$) = (4k − n, ⩾ 6k − n, 12k − 3n) and T1 gives the stated types. These are the results behind the computed conductors 907200, 77485600, 17006400, 555649376 and 327672840715200 for n = 1, …, 5.

4. Root numbers

Theorem (Proved). With $w_\infty = -1$,

$W(E_t) = -w_2 \cdot w_3 \cdot (2/c) \cdot (-1)^{\omega(R_F)} \cdot (ab/R_F)$,

where R_F is the product of the primes p ⩾ 5 dividing F and the symbols are Jacobi symbols. The ingredients are:

- **p | F:** $w_p = -(ab/p)$.

- **p | c:** w_p = (2/p)^{v_p(c)}, and since every prime factor of c is ≡ 1 (mod 4) the product over all p | c is the single Jacobi symbol (2/c), equal to +1 for c ≡ 1 (mod 8) and −1 for c ≡ 5 (mod 8).
- **p = 3:** $w_3 = -1$ exactly when (A, B) ≡ (3, 8) (mod 9). Only 12 classes mod 9 occur.
- **p = 2:** $w_2 = -1$ exactly when (A, B) mod 16 lies in S_2 = {(0,1), (1,0), (4,9), (8,1), (9,4), (12,9), (15,0), (15,8)}. Only 16 classes mod 16 occur, by a finite check over (u, v) mod 16.

Source check. The rows of Rizzo's tables used here were compared with the printed Tables I–III and match: Table I, the rows (3, ⩾5, 9), (2, ⩾3, 6) and (1, ⩾2, 3) with W_p = (−2/p), (−1/p), (−2/p) and Lemma 7; Table II, the row (2, 3, 4) and both branches of (⩾2, 3, 3); Table III, the rows (2, ⩾4, 0), (2, 4, 0), (2, ⩾5, 0) and (⩾7, 8, 10), including the condition "5, 9 (mod 16)" and the distinction between c_6' and c_{6,4}. The normalizations of c_{6,4} for $v_2(c_6) \geqslant 12$ and of c_{4,2} for $v_3(c_4) \geqslant 4$ are the ones used in §4.1, and they are confirmed computationally by the local signs at 2 and 3 for n ⩽ 130.

Observed. The formulas reproduced the conductor and root number of every curve in the 33-parameter scan, the independent Magma table for the orbit n ⩽ 24, PARI's local root numbers at 2, 3 and 5 for n ⩽ 130 and both signs of ε, and the global sign for n ⩽ 14. They also predicted W = −1 for the four rank-3 curves of §5 before any rank was known.

Structure. The sign is a periodic part at 2, 3, 5 times (2/c), times a Liouville-type factor (−1)^{ω(R_F)}(ab/R_F). No closed formula in t alone is expected for the last factor.

4.1 Proof of the root-number theorem

Conventions. The local root numbers at 2, 3 and p ⩾ 5 are read from Rizzo's Tables I–III. A row is indexed by the triple $(v(c_4), v(c_6), v(\Delta))$ reduced mod (4, 6, 12); $x' = x/p^{v(x)}$; the special condition of a row is the condition for W_p = +1. At good reduction W_p = +1; at multiplicative reduction W_p = −1 exactly when the reduction is split, that is, when $-c_6$ is a square mod p; and w∞ = −1. For E_t, $c_4 = -192A$ and $c_6 = 6912B$.

The primes p | F, p ⩾ 5. Here $(-c_6/p) = (-6912bc^5/p) = (-3bc/p)$, and from $27b^2c \equiv -4a^3$ (mod p) we get (3c/p) = (−a/p), so $(-c_6/p) = (ab/p)$ and w_p = −(ab/p).

The primes p | c. By Table I, w_p = (−2/p) for III* and III and (−1/p) for I_0^*. Since p ≡ 1 (mod 4), (−1/p) = 1 and (−2/p) = (2/p), so w_p = (2/p)^{v_p(c)} for every v_p(c). The product over all p | c is the Jacobi symbol (2/c).

The prime 3. Put A = 3α, so $c_6' = c_6/27 = 256B \equiv 4B$ (mod 9) and c_{4,2} = $c_4/9 = -64\alpha$, hence $3c_{4,2} \equiv 6\alpha$ (mod 9). Since $B^2 \equiv 1$ (mod 9) when 3 | A, we have B ≡ ±1 and $c_6'^2 \equiv 7$, so $c_6'^2 + 2 \equiv 0$ (mod 9). The equality $c_6'^2 + 2 \equiv 3c_{4,2}$ (mod 9) therefore holds exactly when α ≡ 0 (mod 3). Table II gives:

- the triple (2, 3, 4) (A ≡ 6 mod 9): type II, $W_3 = +1$, with no condition;
- the triple (⩾2, 3, 3) with equality (A ≡ 0 mod 9): type III, $W_3 = +1$;

- the triple (⩾2, 3, 3) without equality ($A \equiv 3 \bmod 9$): type II, with $W_3 = +1$ exactly when $c_6' \equiv 4, 7, 8 \pmod 9$. Here $B \equiv 1$ gives $c_6' \equiv 4$, so $W_3 = +1$, and $B \equiv 8$ gives $c_6' \equiv 5$, so $W_3 = -1$.

If $3 \mid B$ the reduction is good and $W_3 = +1$. So $w_3 = -1$ exactly when $(A, B) \equiv (3, 8) \pmod 9$.

The prime 2. Put $c_4' = -3A$ and $c_{6,4} = c_6/2^{10} = 27B/4$.

- A odd and $A \equiv 3 \pmod 4$: the row (2, ⩾4, 0) with $c_4' \equiv 3 \pmod 4$ gives $W_2 = +1$ exactly when $v_2(c_6) = 10$, that is, $B \equiv 4 \pmod 8$.
- A odd and $A \equiv 1 \pmod 4$: the rows (2, 4, 0) ($B \equiv 4 \bmod 8$) and (2, ⩾5, 0) ($B \equiv 0 \bmod 8$) with $c_4' \equiv 1 \pmod 4$. With $s = c_4' + 4c_{6,4} \equiv -3A + 27B \pmod{16}$, $W_2 = +1$ exactly when $s \equiv 9, 13 \pmod{16}$ in the first case and $s \equiv 5, 9 \pmod{16}$ in the second.
- A even: the row (⩾7, 8, 10) gives $W_2 = +1$ exactly when $c_6' = 27B \equiv 1 \pmod 4$, that is, $B \equiv 3 \pmod 4$.

Evaluating on the 16 classes gives $w_2 = -1$ exactly on $S_2 = \{(0,1), (1,0), (4,9), (8,1), (9,4), (12,9), (15,0), (15,8)\}$.

How the tables are applied, with two worked checks. The input to Rizzo's tables is always the triple $(v_p(c_4), v_p(c_6), v_p(\Delta))$ of the model $y^2 = x^3 + 4Ax - 8B$ with $A, B \in \mathbb{Z}_p$, reduced mod (4, 6, 12), together with the digits c_4' and c_6' of $c_4 = -192A$ and $c_6 = 6912B$. The finite tables in terms of (A, B) mod 16 and mod 9 are consequences of the rows, not lookups of $(v^2 - u^2, 2uv)$. At $t = 1$ the model $y^2 = x^3 - 8$ has triple $(\infty, 8, 10)$ and $c_6' = 27 \equiv 3 \pmod 4$, so the row (⩾7, 8, 10) gives $W_2 = -1$, as Magma does. At $t = 3/4$ the integral model $y^2 = x^3 + 700x - 120000$ has triple (6, 11, 12) ↦ (2, 5, 0) and $c_4' = -525 \equiv 3 \pmod 4$, so the row (2, ⩾4, 0) applies, with type I_2^*, $v(N) = 6$ and $W_2 = -1$ because $b = 5 \neq 4$; this agrees with Magma's conductor exponent 6 and root number. Rizzo's list of semi-admissible but non-admissible triples at $p = 2$ does not contain (6, 11, 12). A published erratum (Várilly-Alvarado, Remark 4.2) concerns the row (⩾5, 6, 9) of Table II, which our 3-adic analysis never uses; the rows that are used were compared with the printed tables and match.

Assembly. With $w_\infty = -1$ the product of the local factors is $-w_2 w_3 \cdot (2/c) \cdot \Pi_{p \mid F, p \geqslant 5}(-(ab/p)) = -w_2 w_3 (2/c)(-1)^{\omega(R_F)}(ab/R_F)$, because R_F is squarefree and the Jacobi symbol is the product of Legendre symbols over its prime factors.

The orbit $(3 + 4i)^n$, $\varepsilon = -1$. Here $(a, b, c) = (a_n, b_n, 5^n)$, $F = D_n$, and $(2/5^n) = (-1)^n$, so $w_5 = (-1)^n$. For $n = 2k$, $\omega^2 = (-7 + 24i)/25 \equiv 1 + 8i \pmod{16}$ gives $A_n \equiv 1$ and $B_n \equiv 8k \pmod{16}$, hence $c_4' \equiv 13$ and $s \equiv 13 + 8k \pmod{16}$: this is $\equiv 5$ for k odd ($W_2 = +1$) and $\equiv 13$ for k even ($W_2 = -1$). For odd n, $A_n \equiv 7$ and $B_n \equiv 4 \pmod 8$, so $W_2 = +1$. So $w_2 = -1$ exactly when $4 \mid n$. At 3, B_n mod 9 is 8, 1, 8, 1, 8, 1 for $n \equiv 1, 3, 5, 7, 9, 11 \pmod{12}$ and $A_n \equiv 6, 0, 3, 3, 0, 6$, so $(A, B) \equiv (3, 8)$ exactly when $n \equiv 5 \pmod{12}$. For $\varepsilon = +1$ the sign of B flips and the class is $n \equiv 7 \pmod{12}$; the two curves are twists by −1, and since the twist is unramified at 3 and $f_3 = 3$ for $n \equiv 5, 7$ the sign changes there, as the data show.

5. Ranks: certified examples and a scan

Certified bases (exact arithmetic checks plus Magma).

- **t = 4/3 (n = 2), rank 2:** points (64/25, 312/125), (4, 36/5); regulator 9.894.
- **t = 11/2 (n = 3), rank 2:** points (56/25, 24/125), (116/25, 1116/125); regulator 11.294.
- **t = −41/38 (n = 5), rank 1:** generator $(-28919164/251^2, 690221691916/251^3)$ on $y^2 = x^3 - 118500x + 1947500000$; regulator 17.971.
- **Four rank-3 specializations:** t = −46/9, −12/11, −17/31, 15/8, with regulators 87.60, 2776.9, 46.75, 229.31. I recomputed each regulator from the three Magma generators by canonical heights, and the values agree to within 0.1%.

In each case torsion is trivial (checked by point counts) and the points are non-torsion and independent. The upper bounds come from Magma V2.30-1 (no GRH setting was used). So ranks 0, 1, 2 and 3 all occur, and each positive rank is a jump over the generic rank 0.

Scan (Observed). For 33 parameters ($|u| \leqslant 11$, $v \leqslant 7$) ranks from PARI `ellrank` through Sage had equal bounds in every case: 15 curves with W = −1 all have rank 1; of the 18 with W = +1, 11 have rank 0 and 7 have rank 2. Rank parity equals W in all 33 cases and no rank $\geqslant 3$ occurs. Selection towards small conductors biases these counts, so they say nothing about frequencies.

The scanned parameters (Observed). The 33 curves, grouped by sign and rank:

- **W = −1, rank 1 (15 curves):** t = −1, 2, −1/2, 3, −3, 4, 7, 2/3, 3/2, 1/7, −1/7, 3/4, 1/5, −1/5, 1/4.
- **W = +1, rank 0 (11 curves):** t = 0, 1, −2, 1/2, 1/3, 5, −5, −2/3, −3/2, −3/4, −1/4.
- **W = +1, rank 2 (7 curves):** t = −1/3, −4, 11/2, −11/2, −7, 4/3, −4/3.

These parameters reach 15 of the 16 classes (A, B) mod 16 (all except (15, 0)) and 11 of the 12 classes mod 9 (all except (8, 0)). The four further parameters t = ±1/8 and ±1/9 reach the two remaining classes, (15, 0) mod 16 and (8, 0) mod 9, so every residue class of the 2-adic and 3-adic tables has been tested. For all four, Magma gives conductors 32661345600 (t = ±1/8) and 37188238528 (t = ±1/9) and root numbers −1, −1, +1, +1, exactly as predicted. An independent computation reported ranks 2, 1, 1, 2 at t = 8, −7/5, 7/8, 1/9; the signs predicted by the formula (+1, −1, −1, +1) agree with these ranks.

Three further rank-3 parameters (Reported in an independent note; signs checked). The parameters t = −86/27, 55/37 and 11/60 were reported to have rank 3, with rank 2 for E_{-t}, so that the rank over $\mathbb{Q}(i)$ is 5; the equations follow from the family formula (for example $E_{-86/27} : y^2 = x^3 - (26668/8125)x + 37152/8125$). I did not recompute the ranks; the formula of §4 gives $W(E_t) = -1$ and $W(E_{-t}) = +1$ for all three (conductors $4.79 \cdot 10^{15}$, $8.77 \cdot 10^{14}$ and $2.04 \cdot 10^{16}$, equal for t and −t), as rank 3 and rank 2 require. The coefficients

listed for the four earlier rank-3 curves were rechecked from A(t) and B(t); in particular t = −17/31 has a_4 = 1344/625 and a_6 = 4216/625, and the pair 1344/961, −12288/961 that circulated earlier does not belong to the family.

2-Selmer ranks of the scanned curves (Observed, Magma V2.30-1). For 48 parameters (the 33 scanned, ±1/8, ±1/9, 8, −7/5, 7/8, ±3/7, ±5/7 and the four rank-3 curves) Magma's `TwoSelmerGroup` and `RankBounds` give s_2 = dim Sel_2 − dim E(ℚ)[2] and rank bounds [r, r] in every case, so dim Sha[2] = s_2 − r:

W	s_2	rank	dim Sha[2]	curves	parameters
−1	1	1	0	17	−1, 2, −1/2, 3, −3, 4, 7, 2/3, 3/2, 1/7, −1/7, 3/4, 1/5, −1/5, 1/4, 1/8, −1/8
−1	3	1	2	2	−7/5, 7/8
−1	3	3	0	4	−46/9, −12/11, −17/31, 15/8
+1	0	0	0	8	0, 1, −2, 1/2, 1/3, −5, −2/3, −3/4
+1	2	0	2	7	5, −3/2, −1/4, −1/9, 3/7, −3/7, 5/7
+1	2	2	0	10	−1/3, −4, 11/2, −11/2, −7, 4/3, −4/3, 1/9, 8, −5/7

In all 48 cases (−1)^{s_2} = W, as the 2-parity theorem requires, and the root number computed by the formula of §4 equals Magma's. Sha[2] ≠ 0 occurs for 9 of the 48 curves (t = 5, −3/2, −1/4, −1/9, −7/5, 7/8, 3/7, −3/7, 5/7) and always with dimension exactly 2. At t = 3/7 the rank is 0, not 2. Among the 17 curves with W = +1 and s_2 = 2, 10 have rank 2 and 7 have rank 0; all the rank-2 curves arising from the covers C_c of §6 (t = −7, −1/3, 4/3, −11/2) are in the first group. The twist by −1 acts as t ↦ −t, and the pairs t, −t often differ (t = 1/9 has rank 2 and t = −1/9 rank 0 with dim Sha[2] = 2).

6. Where jumps come from

The x-fibration (Proved). Setting x = c gives $y^2(1 + t^2) = (c^3 − 4c)t^2 − 16t + (c^3 + 4c)$, that is the genus-one curve $C_c : w^2 = (1 + t^2)\cdot N_c(t)$. The Jacobian of this fibration is $y^2 = x^3 − 27Ix − 27J$ with $I = 16(c^6 − 12c^2 − 48)$, $J = 128c^3(c^6 − 18c^2 − 72)$ and discriminant proportional to $(c^2 + 4)^2(c^6 − 16c^2 − 64)$. Its fibres are I_8* at ∞, I_2 at c = ±2i and six I_1, with Euler number 24 and trivial lattice rank 16, so its generic Mordell–Weil rank is 0. No rational c gives a degenerate C_c, so every C_c has genus one. Consequently jumps are not explained by a section of either fibration.

An explicit infinite family (Proved). For c = 2 the cover is $w^2 = (1 − t)(1 + t^2)$; with z = 1 − t it is the elliptic curve $w^2 = z^3 − 2z^2 + 2z$. The point (1, 1) is non-torsion (its double has x = 1/4, so Nagell–Lutz applies). For every rational point of this curve, E_t has a rational point $(2, 4w/(1 + t^2))$. The first members are t = 0, −1, 3/4, −7, −40/9, 31/49, −561/400, 161/961, 34440/34969, −69409/529. For each member except t = 0 I checked that the point is not

killed by any $n \leqslant 12$, hence non-torsion by Mazur. For all but finitely many members it is non-torsion, so **infinitely many rational t have rank $\geqslant 1$**. The curves for c = 1 and c = 4 also have positive rank (Jacobian bounds 1), with pairwise non-isogenous Jacobians.

Two further positive-rank covers (Proved). The points of the rank-2 fibre t = −7 have x-coordinates 2, −11/5, 14/45, 94/245, 1001/400 and 229/1805, and each gives a genus-one cover C_c through t = −7. For c = −11/5 the cover has the rational points (t, w) = (−7, 10), (−201/43, 34190/1849), (−87/59, 2210/3481) and their sign-reflected partners, and for c = 14/45 the points (−7, 1400/27), (2/161, 103700/99981) and their partners; I checked that each lies on its curve. Counting points of the quartic over $\mathbb{F}_p$, including points at infinity, for primes $p \geqslant 7$ of smooth reduction gives $\#C(\mathbb{F}_p)$ = 24, 30, 30, 40, 44, 32, 48, … for c = −11/5 and 22, 30, 40, 28, 50, 40, … for c = 14/45, with greatest common divisor 2 over all smooth primes below 100 in both cases. Since C has a rational point, $\#J(\mathbb{F}_p) = \#C(\mathbb{F}_p)$, and torsion injects into $J(\mathbb{F}_p)$ for odd good p, so the torsion of $J_c(\mathbb{Q})$ has order at most 2. The first curve has at least 6 and the second at least 4 distinct rational points, so both Jacobians have rank $\geqslant 1$ and each cover has infinitely many rational points. As for C_2, this gives infinitely many t with a rational point at x = c, non-torsion for all but finitely many t. The new members t = −201/43, −87/59 and 2/161 have predicted root numbers −1 in all three cases (conductors of 19, 21 and 23 digits). Primes of bad reduction must be excluded: including them lowers the gcd to 1.

The six covers through t = −7 (Proved, with Magma's confirmation of infinite order). The fibre $E_{-7} : y^2 = x^3 - (96/25)x + 56/25$ has rank 2. Its points of abscissae 2, −11/5, 14/45, 94/245, 1001/400 and 229/1805 are, up to sign, $P_1 + P_2$, P_1, P_2, $2P_1 + P_2$, $2(P_1 + P_2)$ and $P_1 + 2P_2$, where $P_1 = (-11/5, 1/5)$ and $P_2 = (14/45, 28/27)$ (checked by exact group arithmetic). The canonical heights are $\hat{h}(P_1) = 2.7953$, $\hat{h}(P_2) = 3.7225$ and the pairing is $\langle P_1, P_2 \rangle = -2.3785$, so the regulator is 4.7483 and P_1, P_2 are independent. Magma confirms that P_1 and P_2 have infinite order. The generic sections $P_c = (c, w/(1 + t^2))$ of E over $\mathbb{Q}(C_c)$ are therefore non-torsion for c = −11/5 and c = 14/45: if either were torsion, so would be its specialization at t = −7. The same holds for the other four covers through t = −7. The six covers are different views of the same rank-2 group $E_{-7}(\mathbb{Q})$, and t = −7 is a rational point of each fibre product of two of them, which has genus $\geqslant 2$ and finitely many rational points.

The cover $C_{-11/5}$ has rank 2 (lower bound Proved by exact heights, upper bound by Magma). Taking the point (t, w) = (−7, 10) as origin, the quartic $w^2 = (1 + t^2)((c^3 - 4c)t^2 - 16t + c^3 + 4c)$ with c = −11/5 becomes the Weierstrass model $Y^2 = Z^3 - (114304/125)Z^2 + (173816064/625)Z - 88367865856/3125$. In this model the point over t = −87/59 with w = +2210/3481 is the 2-torsion point (Y = 0); the points over t = −201/43 (both signs of w), over t = −87/59 (w = −2210/3481) and over t = −7 (w = −10) have infinite order, with $\hat{h}$ = 9.446, 8.221, 10.914 and 10.914 (log-height of x normalization) and Gram matrix of rank 2 (singular values 27.41, 12.09, $5 \cdot 10^{-4}$, $3 \cdot 10^{-4}$), the 2 × 2 minors being 66.25 (the points over t = −7 and over t = −87/59 differ by the torsion point). So $C_{-11/5}(\mathbb{Q})$ has rank at least 2;

Magma's RankBounds gives (2, 2), with independent generators of heights 8.2204 and 10.9133, pairing 4.8440 and regulator 66.2470, in agreement with the values above. The rational points of this cover thus form a rank-2 lattice, and the members t = t(mP + nQ) form a two-parameter family of fibres carrying a non-torsion point; each still carries one section, so rank 2 of the base does not give two sections of E.

A cover from a nonconstant x, the curve C_4' (Proved, rank of C_4' reported by Sage). For $x(t) = -2(t + 1)/(t - 1)$, with inverse $t = (w - 2)/(w + 2)$, one has $x^3 + 4Ax - 8B = w^3 + 8$ identically (verified), so E_t has a point with this x whenever $v^2 = w^3 + 8$. The cover of the t-line is $C_4' : v^2 = w^3 + 8$, equivalently $z^2 = (1 - t)(1 + 3t^2)$. This elliptic curve has the point (1, 3) of infinite order and rank 1 (Sage, reported in an independent note), so it has infinitely many rational points, and the section is non-torsion at the specialization w = 1 and hence for all but finitely many members. The first members $t = (w(nR) - 2)/(w(nR) + 2)$ are −1/3, −15, 191/675 and 25665/36481, with signs W = +1, −1, +1, −1 by the formula of §4 (conductors of 7, 13, 29 and 49 digits). The twist $E^{(1-t)(1+3t^2)}$ has fibres $2\cdot III^* + 4\cdot I_0^* + 6\cdot I_1$ (Euler number 48, independent report) and its $\mathbb{Q}(t)$-rank is estimated at 0.84 with 235 primes (the same method gives 0.89 for the cover C_2 and −0.09 for $1 + t^2$), so it is 1 and not 2. The conditional theorem below applies to this cover as well, with hypothesis H for its members; W = +1 occurs at t = −1/3 (rank 2) and t = 191/675.

Two families of genus-one covers (positive rank Proved case by case; estimates Observed). In both families the criterion for positive rank is the one used for c = −11/5: the number of rational points found exceeds the bound on the torsion of the Jacobian given by the greatest common divisor of $\#C(\mathbb{F}_\ell)$ over the primes $\ell < 70$ of smooth reduction.

- **Constant x = c.** For c = p/q with |p|, q ⩽ 45 and t = u/v with |u|, v ⩽ 600, the covers C_c of positive rank are c = 1, 2, 4, 5, 14, 17, 32, 8/5, 13/5, 28/5, 4/9, 10/9, 13/9, 22/9, 37/9, 40/9, 10/17, 16/25, 18/25, 26/25, 14/45, −2/5, −4/25, −11/5, −7/4, −6/5, −18/13, −14/13, −24/25 and −16/25. Those with the most points found are c = 2 (8 values of t), c = −2/5 (t = 18, 57, −1/7, −2/11) and c = −4/25 (t = −103/4, −18/151, −203/396, −315/572). Each of these is the cover through a rational point of some known fibre (for instance c = −18/13 and c = 18/25 contain the rank-3 fibres t = −46/9 and t = −17/31, and c = 22/9 contains t = 15/8).
- **Rotated x.** The function $x_s(t) = 2(t - s)/(1 + st)$, twice the rotation of the circle parameter t by s, satisfies $x_s(\pm i) = \pm 2i$, so $1 + t^2$ divides the numerator M and the cover has genus ⩽ 1 for every rational s. For s = m/n it is $w^2 = -2(mt + n)K(t)$ with $K(t) = n(m^2 - n^2)t^3 + m(m^2 + 5n^2)t^2 + n(n^2 - m^2)t + m(m^2 + n^2)$. The value s = −1 is the cover C_4' and s = 0 gives the rank-0 curve $w^2 = 2t(t^2 - 1)$. For |m| ⩽ 12, n ⩽ 12 and |u|, v ⩽ 300, positive rank occurs for s = 3/4, 7, 1/2, 5/2, 11/2, 1/3, 4/3, 9/5, 1/6, 1/7, 10/9 and 12/11. The cover s = 3/4 has five values of t (−11/2, −4/3, 203/4, −202/39, −58/39), among them both rank-2 fibres t = −11/2 and t = −4/3; the corresponding rotation is by the angle of $7 + 24i = (4 + 3i)^2$.

- **Ranks of the twists.** The estimated $\mathbb{Q}(t)$-rank of the twist attached to each of the 508 rotated covers with $|m|, n \leqslant 20$ is 1 (121 primes; the largest values are 1.26 and 1.16), and the same holds for all c of height $\leqslant 70$ (§8). No cover in either family has a twist of rank 2.

Biquadratic covers over C_4' (Proved genus statement, Observed ranks). Let C be the fibre product of $v^2 = (1 - t)(1 + 3t^2)$ with $z^2 = f(t)$, a $(\mathbb{Z}/2)^2$-cover of the t-line. If it is branched at b points, each with two points above it, Riemann–Hurwitz gives $g = b - 3$. The first cover is branched at $t = 1, \pm i/\sqrt{3}$ and ∞, so $g \leqslant 1$ requires the branch points of f to lie in this set, that is $f = k$, $k(1 - t)$, $k(1 + 3t^2)$ or $k(1 - t)(1 + 3t^2)$ with $k \in \mathbb{Q}^*$; every other f gives $g \geqslant 2$ and finitely many rational points (Faltings). For example $f = (1 - t)(1 + t^2)$ (the cover C_2) has branch set $\{1, \pm i/\sqrt{3}, \pm i, \infty\}$ and $g = 3$. In the genus-$\leqslant 1$ cases the rank over $\mathbb{Q}(C)$ is rank $E^{d_1} + \text{rank } E^f + \text{rank } E^{fd_1}$ with $d_1 = (1 - t)(1 + 3t^2)$, and the estimated $\mathbb{Q}(t)$-ranks are: 0.93 for $k \cdot d_1$ with $k = 1$ and at most 0.13 for the other squarefree k with $|k| \leqslant 30$, and at most 0.19 for k, $k(1 - t)$, $k(1 + 3t^2)$ and $k(1 + t^2)$ in the same range. So no such cover has rank 2 over $\mathbb{Q}(C)$.

Fibre products of the known covers (Proved rule, Observed list). For two double covers $w^2 = d_1(t)$ and $z^2 = d_2(t)$ the fibre product has genus $b - 3$, where b is the number of points of $\mathbb{P}^1$ at which at least one of d_1, d_2 has odd order of vanishing (the point ∞ counts when the degree is odd), $b = |\mathrm{Br}(d_1) \cup \mathrm{Br}(d_2)|$. All the known positive-rank covers have $|\mathrm{Br}| = 4$. Two of them that are not constant multiples of each other therefore have $b \geqslant 5$ and genus $\geqslant 2$, and genus $\leqslant 1$ would need $\mathrm{Br}(d_1) = \mathrm{Br}(d_2)$, that is $d_2 = k \cdot d_1$, where the fibre product is not geometrically irreducible ($z = \pm\sqrt{k} \cdot w$) and its rational points are the finitely many zeros of d_1. For the 44 covers listed above (30 constant-x covers and 14 rotated covers, 945 pairs) the genus of the fibre product is 3 for 437 pairs and 5 for 508 pairs; no pair has genus $\leqslant 2$. A pair with genus 1 needs covers with $|\mathrm{Br}| = 2$ (a linear or irreducible quadratic d, a conic), and then the three quotients of the biquadratic cover are the twists by d_1, d_2 and $d_1 d_2$, so the rank over its function field is the sum of the ranks of E^{d_1}, E^{d_2} and $E^{d_1 d_2}$; among the 12672 linear and quadratic twists tested, none has a section, so the rank is at most 1.

Covers of C_4' itself, and genus-zero covers (Observed). A double cover of C_4' of genus $\leqslant 1$ is unramified (Riemann–Hurwitz), hence an isogeny cover; over $\mathbb{Q}$ the only one up to twisting by constants is $v(k(w + 2)) = v(k \cdot 4/(1 - t))$, the case $f = k(1 - t)$ above. For the rational 3-isogeny from $y^2 = x^3 - 216$ to C_4': $v^2 = w^3 + 8$, given by $(x, y) \mapsto ((x^3 - 864)/(9x^2), y(x^3 + 1728)/(27x^3))$, and for the multiplication maps [2] and [3] of C_4', the estimated rank of E over the function field of the cover is 0.90, 0.93 and 0.84 (against 0.92 for C_4' itself), computed as the average of $-N_1/p$ over the primes below 500 (3-isogeny) and 300 ([2], [3]), with N_1 the sum of a_{t(R)} over the points R of the cover over $\mathbb{F}$_p; so no new section

appears on these covers. Among the 12672 linear and quadratic twists d(t) with coefficients of absolute value at most 12 (the genus-zero double covers) the largest estimated $\mathbb{Q}(t)$-rank is 0.31, so none carries a section.

Rotated power maps (Observed). A cover $t = \varphi(s)$ of the t-line by a rational s-line that is totally ramified over both III* fibres $t = \pm i$ has the form $\varphi_{n,a} = R_a \circ \varphi_n$, with $\varphi_n(s) = \cot(n \cdot \operatorname{arccot} s)$ the n-th power map of the circle $t = \tan(\theta/2)$ and $R_a(\tau) = (\tau - a)/(1 + a\tau)$ the rotation by a rational a (the class of the rotation point of the circle modulo n-th powers is the Kummer class of the cover; for n = 2 the cover is the twist by $(1 + a^2)(1 + t^2)$). The estimated rank of E over $\mathbb{Q}(s)$ (average of $-N_1(p)/p$ over the primes below 700, with $n \in \{1, 2, 3, 5, 6, 7\}$ so that the fibres over $t = \pm i$ stay additive) is at most 0.14 for n = 1 and n = 2 (44 values of a with height $\leqslant 8$), at most 0.08 for n = 3 and 0.2 for n = 5 (64 values with height $\leqslant 10$), and at most 0.31 for n = 6 and 0.1 for n = 7 (44 values with height $\leqslant 8$). So no rotation produces a rational section over these covers; for n = 4 and 8 the fibres over $t = \pm i$ become smooth and the test needs the minimal model, which I did only for a = 0 (estimated rank −0.23 for n = 4).

The twist for C_4' at p = 5 (Observed; the bound is conditional on good reduction at 5). Let T be the twist of E by $d_1 = (1 - t)(1 + 3t^2)$ (equivalently by $w^3 + 8$ over $\mathbb{Q}(w)$). Its fibres are III* at $t = \pm i$ (untwisted), I_0^* at $t = 1, \pm i/\sqrt{3}$ and ∞, and six I_1 (Euler number 48, $\chi = 4$). The pairwise resultants and discriminants of D, $1 - t$, $1 + t^2$ and $1 + 3t^2$ are divisible only by the primes 2, 3, 7 and 23 ($\operatorname{Res}(D, 1 + 3t^2) = 9604 = 2^2 \cdot 7^4$), so the configuration is preserved at every $p \notin \{2, 3, 7, 23\}$, in particular at p = 5. Counting the points of all fibres of the model $Y^2 = X^3 + a_4 d_1^2 X + a_6 d_1^3$ over $\mathbb{F}_{5^r}$ gives $S_1, \ldots, S_7 = -2, -42, -104, 1146, -13112, 5994, -17026$ (S_1 to S_6 agree with an independent report; S_7 is new). With $L(T) = \exp(\sum S_r T^r/r)$ of degree 14 (the conductor has degree 4 + 8 + 6 = 18, minus 4), the coefficients $c_0, \ldots, c_7$ are 1, −2, −19, 6, 535, −2950, −1125, 42500; since $c_7 \neq 0$ the sign of the functional equation is +1, and $L(T) = (1 - 5T)^2 \cdot Q_{12}(T)$ with $Q_{12} = 1 + 8T + 36T^2 + 166T^3 + 1295T^4 + 5850T^5 + 25000T^6 + 146250T^7 + 809375T^8 + 2593750T^9 + 14062500T^{10} + 78125000T^{11} + 244140625T^{12}$. All reciprocal roots have modulus 5 and Q_{12} has no cyclotomic factor, so the eigenvalues of the form $5 \cdot$(root of unity) are exactly the double eigenvalue +5. Hence, if the surface has good reduction at 5, $\rho(T_{\bar{\mathbb{Q}}}) \leqslant 32 + 2 = 34$ and the geometric rank is 1 or 2. A second prime would be needed to decide: the counts $S_1, \ldots, S_6$ at p = 11 require about 10^{12} operations.

The twist for C_4' has geometric rank exactly 1 (Conditional on the Tate conjecture for this surface at p = 5 and p = 13). Counting the points of all fibres by baby-step giant-step in each fibre (with Frobenius-orbit reduction in t; the program reproduces every count above for p = 5, the K3 counts N_1 to N_5 at p = 7, 11, 13 and the claimed S_1 to S_4 at p = 11) gives $S_1, \ldots, S_6 = -12, -570, -3198, 922, 174738, -5237286$ at p = 11 and 2, −96, 980, −6424, 129002, 11060130 at p = 13. Writing $L = (1 - pT) \cdot L_{13}$ with L_{13} of degree 13, the coefficients of L_{13} up to T^6 are determined by $S_1, \ldots, S_6$, and the functional equation leaves one sign, which the Weil

bounds fix (at p = 5 this reproduces $(1 - 5T)^2 Q_{12}$ from $S_1, \ldots, S_6$ alone). At p = 13 the eigenvalues of the form 13·(root of unity) are +13 and −13, so $\rho(T_{\bar{\mathbb{F}}_{13}}) \leqslant 34$; at p = 5 they are +5 twice, so $\rho \leqslant 34$; at p = 11 they are +11 three times and −11, so $\rho \leqslant 36$, and p = 11 gives no information. Over $\mathbb{F}_{p^{12}}$ all 34 classes are rational (the Frobenius action on the components has order dividing 6), and the Artin–Tate formula (with $\alpha = \chi(O) - 1 = 3$, and $Q = p^{12}$ a square) gives $|\mathrm{disc}\, NS(T_{\bar{\mathbb{F}}_p})| \equiv \Lambda_p = \Pi(1 - \alpha^{12}/p^{12})$ over the other 12 eigenvalues, modulo squares. The squarefree classes are 79 at p = 5 and 10 at p = 13 (the value 79 agrees with an independent report). If $\rho(T_{\bar{\mathbb{Q}}})$ were 34, these would coincide (van Luijk's argument). They do not, so $\rho(T_{\bar{\mathbb{Q}}}) = 33$ and the geometric rank of the twist is exactly 1, hence so is its $\mathbb{Q}(t)$-rank. The result uses the Tate conjecture for T over $\mathbb{F}_5$ and $\mathbb{F}_{13}$ (equivalently the Birch–Swinnerton-Dyer conjecture for the generic fibre over $\mathbb{F}_p(t)$, known for K3 surfaces but not in general), the good reduction at 5 and 13 (the configuration is preserved at every $p \notin \{2, 3, 7, 23\}$), and the square order of the Brauer group.

Evidence on coverage (Observed). Among the 22 positive-rank curves of the scan, 12 have a rational point with constant x of small height. The rank-0 curves have such points only when they are torsion.

Parity (Conditional on the parity conjecture). If t lies in one of these families and $W(E_t) = +1$, one found point plus parity forces rank ⩾ 2. This explains the W = +1 rank-2 curves with a constant-x point.

An unconditional version through the 2-Selmer group (Proved). For t with $E_t(\mathbb{Q})[2] = 0$ (all t except finitely many, see §8) let $s_2 = \dim \mathrm{Sel}_2(E_t) = \mathrm{rank}\, E_t(\mathbb{Q}) + \dim Ш(E_t)[2]$. By the 2-parity theorem (Monsky; Dokchitser–Dokchitser), $(-1)^{s_2} = W(E_t)$, with no hypothesis on Sha: the Cassels–Tate pairing makes the finite part of $Ш[2^\infty]$ of square order, so its 2-torsion has even dimension. Hence if E_t has a non-torsion point and $W(E_t) = +1$, then $s_2 \geqslant 1$ and s_2 is even, so $s_2 \geqslant 2$. Moreover the 2^∞-Selmer corank rank + k, with k the corank of the divisible part of $Ш[2^\infty]$, is even by the p-parity theorem, so rank 1 would force k to be odd, that is, $Ш(E_t)$ infinite. Hence **rank $E_t(\mathbb{Q}) \geqslant 2$ or $Ш(E_t)$ is infinite.** This applies to every member t of the families of this section with W = +1 and a non-torsion point at the fixed x, for example t = −7 (where the rank is 2) and t = −2047837/2965284 (W = +1 by the formula of §4). It replaces the parity conjecture in the conditional statement above, at the price of the weaker conclusion $\dim \mathrm{Sel}_2 \geqslant 2$. It still needs infinitely many members with W = +1, which is open. At t = 3/7 Sage reported rank bounds (0, 2) and the formula gives W = +1 (conductor 20307956672); Magma certifies rank 0 with $\dim \mathrm{Sel}_2 = 2$, so $\dim Ш[2] = 2$ (§5).

A barrier for the sign (Proved). The p-parity theorem, $(-1)^{r_p(E)} = W(E)$ for the p^∞-Selmer corank r_p, is known for all elliptic curves over $\mathbb{Q}$ (Dokchitser–Dokchitser, Monsky, Nekovář); only the passage from r_p to the rank needs finiteness of $Ш[p^\infty]$. The remaining obstruction is the distribution of the sign. The unconditional equidistribution

results for root numbers in one-parameter families (Helfgott, Desjardins) apply when every irreducible factor of the polynomial M(t) marking multiplicative reduction has degree at most 3. For E_t this polynomial is D(t) = t⁶ − 30t⁴ − 24t² − 1, which is irreducible of degree 6 (D(t) = G(t²) with G an irreducible cubic, and Gal(D) = S_4). Moreover this cannot be repaired by a rational base change t = φ(s): if s_0 is a root of D(φ(s)) then φ(s_0) is a root of D, so [ℚ(s_0) : ℚ] is divisible by 6. Hence after any such change every irreducible factor of the pulled-back polynomial has degree divisible by 6, and the unconditional results never apply. A sign statement for E_t requires a Chowla-type input for values of a degree-6 form.

Conditional theorem (Conditional on H). Let c ∈ {2, 1, 4, −11/5, 14/45}, so that the genus-one curve C_c has a positive-rank Jacobian and infinitely many rational points. For P = (t, w) ∈ C_c(ℚ) with E_t nonsingular let P_t = (c, w/(1 + t²)) ∈ E_t(ℚ). Let 𝔇_c be the set of t = t(P) for which P_t is non-torsion and E_t(ℚ)[2] = 0. Then 𝔇_c contains all but finitely many t(P): the section of the twist E^{f_c} has Shioda height 1, hence is non-torsion over ℚ(C_c), and by Silverman's specialization theorem the points of C_c(ℚ) at which it specializes to a torsion point have bounded height, hence are finite in number; the t with 2-torsion are finite in number by §8. Consider the hypothesis

H. W(E_t) = +1 for infinitely many t ∈ 𝔇_c.

By the formula of §4 this means that −$w_2 w_3$(2/c′)(−1)^{ω(R_F)}(ab/R_F) = +1 for infinitely many of these t, where c′ is the odd part of the hypotenuse (for c = 2, c′ is a perfect square, so (2/c′) = 1); the factors w_2 and w_3 take finitely many values, so H asks for a prescribed value of the Liouville-type term. Then:

1. **(Unconditional consequence of H.)** For these t, two statements hold, with different proofs. (a) dim Sel_2(E_t) ⩾ 2, hence rank E_t(ℚ) ⩾ 2 or Sha(E_t)[2] ≠ 0: indeed dim Sel_2 = rank + dim E_t(ℚ)[2] + dim Sha[2] with E_t(ℚ)[2] = 0, and by the parity theorem of Monsky and Dokchitser–Dokchitser dim Sel_2 − dim E_t(ℚ)[2] has the parity of W = +1, while the point gives dim Sel_2 ⩾ 1. (b) rank E_t(ℚ) ⩾ 2 or Sha(E_t)[2^∞] is infinite (contains a copy of $ℚ_2/ℤ_2$): the 2^∞-Selmer corank equals rank + k, where k is the corank of the divisible part of Sha[2^∞], it has the parity of W = +1, and it is at least 1 because of the point, so rank 1 would force k odd, hence k ⩾ 1. Statement (b) is stronger than the one obtained from (a) alone, since a finite nonzero Sha[2] does not make Sha[2^∞] infinite; it uses the corank parity, not the dimension of Sel_2.
2. **(If Sha(E_t)[2^∞] is finite for these t.)** rank E_t(ℚ) ⩾ 2, since the 2^∞-Selmer corank then equals the rank and has the parity of W.

In either case infinitely many distinct t arise, because the map C_c → ℙ¹_t has finite fibres. H is not proved. Its content is a prescribed parity of the number of primes ⩾ 5 dividing F(t(P)), twisted by a Jacobi symbol: a Liouville-type statement for values of a sextic form along a thin sequence of height growing like e^{n²}. The data are consistent with H for c =

2: among the 11 members of $C_2(\mathbb{Q})$ other than t = 0, 1, the sign is +1 at t = −7 and t = −2047837/2965284 (rank 2 at t = −7, rank ⩾ 2 at the other member only under the parity conjecture).

Signs along the covers (Proved structure, Observed data). For t = u/v on $C_2(\mathbb{Q})$, the numbers v, v − u and $v^2 + u^2$ are pairwise coprime up to a factor 2 and $v(v - u)(v^2 + u^2)$ is a square, so each is a square up to a unit or a factor 2. Hence c is a perfect square, (2/c) = +1, and every prime dividing c occurs to an even power. The root numbers computed from the formula in §4 for the members t = 0, 1, −1, 3/4, −7, −40/9, 31/49, −561/400, 161/961, 34440/34969, −69409/529 are +1, +1, −1, −1, +1, −1, −1, −1, −1, −1, −1: after t = −7 the next eight members have W = −1. The following two members, t = −2047837/2965284 (6P) and t = −917447/5013121 (6P + T), give +1 and −1, so the sign +1 occurs at 4 of the 13 members and the streak of −1 is broken. The next members give W = +1 at both 7P and 7P + T and W = −1 at 8P + T (Appendix B; the sign at 8P needs a factorization still missing), so +1 occurs at 8 of the 19 members t = 0, 1, −1, 3/4, −7, … , 9P, 9P + T (W = −1 at 8P and 8P + T and W = +1 at 9P and 9P + T), and under the parity conjecture t(7P), t(7P + T), t(9P) and t(9P + T) have rank ⩾ 2 (the point (2, y) is non-torsion there). At both of these members the point (2, 15422232/3603685) is non-torsion (no multiple up to 12 is the point at infinity). Under the parity conjecture t = −2047837/2965284 therefore has rank ⩾ 2; its conductor has 50 digits, so no rank computation is possible. Along C_1 (x = 1) the members t = −1, −1/3, −1399/597, 41/679 have W = −1, +1, +1, −1, and along C_4 the members t = −11/2, 4/3 have W = +1. For C_1 the next member (the point 2G, see below) has numerator and denominator of about 16 digits, so F is of order 10^{96}, too large to factor, and its sign cannot be computed. For C_2 the members 6P and 6P + T have numerators and denominators of 6 to 7 digits (F has 41 digits and could be factored); the following ones have about 10 digits and F is too large to factor.

A conditional example at large height (Certified inputs, Conditional conclusion). At t = −1399/597 the point (1, 963/481) lies on E_t and is non-torsion (no multiple up to 12 is the point at infinity; canonical height about 8.0). On the minimal model $y^2 = x^3 - 92590209478000x + 1117666823855904075000$ it is (5784025, 27850080375). Sage independently returns root number +1 and conductor 730600573632065586208900960 0, which equals the conductor predicted by the theorem of §3. Under the parity conjecture, rank $E_t(\mathbb{Q}) \geqslant 2$. A search of the minimal model for a second point (q = 1 up to $X = 2 \cdot 10^9$, q = 2 up to $5 \cdot 10^8$, q = 3 up to $2 \cdot 10^8$) found none. The curve C_1 has the Weierstrass model $Y^2 = Z^3 + 128Z^2 + 10496Z + 319488$, on which the point of t = −1/3 has order 2 (call it T). Writing G for the point of t = −1399/597, the t-coordinate of G + T is 41/679 and that of 2G has a 16-digit numerator.

7. The quadratic twist

Let $d = (1 - t)(1 + t^2)$ and $E^d : Y^2 = X^3 + 4(1 - t^2)(1 - t)^2(1 + t^2)X - 16t(1 - t)^3(1 + t^2)^2$. Its discriminant is $4096(1 - t)^6(1 + t^2)^3 D(t)$, so the fibres are $2 \cdot \mathrm{III}$, $2 \cdot \mathrm{I}_0^*$ (at $t = 1$ and ∞) and $6 \cdot \mathrm{I}_1$, with trivial lattice rank 12 and a K3 surface S^d. For rational t in the C_2 family $d(t)$ is a square, so sections of E^d specialize to points of E_t. It has the explicit section $P_d = (2d, 4(1 - t)^2(1 + t^2))$; P_d is non-torsion (its specializations have positive height), so $\rho(S^d) \geqslant 13$ (**Proved**).

The L-polynomial at p = 5 (Certified). The degree-10 polynomial $1 - 2T - 25T^2 + 150T^3 - 5000T^5 + 93750T^7 - 390625T^8 - 781250T^9 + 5^{10}T^{10}$ has the coefficients $l_1, \ldots, l_5 = -2, -25, 150, 0, -5000$, which I recomputed independently by counting the points of all fibres over $\mathbb{F}_5$, …, $\mathbb{F}_{5^5}$; since $l_5 \neq 0$ the functional equation has sign +1 and determines the rest. It factors as $(5T - 1)^2(5T + 1)^2$ times an irreducible sextic. **Conditional** on good reduction at 5, which the lemma ($p > 5$) does not cover, this gives $\rho(S^d) \leqslant 16$ and rank $\leqslant 4$. The bound from $p = 7$ below is stronger.

The L-polynomial at p = 7 (Proved). The prime 7 is a prime of good reduction of the twist: the radicals of its a_4, a_6, Δ, f and g have discriminants involving only 2, 3 and 23, so the lemma applies. Independent point counts over $\mathbb{F}_7$, …, $\mathbb{F}_{7^5}$ give $l_1, \ldots, l_5 = -2, -35, 0, 1372, 0$. Since $l_5 = 0$, the sign ε of the functional equation is not fixed by the coefficients. For $\varepsilon = +1$ no zero of the L-polynomial has the form $7 \cdot$(root of unity), so Tate's inequality would give $\rho(S^d_{\{\bar{\mathbb{F}}_7\}}) \leqslant 12$. This contradicts the section P_d, which reduces to a class independent of the rank-12 trivial lattice and gives $\rho \geqslant 13$. Hence $\varepsilon = -1$, and then exactly two zeros have this form (the reciprocal roots 7 and −7). Therefore $\rho(S^d) \leqslant \rho(S^d_{\{\bar{\mathbb{F}}_7\}}) \leqslant 14$, and **rank $E^d(\bar{\mathbb{Q}}(t))$ is 1 or 2.** The eigenvalue −7 of a possible second class means that a second geometric section, if it exists, is not defined over $\mathbb{Q}(t)$ but over an extension in which 7 is inert. A search over $\mathbb{Q}(i)(t)$ with Gaussian-integer coefficients $|\mathrm{Re}|, |\mathrm{Im}| \leqslant 2$ and $\deg X \leqslant 4$ found no section other than P_d (Observed).

The twist has rank exactly 1 over $\mathbb{Q}(t)$ (Proved). At $p = 7$ the reciprocal roots 7 and −7 of the L-polynomial each occur once, so the zero at $T = 1/7$ has multiplicity one. The Tate conjecture holds for K3 surfaces over finite fields, so rank $E^d(\mathbb{F}_7(t))$ equals this multiplicity, namely 1. The surface has good reduction at 7, so reduction gives a homomorphism $E^d(\mathbb{Q}(t)) \to E^d(\mathbb{F}_7(t))$ (Néron mapping property at the discrete valuation ring $\mathbb{Z}_{(7)}[t]_{(7)}$), and it is injective: a section reducing to the zero section would have the same class in $\mathrm{NS}(S^d_{\{\bar{\mathbb{F}}_7\}})$ as the zero section, and injectivity of Néron–Severi specialization then gives equality of the two sections. Hence rank $E^d(\mathbb{Q}(t)) \leqslant 1$, and P_d gives equality. Consequently the original curve over the function field of the genus-one cover C_2 has rank rank $E(\mathbb{Q}(t))$ + rank $E^d(\mathbb{Q}(t)) = 0 + 1 = 1$, so no cover of this type carries a second independent section. That the geometric rank is also 1 is shown next. (For $c = -11/5$ the reduction at 7 coincides with that of $c = 2$, since $-11/5 \equiv 2 \bmod 7$, but the good-

reduction hypotheses were not checked for that twist, so I make no claim for it. For c = 1 and c = 4 the prime 7 is a prime of bad reduction because D(3) ≡ 0 and D(4) ≡ 0 (mod 7) meet a root of q_c.)

Good reduction of the twist at p ⩾ 5, p ≠ 23 (Proved). The model $Y^2 = X^3 + a_4X + a_6$ with $a_4 = 4(1 - t^2)(1 - t)^2(1 + t^2)$ and $a_6 = -16t(1 - t)^3(1 + t^2)^2$ has $\Delta = 4096(1 - t)^6(1 + t^2)^3D(t)$ and $c_4^3 - c_6^2 = 1728\Delta$ exactly, degrees 6, 8, 18 with leading coefficients −4, 16, 4096, and in the chart s = 1/t the orders at s = 0 are (2, 4, 6). Over $\mathbb{Q}$ the fibres are I_0^* at t = 1 (orders (3, 3, 6)) and at ∞, III along $1 + t^2 = 0$ (orders (1, 2, 3)), and I_1 at the six simple roots of D. Since disc $D = 2^6 \cdot 3^{18} \cdot 23^2$, D(1) = −54 and $\mathrm{Res}(D, 1 + t^2) = 64$, for p ⩾ 5 and p ≠ 23 the same orders occur modulo p, D is squarefree mod p and coprime to $(1 - t)(1 + t^2)$, and the leading coefficients are units. The Weierstrass surface over $\mathbb{Z}_{(p)}$ is then flat, with singular locus a finite étale scheme (the sections t = 1 and t = ∞ and the pair t = ±i) along which the rational double point has constant type (D_4, D_4, A_1). Blowing up this locus (repeatedly) resolves all fibres simultaneously, so the minimal resolution is a smooth projective K3 surface over $\mathbb{Z}_{(p)}$ whose fibres over $\mathbb{Q}$ and over $\mathbb{F}_p$ are the minimal resolutions of the Weierstrass fibres. This is also the situation of Lemma 3.11 of Kazalicki–Naskręcki, whose hypotheses as used in §2.1 hold: the radicals of a_4, a_6, Δ, f and g have discriminants divisible only by 2, 3 and 23. The primes used below, 7, 11, 17 and 19, satisfy these conditions.

The twist has geometric rank exactly 1 (Proved, with computed L-polynomials). For the twist E^d with $d = (1 - t)(1 + t^2)$ I counted the points of all fibres over $\mathbb{F}_{p^k}$, k ⩽ 4, for p = 7, 11, 13, 17, 19 (all primes of good reduction: p > 5, p ≠ 23). The counts (N_1, N_2, N_3, N_4) are (−2, −74, −218, 2462), (12, −166, −1506, −11826), (−6, −140, −3180, 66252), (−12, −382, −810, −270194) and (−36, −574, 5328, −268106). Removing the factor (1 − pT) of the section P_d, the remaining factor of degree 9 is determined by the first four coefficients and the functional equation; its sign is fixed by the Weil bounds and by the evenness of ρ over $\bar{\mathbb{F}}_p$. The eigenvalues of the form p·(root of unity) give $\rho(S^d_{\bar{\mathbb{F}}_p}) = 14$ for p = 7, 11, 17, 19 (the roots +p and −p) and ρ = 16 for p = 13. Over $\mathbb{F}_{p^2}$ all 14 classes are rational, and the Artin–Tate formula gives $|\mathrm{disc}\, NS(S^d_{\bar{\mathbb{F}}_p})| \equiv \Lambda_p = L_8(1/p) \cdot L_8(-1/p)$ modulo squares, where $L_8 = L/(1 - p^2T^2)$ is the remaining factor of degree 8 (the Brauer group has square order and $q = p^2$ is a square). The values are $\Lambda_7 = 512/49$, $\Lambda_{11} = 1152/121$, $\Lambda_{17} = 1408/289$ and $\Lambda_{19} = 484/361$, of squarefree classes 2, 2, 22 and 1 (the last is corrected to 11 in F.45; the conclusion is unchanged). If $\rho(S^d_{\bar{\mathbb{Q}}})$ were 14, then $NS(S^d_{\bar{\mathbb{Q}}})$ would be a sublattice of finite index in each of these lattices, forcing equal square classes (van Luijk's argument). The classes at p = 7 and p = 19 differ, so $\rho(S^d_{\bar{\mathbb{Q}}}) = 13$ and $\mathrm{rank}\, E^d(\bar{\mathbb{Q}}(t)) = 1$ exactly. Consequently the original curve over the function field of the cover C_2 has geometric rank exactly 1 as well.

A second elliptic fibration on the twist (Proved; consistency with ρ = 13). With $h = (1 - t)(1 + t^2)$ and u = x/h, v = y/h, the surface Y of this twist becomes $v^2 = (1 - t)[(u^3 - 4u)t^2 - 16t + (u^3 + 4u)]$, a cubic in t with the rational point (t, v) = (1, 0). It is the elliptic curve $Y^2 = X^3 +$

$c_2X^2 + c_1c_3X + c_0c_3^2$ over $\mathbb{Q}(u)$ with $c_3 = -u^3 + 4u$, $c_2 = u^3 - 4u + 16$, $c_1 = -u^3 - 4u - 16$, $c_0 = u^3 + 4u$, and it has the rational 2-torsion point $(c_3, 0)$. I verified the identities exactly: $c_4^3 - c_6^2 = 1728\Delta$ and $\Delta = -256u^2(u-2)^4(u+2)^2(u^2+2u+4)^2(u^6-16u^2-64)$, with c_4 coprime to every finite factor of Δ (multiplicative reduction) and orders (2, 3, 6) at $u = \infty$. The fibres are I_4 ($u = 2$), I_2 ($u = 0$, $u = -2$ and the two roots of $u^2 + 2u + 4$), six I_1 (the roots of $u^6 - 16u^2 - 64$, the same sextic as for the x-fibration of E) and I_0^* at ∞; the Euler number is $4 + 4\cdot2 + 6 + 6 = 24$. The trivial lattice has rank $2 + 3 + 4 + 4 = 13$ and absolute discriminant $4\cdot2^4\cdot4 = 256$, which becomes $256/2^2 = 64$ after the 2-torsion section. This equals the discriminant $64\cdot1$ of the rank-13 lattice $U \oplus A_1^2 \oplus D_4^2 \oplus \langle P_d \rangle$ of the first fibration, as it must if both lie in the same Néron–Severi lattice. Since $\rho(Y_{\bar{\mathbb{Q}}}) = 13$, the Mordell–Weil group of this fibration over $\bar{\mathbb{Q}}(u)$ has rank $13 - 13 = 0$, so it is the group of order 2.

What the counts are (Proved). For $t \in \mathbb{F}_Q \cup \{\infty\}$ put $a_t = Q + 1 - \#$(the projective Weierstrass cubic of the fibre of the model above), allowing singular fibres. This is the trace in the local Euler factor in every case: $Q + 1 - \#E_t$ for good fibres, +1 for a split node, −1 for a non-split node and 0 for a cusp. The model is minimal at every place, since the orders of Δ are 6, 3, 1 and 6 (all < 12), and the fibre at ∞ in the chart $s = 1/t$ is the cusp $y^2 = x^3$, so $a_\infty = 0$. Then $N_k = \Sigma_{t \in \mathbb{P}^1(\mathbb{F}_{p^k})} a_t$ and $L(T) = \exp(\Sigma\, N_k T^k/k)$ is the L-function of E^d over $\mathbb{F}_p(t)$, of degree $10 = (2\cdot2 + 2\cdot2 + 6) - 4$ by the Grothendieck–Ogg–Shafarevich formula; it is the characteristic factor of Frobenius on the part of H^2 complementary to the 12 classes of the trivial lattice (Shioda–Tate). No point counts of the resolved surface are used, so no correction for exceptional curves arises. Checks of the counts: (i) N_1, N_2, N_3 at $p = 7, 11, 13, 17, 19$ were recomputed in Magma V2.30-1 by a different implementation (a loop over $\mathbb{F}_{p^k}$ with the quadratic character) and agree exactly with the values above; (ii) N_4 at $p = 7, 11$ and 13 was recomputed by a second program (logarithm tables in Python, against the first written in C) and agrees (2462, −11826, 66252); (iii) at $p = 7$ the polynomial built from $N_1, \ldots, N_4$ and the functional equation predicts $N_5 = 38$, in agreement with the directly counted $N_5 = 38$. The values N_4 at $p = 17$ and 19 come from the C program alone.

Short sections (Observed). Sections with $\deg X \leqslant 4$ over $\mathbb{F}_p(t)$ are, for $p = 7, 11$ and 13, exactly $\pm P_d$ and $\pm 2P_d$ (searched among 7^5, 11^5 and 13^5 candidates); for $p = 5$ there are two further ones. A second rational section independent of P_d and disjoint from the zero section, with 7-integral coefficients, cannot exist, by injectivity of specialization. A section with $\deg X \leqslant 4$ found by integer-coefficient search ($X = t^4 + 4t^3 + 4t$) is exactly $2P_d$.

8. Negative results (bounded searches, not theorems)

- Among about 19000 nonconstant $x(t)$ of low degree (linear fractions and quadratic over linear with coefficients $\leqslant 3$, quadratics $\leqslant 4$, linear $\leqslant 6$), exactly three classes of double covers of the t-line of genus $\leqslant 1$ occur: $w^2 = (1 - t)(1 + 3t^2)$ from $x = -2(t + 1)/(t -$

1) (the cover C_4' of §6), its mirror $t \mapsto -t$, and $w^2 = 2t(t^2 - 1)$ from $x = 2t$ (rank 0). An earlier version of this search computed the radical of the polynomial instead of its part of odd multiplicity and wrongly reported none.

- Fibre products of pairs of constant-x covers tested have genus 2 or 3, so by Faltings each has finitely many rational points.
- No second section of the twist with integer coefficients $|c_i| \leqslant 6$ and $\deg X \leqslant 4$.

Rational 2-torsion (Proved finiteness, Observed list). E_t has a rational point of order 2 exactly when $x^3 + 4Ax - 8B$ has a rational root c, that is, when (t, c) is a rational point of the curve $c^3(1 + t^2) + 4c(1 - t^2) - 16t = 0$. Solving for t gives the hyperelliptic curve $y^2 = 64 + 16c^2 - c^6$, which has genus 2 because $c^6 - 16c^2 - 64$ is irreducible. By Faltings only finitely many t have a rational 2-torsion point. Among rational c of height $\leqslant 400$ the only points are $c = 0$, ± 2 ($y = \pm 8$), giving $t = 0, 1, -1$. None of these curves has full rational 2-torsion. Hence, within this height bound, no E_t is $\mathbb{Q}$-isomorphic to a curve $y^2 = x(x - a^2)(x - b^2)$ of Naskręcki's family.

Base changes and twists tested for rank $\geqslant 2$ over $\mathbb{Q}(t)$ (Observed; one case Proved). A rank of E^d over $\mathbb{Q}(t)$ can be estimated by the Nagao-type average of $-N_1(p)/p$ over primes p, where $N_1(p) = \Sigma_{t \in \mathbb{F}_p} a_t(E^d) + a_\infty$; the method was calibrated on the known twists ($d = 1 + t^2$ gives -0.05 and $d = (1 - t)(1 + t^2)$ gives 1.08).

- **Geometric rank jumps by ramification at $t = \pm i$.** The base change $t = (w^2 - 1)/(2w)$ (the doubling map on the circle $t = \tan(\theta/2)$) keeps the surface a K3 with fibres $2 \cdot I_0^* + 12 \cdot I_1$ and has geometric Mordell–Weil rank 6, equivalently the twist E^{1+t^2} is a rational elliptic surface with fibres $2 \cdot \mathrm{III} + 6 \cdot I_1$ and Mordell–Weil lattice D_6^*. The Galois representation on it has traces $0, \pm 2, \pm 6$ only. At $p = 13$ the L-polynomial is $(1 + 13^3T^3)^2$, which has no eigenvalue $+13$, so E^{1+t^2} has rank 0 over $\mathbb{F}_{13}(t)$, hence over $\mathbb{Q}(t)$ (Proved). The multiplicity of every quadratic character χ_c with $|c| \leqslant 300$ in this representation is at most 0.13, so $E^{c(1 + t^2)}$ has rank 0 over $\mathbb{Q}(t)$ for these c. For the n-th power maps $t = \cot(n \cdot \operatorname{arccot} w)$ the averages are -0.001 ($n = 3$) and -0.23 ($n = 4$, geometric rank 14), so the rank over $\mathbb{Q}(w)$ is 0 in these cases.
- **Polynomial twists.** Among the 55634 twists by primitive polynomials $d(t)$ of degree 2 to 4 with coefficients of absolute value at most 4, estimated with the 110 primes below 600, exactly six classes have an estimated rank near 1 (0.82 to 0.94): $\pm(1 - t)(1 + t^2)$, $\pm(1 - t)(1 + 3t^2)$ (the cover C_4' of §6) and $-t(1 + t^2)$, $t(1 - t^2)$ (covers of rank 0). All other values are at most 0.37 and none is near 2. An earlier version of this scan used 30 primes and missed $(1 - t)(1 + 3t^2)$.
- **The family f_c.** For the 5975 rational numbers c of height at most 70, the twist by $f_c = (1 + t^2)q_c(t)$ has estimated rank near 1. The two largest initial values ($c = \pm 38/3$, estimate 1.63) fell to 1.11 with 260 primes, and the other top candidates to between 0.82 and 1.03.

9. Open problems

1. **Problem H.** Infinitely many n with W(E_{t_n}) = +1 along the orbit t_n = 1 − x(nP). Everything else about the sign is proved; what remains is the parity of the number of prime factors of F_n, which needs a Chowla-type input along a thin orbit (Appendix B, F.24).
2. Infinitely many rational t with rank ⩾ 2 (and ⩾ 3). This follows from H together with the finiteness of Sha[2^∞]; unconditionally a second section on a cover would suffice, but none exists on the 18 covers examined (F.44–F.46).
3. Whether every twist E^{d_c} has geometric rank exactly 1, that is ρ = 13 for every rational c. Proved for 18 values; a failure would be a rational Noether–Lefschetz point, which is expected to be rare, and a proof by two primes cannot be uniform (F.46).
4. The parity conjecture for these curves (finiteness of Sha[2^∞]), on which the rank statements deduced from root numbers depend.
5. Whether the extra characteristic-5 sections of the twist lift to characteristic 0.
6. The frequency of jumps and a theorem that explains jumps not coming from constant-x covers.
7. Arithmetic identities between the values F_n that are not identities of functions, and direct constructions of a second point on E_{t_n} (F.34).

Appendix A. Reproducing the predictions

The following Sage script predicts the conductor and the root number of E_t from the formulas of §§3–4, then compares them with Sage's conductor and root number and computes rank bounds through PARI.

```
from math import gcd

def trip(u, v):
    d = gcd(gcd(abs(v*v - u*u), abs(2*u*v)), v*v + u*u)
    return (v*v - u*u)//d, (2*u*v)//d, (v*v + u*u)//d

def w2(A, B):
    if A % 2 == 1:
        if A % 4 == 3:
            return 1 if B % 8 == 4 else -1
        s = (-3*A + 27*B) % 16
        if B % 8 == 4:
            return 1 if s in (9, 13) else -1
        return 1 if s in (5, 9) else -1
    return 1 if B % 4 == 3 else -1
```

```
def w3(A, B):
    if B % 3 == 0:
        return 1
    if A % 9 in (0, 6):
        return 1
    return 1 if (4*B) % 9 in (4, 7, 8) else -1

def predict(u, v):
    a, b, c = trip(u, v)
    Fv = 4*a**3 + 27*b*b*c
    A16 = (a*inverse_mod(c, 16)) % 16
    B16 = (b*inverse_mod(c, 16)) % 16
    A9 = (a*inverse_mod(c, 9)) % 9
    B9 = (b*inverse_mod(c, 9)) % 9
    f2 = 5 if (A16 % 4 == 1) else 6
    f3 = 0 if B9 % 3 == 0 else {0: 2, 3: 3, 6: 4}[A9]
    N = 2**f2 * 3**f3
    W = -w2(A16, B16) * w3(A9, B9)
    if c > 1:
        for p, m in factor(c):
            if m % 4 != 0:
                N *= p**2
        W *= jacobi_symbol(2, c)
    R = 1
    omega = 0
    for p, m in factor(abs(Fv)):
        if p >= 5:
            N *= p
            R *= p
            omega += 1
    W *= (-1)**omega
    if R > 1:
        W *= jacobi_symbol((a*b) % R, R)
    return N, W

for (u, v) in [(0,1),(1,1),(-1,1),(2,1),(1,2),(-1,3),(4,3),(11,2)]:
    A = QQ(v*v - u*u) / QQ(v*v + u*u)
    B = QQ(2*u*v) / QQ(v*v + u*u)
    E = EllipticCurve([4*A, -8*B]).minimal_model()
    Np, Wp = predict(u, v)
    print(u, v, "N", E.conductor(), Np, "W", E.root_number(), Wp)
    try:
        alarm(120)
        r = E.pari_curve().ellrank()
```

```
        cancel_alarm()
        print("   rank bounds", r[0], r[1])
    except AlarmInterrupt:
        print("   rank computation timed out")
```

The 2-adic and 3-adic rules are the finite tables of §4; the structure of the tables was cross-checked against the Magma table for the orbit ($n \leqslant 24$) and the 33 scanned curves.

Appendix B. Problem H, stated independently of the rest of the paper

This appendix defines the hypothesis H of §6 in elementary terms, so that it can be studied without elliptic curves of rank $\geqslant 2$ or K3 surfaces.

Objects. Let $E' : v^2 = z^3 - 2z^2 + 2z$, with the rational points $P = (1, 1)$ (infinite order) and $T = (0, 0)$ (order 2); $E'(\mathbb{Q}) = \langle P \rangle \times \langle T \rangle$. For $R = (z, v) \in E'(\mathbb{Q})$ put $t(R) = 1 - z(R)$ (these are exactly the rational t with $(1 - t)(1 + t^2)$ a square). Write $t_n = t(nP)$ and $t'_n = t(nP + T)$: 0, −1, 3/4, −7, −40/9, 31/49, −561/400, 161/961, 34440/34969, −69409/529, −2047837/2965284, −917447/5013121, …

For $t = u/v$ in lowest terms put $d = \gcd(v^2 - u^2, 2uv, v^2 + u^2) \in \{1, 2\}$, $a = (v^2 - u^2)/d$, $b = 2uv/d$, $c = (v^2 + u^2)/d$, so $a^2 + b^2 = c^2$, and

$F(t) = 4a^3 + 27b^2c$.

For t on this family c is a perfect square (§6) and $F(t) \neq 0$ for all rational t. Define

$s(t) = \#\{$ primes $p \geqslant 5 : p$ divides $F(t)$ and $(ab/p) = +1 \}$

(the Legendre symbol). By §4 these are the primes at which E_t has split multiplicative reduction, and $w_p = -1$ exactly there.

The sign. By the theorem of §4, $W(E_t) = f(t) \cdot (-1)^{s(t)}$, where $f(t) = -w_2(t) w_3(t) (2/c)$ takes finitely many values and depends only on $(A, B) = (a/c, b/c)$ modulo 16 and modulo 9 (the rules are in §4.1 and Appendix A). Along t_n the value $f(t_n)$ equals +1 for every $n \leqslant 534$ ($w_2 = -1$ and $w_3 = +1$ throughout), and along t'_n it follows the pattern (period 18, observed for $n \leqslant 534$)

$n = 1, \ldots, 18 :$ − − − + − + + + − + + + − + − − − +

($f(t'_n) = +1$ for $n = 4, 6, 7, 8, 10, 11, 12, 14, 18$). By compactness of $E'(\mathbb{Q}_2)$ and $E'(\mathbb{Q}_3)$ the sequence f is periodic in n.

Problem H. Show that $W(E_{t_n}) = +1$ for infinitely many n, that is, that $s(t_n)$ is even for infinitely many n (equivalently, for the other sequence, that $(-1)^{s(t'_n)} = f(t'_n)$ for infinitely many n). A weaker version allows any infinite subset of $E'(\mathbb{Q})$ in place of $\{nP\}$.

Data. The primes ⩾ 5 dividing F all occur to the first power in the examples below.

member	t	F ⩾5-primes (S = split, N = non-split)	s	W
2P	3/4	23 N, 4241 S	1	−1
3P	−40/9	307 S, 7743563 N	1	−1
4P	−561/400	4372373 S, (12 digits) N	1	−1
5P	34440/34969	2851 N, 12281 S, (22 digits) N	1	−1
6P	−2047837/2965284	631 N, 6907 S, 69736487 N, (26 digits) S	2	+1
2P + T	−7	823 S	1 (f = −1)	+1
3P + T	31/49	23 N, 83 N, 2060749 N	0 (f = −1)	−1
4P + T	161/961	1379423 N, (10 digits) S	1 (f = +1)	−1
5P + T	−69409/529	32079199 S, (20 digits) S	2 (f = −1)	−1
6P + T	−917447/5013121	163 N, 307191923 N, (29 digits) S	1 (f = +1)	−1

New factorizations. The members 7P, 7P + T, 8P and 8P + T have been factored completely (7P, 7P + T and 8P + T by elliptic-curve factorization with a program written for this purpose, 8P with Magma; in each case the product of the primes was checked to equal F and each factor was checked to be a probable prime). F has 56 and 57 digits for 7P and 7P + T and 73 digits for 8P + T. At 7P and 7P + T the hypotenuse c = 2178834605825935201 is a perfect square, and so is the hypotenuse 6665309504721675412012 81 of 8P + T, as on every member of C_2.

member	t	primes ⩾ 5 of F (S = split, N = non-split)	s	f	W
7P	−1471207920/119924401	2479847 N, 6272027 N, 3464148854674783947 13 S, 1504836637469463028067 S	2	+1	+1
7P + T	1351283519/1591132321	23 N, 62653 N, 238858259 S, 138886167672809043791 N, 210834378622780780591 S	2	+1	+1
8P	365693412159/729931009600	23 N, 541 N, 55102940955616303821050453 S, 19628658192904209174397663207591571383153 21 N	1	+1	−1
8P + T	−1095624421759/364237597441	181 N, 8456963754372181 S, 1143775645981565351 N, 47753800121867236572402232971235631 N	1	+1	−1
9P	360706316862320/1115133122044881	599 N, 27269502419 N, 505725298418800171 N, 941088050120417071 N, 94942246972951319670038570498977297875436 1 N	0	+1	+1
9P + T	−1475839438907201/754426805182561	83 S, 412824010409 S, 472497673951447 S, 10041745994154844904018294640310451555205 918563419704772687390 9 N	3	−1	+1

The prescribed parity (Proved identity, Observed values). Let F″ = |F|/(2^α3^β) and suppose F″ squarefree, as it is at every member tested (this fails in general, since 23^2 divides F_{32} (see Appendix B); without it the identity below holds with the radical R_F in place of F″, and with λ replaced by (−1)^{ω(R_F)}). In the notation $F(a, b) = a^6 − 30a^4b^2 − 24a^2b^4 − b^6$ with t = a/b one has $4a_Δ^3 + 27b_Δ^2c_Δ = −4F(a, b)/d^3$ for the primitive triple (a_Δ, b_Δ, c_Δ) and d ∈ {1, 2} (checked on all 22 members listed here), so F″ is also the part of |F(a, b)| prime to 6. The sign at a prime p ⩾ 5 dividing F is w_p = −(ab/p) for every t, and not only on the members tested: p | F forces p ∤ abc, the integral model $Y^2 = X^3 + 4ac^3X − 8bc^5$ has $v_p(c_4) = 0$, so it is minimal with multiplicative reduction, and the reduction is split exactly when $−c_6$ is a square modulo p, which equals (ab/p) by $27b^2c ≡ −4a^3$ (§4.1). Then Π_{p | F″}(−(ab/p)) = (−1)^{ω(F″)}(ab/F″), where (ab/F″) is a Jacobi symbol that can be computed from t without factoring F″, and so W(E_t) = f·(ab/F″)·λ(F″), with λ the Liouville function. Putting ε = f·(ab/F″), which is computable, W(E_t) = +1 exactly when λ(F″(t)) = ε, that is when ω(F″) is even if ε = +1 and odd if ε = −1. Along the 16 members 2P, …, 9P + T the values (ω, ε, W) are (2, −1, −1), (1, −1, +1), (2, −1, −1), (3, +1, −1), (2, −1, −1), (2, −1, −1), (3, +1, −1), (2, −1, −1), (4, +1, +1), (3, +1, −1), (4, +1, +1), (5, −1, +1), (4, −1, −1), (4, −1, −1), (5, −1, +1), (4, +1, +1) for 2P, 2P + T, 3P, 3P + T, …, 9P, 9P + T in this order, and W = +1 exactly when λ = ε in all 16 cases; ε takes the value +1 six times and −1 ten times, so the required parity of ω(F″) is not constant: it is even only when ε = +1. A statement of the form "ω is even infinitely often" therefore has to be replaced by "λ(F″(t_n)) = ε_n infinitely often". For a member whose F″ is known only up to a composite cofactor C of unknown number of prime factors, the identity shows that W depends on the parity of ω(C) and on (ab/C): at 10P (cofactor of 94 digits, no prime factor below about 25 digits found by elliptic-curve factorization) and 10P + T (78 digits) one gets W = +1 if C is a product of two primes and W = −1 if it is a product of three.

A correlation form of the problem (Identity; the involution of C_2 Observed). With ε_n = f(t_n)(a_nb_n/N_n) and L_n = λ(N_n) for the prime-to-6 part N_n of |F(a_n, b_n)|, $F(a, b) = a^6 − 30a^4b^2 − 24a^2b^4 − b^6$, one has W(E_{t_n}) = ε_nL_n when N_n is squarefree, so #{n ⩽ X : W(E_{t_n}) = +1} = X/2 + S(X)/2 with S(X) = Σ_{n ⩽ X} ε_nλ(N_n). Positive upper density of positive signs is limsup S(X)/X > −1, and infinitely many positive signs is X + S(X) → ∞; the latter is only a restatement of the claim, not a weaker statement. The Liouville sign depends only on the parity of the valuations: λ(N) = (−1)^{ω(sqf(N))}, whereas the root number involves all primes dividing F, through (−1)^{ω(rad N)}(ab/rad N), so the two agree only for squarefree N; defining L_n with the radical removes the hypothesis. The two members nP and nP + T of C_2 are related by t ↦ (t + 1)/(t − 1), (a, b) ↦ (a + b, a − b) with $a'^2 + b'^2 = 2(a^2 + b^2)$ (the involution M(t) = (10 − 11t)/(10t + 11) with multiplier 221 belongs to C_{−11/5}, not to C_2). Put $G_2(a, b) = F(a + b, a − b) = −54a^6 + 54a^4b^2 + 64a^3b^3 + 54a^2b^4 − 54b^6$. Then gcd(F, G_2) = 1 and $\mathrm{Res}(F(t, 1), G_2(t, 1)) = 2^{17}·7·103·519521^2$, so F(a, b) and $G_2(a, b)$ have no common prime factor outside {2, 7, 103, 519521}, and the numerator of D((t + 1)/(t − 1)) is $27t^6 − 27t^4 − 32t^3 − 27t^2 + 27$ up to a factor −2, not a multiple of D. On the 3093 pairs

(t, (t + 1)/(t − 1)) with t = a/b, |a|, b ⩽ 50 (recomputed directly from the definition Π(t) = (−1)^{ω(R_t)}(C_t/R_t) of Appendix D; an earlier figure, mean 0.024 on 4406 pairs, used a separate implementation of f(t) that had not been validated against the table and is superseded), the global factors Π(t) = W(E_t)/f(t) have joint counts (+, +), (−, −), (−, +), (+, −) equal to 776, 783, 769, 765 and mean product 0.008 (standard error 0.018), so the global factors of the two members of a pair look independent; coprimality of F and G_2 gives disjoint supports of the primes but, as for any such statement, does not by itself give cancellation in Σ λ(N_n)λ(N_n′).

Exact definition of ε (notation). For t = U/V in lowest terms put $F(U, V) = U^6 − 30U^4V^2 − 24U^2V^4 − V^6$ and let F″ = N be the part of |F(U, V)| prime to 6 (the same number as the part prime to 6 of $|4a^3 + 27b^2c|$ for the triple below, since $4a^3 + 27b^2c = −4F(U, V)/d^3$ with d ∈ {1, 2}). In ε(t) = f(t)(ab/F″) the letters a, b denote the **legs of the primitive Pythagorean triple** (a, b, c) = ((V² − U²)/d, 2UV/d, (U² + V²)/d), not the numerator and denominator of t. In t-coordinates, $ab = 2UV(V^2 − U^2)/d^2$ and (ab/F″) = (2UV(V² − U²)/F″), because d^2 is a square. The symbols (U/F″) and (UV/F″), which are what one obtains by reading a, b as the numerator and denominator of t, give the same values as (b/F″) and are **not periodic in n** (no period up to 29 for 2 ⩽ n ⩽ 60), whereas (ab/F″) has period 6; the modulus itself is not the issue: the part of |F| prime to 6, prime to 2, or the part of $|4a^3 + 27b^2c|$ prime to 6 give identical symbols on n = 2, …, 60.

The computable sign ε_n is periodic in n (Observed for 2 ⩽ n ⩽ 202; the proof is reduced to two local statements, see the next paragraphs). An implementation of f(t) = $−w_2w_3(2/c)$ and of ε(t) = f(t)(ab/F″) from the local rules of §4 reproduces the table of this appendix: f agrees in 16 of 16 members and ε = W·λ in 16 of 16. With it, ε_n and the Jacobi symbol (ab/F″), which needs no factorization, were computed for 2 ⩽ n ⩽ 202 on both sequences (numerators of up to about 7000 digits). In all 201 values of each sequence,

ε(t(nP)) = +1 if and only if n ≡ 0, 1, 5 (mod 6), and ε(t(nP + T)) = +1 if and only if n ≡ 0 (mod 3),

with no exception. (Since f(t(nP)) = +1, the first statement is a statement about the Jacobi symbol alone; the second combines a Jacobi symbol of period 18 with the finite factor f.) The count of six values ε = +1 among the 16 tabulated members is the count of these classes for n = 2, …, 9. Consequences, when F″ is squarefree: (a) W(E_{t(nP)}) = ε_n λ(F″), so the root number is +1 exactly when λ(F″(t_n)) = +1 for n ≡ 0, ±1 (mod 6) and when λ(F″(t_n)) = −1 for n ≡ 2, 3, 4 (mod 6), and similarly along nP + T with the classes n ≡ 0 and n ≢ 0 (mod 3); (b) since the product of the two periodic signs is $(−1)^n$, W(E_{t_n})W(E_{M(t_n)}) = $(−1)^n$ λ(R_n), where R_n is the radical of $N(F·G_2)$ (the two values are coprime by Appendix C), which agrees with the eight tabulated members n = 2, …, 9; (c) Problem H fails only if λ(F″(t_n)) follows, from some point on, the prescribed periodic pattern of (a) on both sequences. Thus the required parity is not varying irregularly: it is a fixed function of n modulo 6. A proof route: write t_n = U/V; V is a perfect

square (the denominator of x(nP) on an integral model), and Jacobi reciprocity reduces (ab/F'') to characters of the odd parts of U and $V^2 - U^2$, because $F \equiv -V^6$, U^6 or $-54V^6$ modulo the primes dividing U, V or $V^2 - U^2$ respectively; periodicity then requires 2-adic and 3-adic control of these numbers along nP, in particular of the 2-adic valuation of U, which has not been carried out.

Reduction of the periodicity to local statements (Proved reduction; two local checks not yet written out). (1) *Reciprocity formula (proved).* Let $t = U/V$ in lowest terms with V a perfect square, as for every t_n (the denominator of x(nP) on an integral model is a square). Without any squarefreeness assumption,

$$\varepsilon(t) = f(t) \cdot \Pi_{v \in \{2, 3, \infty\}} (2t(1 - t^2), F(t))_v \cdot (-1, t)_v \cdot (-6, 1 - t^2)_v,$$

where $(,)_v$ is the Hilbert symbol and $F(t) = f(t)$ is the sextic. Proof: Hilbert reciprocity for the pair (ab, F) turns $(ab/N) = \Pi_{p \mid N}(ab, F)_p$ into the product of the symbols at 2, 3, ∞ and at the primes $p \geqslant 5$ dividing ab; at those primes $F \equiv -V^6$, U^6 or $-54V^6$ modulo p according as p divides U, V or $V^2 - U^2$, so the symbol equals $(-1/p)$, 1 or $(-6/p)$ to the power v_p; a second application of reciprocity to $(-1, U)$ and $(-6, V^2 - U^2)$ removes these primes, and since V is a square the pair (U, V) may be replaced by (t, 1) in every local symbol. The formula agrees with the computed ε in 1014 of 1014 cases ($t = U/V$, $V = D^2$, $D \leqslant 7$, all real positions of t). (2) *Real place (proved).* The product of the three real symbols is +1 for $t \leqslant 1$ and −1 for $t > \rho \approx 5.55$, the real root of f; for the integers $t = 2, \ldots, 5$ it is +1 again. So the sequence $t_n \leqslant 1$ has no real contribution, and the only exceptions to dependence on residues for integer t in $|t| \leqslant 6000$ are exactly $t = 2, 3, 4, 5$, as predicted. (3) *Singular points (proved at 0, ±1).* f has no root in $\mathbb{Q}_2$ (none modulo 4) and none in $\mathbb{Q}_3$ (none modulo 3^5), so F has no zero on the local points. Writing the product as $(t, -F)(1 - t, -6F)(1 + t, -6F)(2, F)$ shows that the residues at $t = 0, 1, -1$ are $-F(0) = 1$, $-6F(1) = 324 = 18^2$ and $-6F(-1) = 324$, all squares, so the symbols are locally constant there; near $t = \infty$ at 2 the symbols reduce to $(-1, -6)$ and $(1 - s^2, -6)$ with $s = 1/t$, and $1 - s^2 \equiv 1 \pmod 8$ because $s \in 4\mathbb{Z}_2$ when V is a square. (4) *Local constancy (verified numerically).* Hence ε is locally constant on the points of C_2 over $\mathbb{Q}_2$ and $\mathbb{Q}_3$, and by compactness ε(t) depends only on t modulo 2^{k_2} and 3^{k_3}. On 3765 points $t = U/V$ with $V = D^2$ ($D \leqslant 16$) and $(1 - t)(1 + t^2)$ a square in both $\mathbb{Q}_2$ and $\mathbb{Q}_3$, ε is a function of (t mod 8, t mod 9) with no conflict among 40 classes (and none at the finer levels mod 32, 27); without the condition that t be a local point of C_2 there are conflicts at the 2-adic poles, so that condition is used. (5) *Periodicity.* J_2 has good reduction at 3, so $n \mapsto nP \bmod 9$ is periodic, and $t_n \bmod 9$ has period 9; the class of t_n modulo 8 at 2 has period 4 (both checked for $2 \leqslant n \leqslant 202$). Therefore ε_n is periodic with period dividing 36, and the pattern of period 6 seen on 201 values (more than five full periods) is its exact form. *What remains:* a written verification of the local constancy at the 2-adic points over $t = \infty$ for the factors f(t) (the rule for w_2 and the symbol (2/c)), and a proof that $t_n \bmod 8$ has period 4 from the structure of $J_2(\mathbb{Q}_2)$.

Valuation laws along nP (Observed for 2 ⩽ n ⩽ 200; consistent with the elliptic divisibility sequence computed to n = 31 in Magma). Write t_n = U_n/V_n in lowest terms. By the classical identity x(nP) − x(P) = −ψ_{n−1}ψ_{n+1}/ψ_n² (here x(P) = 1), U_n is, up to a unit and a power of 2, the product W_{n−1}W_{n+1} of terms of the elliptic divisibility sequence of P, and V_n = W_n². The exact data show, in all 199 cases n = 2, …, 200: $v_2(V_n) = 2v_2(n)$ for n even and 0 for n odd; $v_3(V_n) = 2v_3(n)$ if 3 | n and 0 otherwise; $v_2(U_n) = v_2(n^2 - 1)$ for n odd and 0 for n even; $v_3(U_n) = v_3(n^2 - 1)$ if 3 ∤ n and 0 otherwise (the Magma table of $v_2(W)$, $v_3(W)$, $v_2(U)$, $v_3(U)$ for n ⩽ 30 agrees: v_p(W_n) = v_p(n) for p = 2, 3). Two consequences. (a) The 2-adic valuation of U_n is unbounded along the odd n (it is $v_2(n - 1) + v_2(n + 1)$), so t_n approaches 0 in $\mathbb{Q}_2$ when n ≡ ±1 modulo a high power of 2; this is exactly the situation covered by the vanishing residue at t = 0 in item (3), so the numerically observed period is not an accident of small n. (b) The class of t_n modulo 8 at 2 depends only on n modulo 4: for n odd, $v_2(U_n) \geqslant 3$ gives t_n ≡ 0 (mod 8); for n ≡ 2 (mod 4), $v_2(V_n) = 2$ gives 1/t_n ≡ 4 (mod 8); for n ≡ 0 (mod 4), $v_2(V_n) \geqslant 4$ gives 1/t_n ≡ 0 (mod 8). This settles the second open item above, period 4 of t_n modulo 8, given the valuation law for p = 2, which is verified here but not proved.

Splitting ε by place (Observed for 2 ⩽ n ⩽ 200). Collect all 2-adic data (the sign −1 of f, the rule w_2, the symbol (2/c) and the Hilbert symbols at 2) into Ψ_2, all 3-adic data (w_3 and the Hilbert symbols at 3) into Ψ_3, and use that the real factor is +1 for t ⩽ 1; then ε_n = $\Psi_2(n)\Psi_3(n)$ (checked against the Jacobi-symbol value of ε_n for 2 ⩽ n ⩽ 59, 58 of 58). Along nP, $\Psi_3(n) = -1$ if 3 | n and +1 otherwise, and $\Psi_2(n) = -(-1)^n$; along nP + T, $\Psi_3(n)$ is the same and $\Psi_2(n) = -1$ for every n. So each factor depends only on n modulo 2 or modulo 3, with minimal periods 2 and 3 (and 1 and 3 on nP + T), and the product has periods 6 and 3, as observed. This is the statement that the remaining factors depend only on n modulo 6; its proof is the local analysis of items (3)–(5) above.

F(U_n, V_n) is prime to 6, and the elementary reciprocity identity (Proved from the valuation laws; verified for 2 ⩽ n ⩽ 200). Modulo 2, F(U, V) ≡ U⁶ + V⁶, which vanishes only if U and V are both odd; by the valuation laws above V_n = W_n² is odd only for n odd, where $v_2(U_n) \geqslant 3$, so U_n and V_n are never both odd and F_n is odd. Modulo 3, F(U, V) ≡ U⁶ − V⁶, which vanishes only if neither U nor V is divisible by 3; but exactly one of them is (3 | n gives 3 | V_n, and 3 ∤ n gives 3 | U_n because $v_3(U_n) = v_3(n^2 - 1) \geqslant 1$), so F_n ≡ ±1 (mod 3). Hence F″_n = |F_n| exactly (checked for all 199 values 2 ⩽ n ⩽ 200; the Magma output for n ⩽ 20 shows $v_2(F) = v_3(F) = 0$ as well). With N = |F_n|, σ_F the sign of F_n, and each X ∈ {U, V − U, V + U} written X = ±2^{i}X′ with X′ odd and positive, quadratic reciprocity gives (X/N) = (±1/N)(2/N)^{i}(σ_F/X′)(F/X′)(−1)^{((X′−1)/2)((N−1)/2)}, where (F/U′) = (−1/U′) and (F/X′) = (−6/X′) for X = V ∓ U because F ≡ −V⁶ modulo the primes of U and F ≡ −54V⁶ modulo the primes of V ∓ U (and 3 ∤ V ∓ U). Since (V/N) = 1, (ab/F″) = (2/N)(U/N)((V − U)/N)((V + U)/N). This elementary formula equals the directly computed Jacobi symbol in

199 of 199 cases. The single factors are not periodic in n (for example (U/N) is not); only their product is, and that is the statement proved by the Hilbert-symbol analysis above, which groups all the 2-adic and all the 3-adic quantities.

Closed form of ε_n along nP and the period-six law (Proved from two facts that are verified for n ⩽ 200; the closed form is verified in 199 of 199 cases). Write m = V_n² − U_n². An independent review derived the exact formulas for the symbol J_n = (U_n/|F_n|) (n even: J_n = (−1/|U_n|)·sgn(U_n)^{[F_n<0]}; n odd: J_n = sgn(U_n)^{[F_n>0]}) from F_n ≡ 1 (mod 8) for n even and F_n ≡ 7 (mod 8) for n odd (both verified, 199 of 199, and the formulas themselves 199 of 199; the congruences follow from the valuation laws since for n odd 8 | U_n and V_n is odd, and for n even U_n is odd and 4 | V_n). The symbol J_n is not ε_n: the correction is K_n = (2m/|F_n|), so that ε_n = J_nK_n (199 of 199). Apply reciprocity to K_n as well: N = |F_n| ≡ ±1 (mod 8) gives (2/N) = 1; both V − U and V + U are odd (valuation laws), prime to 3, and F ≡ −54V⁶ modulo their primes, which gives (X/N) = (±1/N) (σ_F/|X|)(−6/|X|)(−1/|X|)^{δ} with δ = 1 if N ≡ 7 (mod 8) and 0 otherwise. Writing (−1/|x|) = sgn(x)χ_4(x) and (−6/|x|) = sgn(x)χ_{−24}(x) for the odd characters χ_4 and χ_{−24}, and (6/|x|) = χ_{24}(x) for the even character, all the real signs (those of U, V ∓ U and F) cancel in the product, as the real symbols are +1 for t ⩽ 1, and one is left with

ε_n = χ_4(U_n)·χ_{−24}(m) for n even, and ε_n = χ_{24}(m) for n odd.

Now m ≡ 7 (mod 8) for n even and m ≡ 1 (mod 8) for n odd, and m ≡ 2 (mod 3) if 3 | n, m ≡ 1 (mod 3) otherwise (all three verified, 199 of 199, and consequences of the valuation laws: for n odd m ≡ V² ≡ 1 (mod 8); for n even m ≡ −U² ≡ 7 (mod 8); modulo 3, 3 | n gives 3 | V, m ≡ −U² ≡ 2, and 3 ∤ n gives 3 | U, m ≡ V² ≡ 1). For m ≡ 1 (mod 8) the character χ_{24}(m) equals (m/3), and for m ≡ 7 (mod 8) the character χ_{−24}(m) equals (m/3). Hence for n odd ε_n = −1 if 3 | n and +1 otherwise, and for n even ε_n = χ_4(U_n)·(m/3) = −(m/3) provided U_n ≡ 3 (mod 4), i.e. +1 if 3 | n and −1 otherwise. This is exactly ε_n = +1 if and only if n ≡ 0, 1, 5 (mod 6). The proof therefore rests on two facts that remain to be proved: (i) the valuation laws for the elliptic divisibility sequence at 2 and 3, and (ii) U_n ≡ 3 (mod 4) for n even (verified for all 100 even n ⩽ 200; more precisely U_n ≡ 3 (mod 8) for n ≡ 2 (mod 4) and U_n ≡ 7 (mod 8) for 4 | n, 50 cases each, so the assertion U_n ≡ 7 (mod 8) for all even n is false, as U_2 = 3, and only the congruence modulo 4 is needed; a 2-adic proof would use that for n even the point nP lies in the formal group, where x = w^{−2} − a_2 + O(w²) with a_2 = −2, so that the numerator φ_n of x(nP) is ≡ 1 (mod 8) and U_n = V_n − φ_n ≡ −1 (mod 4)). The sequence nP + T has not been treated this way.

Two points of precision for the closed form. (a) The reduction argument at the additive prime 2 must be stated for the subgroup of points with nonsingular reduction, not for good reduction: modulo 2 the curve is y² = x³ with its cusp at (0, 0), the point P = (1, 1) reduces to a nonsingular point, the group of nonsingular points over $\mathbb{F}_2$ has order 2, so nP lies in the formal-group subgroup $E_1(\mathbb{Q}_2)$ exactly when n is even (x(2P) = 1/4 has a pole at 2). The valuation laws v_2(W_n) = v_2(n) for n even and v_3(W_n) = v_3(n) for 3 | n (the latter

from the good reduction at 3 and $v_3(W_3) = 1$, $W_3 = 3$) are verified for $n \leqslant 200$ and are proved in the theorem that follows. (b) The identity W(E_{t_n}) = ε_nλ(F″_n) holds when F″_n is squarefree; in general the root number is f(t_n)(−1)^{ω(R)}(ab/R) with R the radical, which differs from ε_nλ(F″_n) when some prime divides F″_n to a power at least 2. Squarefreeness of F″_n is false in general (23^2 divides F_{32}, see the next paragraph), so the period-six law is a statement about ε_n, and the exact link to W(E_{t_n}) is the formula given there.

Theorem (period six along nP; proved). Let $J_2 : y^2 = x^3 - 2x^2 + 2x$, P = (1, 1), t_n = 1 − x(nP) = U_n/V_n in lowest terms, F_n = F(U_n, V_n), N_n = |F_n| and ε_n = f(t_n)(ab/F″_n) with F″_n the part of N_n prime to 6. For every $n \geqslant 2$: F″_n = N_n, f(t_n) = +1, and ε_n = +1 if and only if $n \equiv 0, 1, 5 \pmod 6$. The proof is in the following lemmas; everything is elementary except the standard facts that the denominator of x(nP) on an integral model is a square and that the points of $J_2(\mathbb{Q}_2)$ with v(x) < 0 together with O form a subgroup E_1 (the kernel of reduction to the nonsingular points).

Lemma A (the point nP at 2). (a) For n odd, V_n is odd and $v_2(U_n) = v_2(n^2 - 1) \geqslant 3$. (b) For n even, $v_2(V_n) = 2v_2(n)$, U_n is odd, $U_n \equiv 3 \pmod 8$ if $n \equiv 2 \pmod 4$ and $U_n \equiv 7 \pmod 8$ if 4 | n; in particular $U_n \equiv 3 \pmod 4$. *Proof.* Modulo 2 the curve is $y^2 = x^3$ with its cusp at (0, 0); P = (1, 1) reduces to a nonsingular point, the group of nonsingular points over $\mathbb{F}_2$ has order 2, so $nP \in E_1$ exactly when n is even. For $Q \in E_1$ put v(x(Q)) = −2k (it is even since $y^2 = x^3(1 + \ldots)$), $k \geqslant 1$; then v(y) = −3k. *Doubling:* $x(2Q) = (x^2 - 2)^2/(4x(x^2 - 2x + 2))$; here $v((x^2 - 2)^2) = -8k$ and $v(4x(x^2 - 2x + 2)) = 2 - 2k - 4k$, so v(x(2Q)) = −2k − 2, that is k increases by exactly 1. *Odd multiples:* [m]Q for m odd has the same k, by induction on m using the formal group law $F(z_1, z_2) = z_1 + z_2 + z_1z_2(\ldots)$ over $\mathbb{Z}_2$ and v(z(Q)) = k (here z = −x/y): [m]Q = Q + [m − 1]Q with v(z([m − 1]Q)) > k. Now x(2P) = 1/4 has k = 1. For $n = 2^s m$ with m odd, nP = 2^{s−1} has k = s, so $v_2(V_n) = 2s$, and U_n is odd because gcd(U_n, V_n) = 1. To find U_n mod 8, use the expansion $x = z^{-2} + \Sigma_{j \geqslant 0} c_j z^j$ with $c_j \in \mathbb{Z}_2$ ($c_0 = -a_2 = 2$): write $V_n = 2^{2s}D'^2$ with D′ odd, $z = 2^s u$ with u a unit; then $\varphi_n := x(nP)V_n = u^{-2}D'^2 + (\text{terms of 2-adic valuation} \geqslant 3) \equiv 1 \pmod 8$, because the square of a 2-adic unit is ≡ 1 (mod 8). Since U_n = V_n − φ_n, $U_n \equiv -1 \pmod 4$; and $V_n \equiv 4 \pmod 8$ for s = 1, $V_n \equiv 0 \pmod{16}$ for $s \geqslant 2$ gives the residues modulo 8. (a) For n odd $nP \notin E_1$ and V_n is odd. For $R \in E_1$ with v(x_R) = −2k, the chord formula gives $x(P + R) - 1 = (x_R - 2y_R + 1)/(x_R - 1)^2$ (checked on 38 cases). If $k \geqslant 2$, $v(x_R - 2y_R + 1) = 1 - 3k$ because 1 − 3k < −2k, and $v((x_R - 1)^2) = -4k$, so v(x(P + R) − 1) = k + 1. For $n \equiv 1 \pmod 4$ take R = (n − 1)P, $k = v_2(n - 1) \geqslant 2$; for $n \equiv 3 \pmod 4$ use nP = (n + 1)P − P, so x(nP) = x(P + R′) with R′ = −(n + 1)P and $k = v_2(n + 1) \geqslant 2$. Since t_n = 1 − x(nP) and V_n is odd, $v_2(U_n) = v_2(t_n) = k + 1 = v_2(n^2 - 1) \geqslant 3$. □

Lemma B (the point nP at 3). J_2 has good reduction at 3 (its discriminant is -2^8) and P has order 3 modulo 3 ($2P \equiv (1, 2) = -P$ over $\mathbb{F}_3$). Hence 3 | n if and only if 3 | V_n, and if 3 ∤ n then $x(nP) \equiv 1$ and 3 | U_n; exactly one of U_n, V_n is divisible by 3. □

Lemma C. Modulo 2, $F \equiv U^6 + V^6$, and U, V are never both odd by Lemma A; modulo 3, $F \equiv U^6 - V^6$, nonzero by Lemma B; so F_n is prime to 6 and $F''_n = N_n$. By Lemma A, $F_n \equiv -V^6 \equiv -1 \equiv 7 \pmod 8$ for n odd (as $8 \mid U_n$) and $F_n \equiv U^6 \equiv 1 \pmod 8$ for n even (as $4 \mid V_n$), so $N_n \equiv \pm 1 \pmod 8$ and $(2/N_n) = 1$. Also $V_n \pm U_n$ is odd and prime to 3, V_n is a square, and $t_n \leqslant 1$ since $x \geqslant 0$ on the real curve. □

Lemma D (f = +1). In the triple, $3 \mid U_nV_n$ gives $3 \mid b$, so $B \equiv 0 \pmod 3$ and $w_3 = +1$. For n odd, $8 \mid U_n$ gives $(A, B) \equiv (1, 0) \pmod{16}$ and $c \equiv 1 \pmod 8$; for $n \equiv 2 \pmod 4$, $(A, B) \equiv (15, 8) \pmod{16}$; for $4 \mid n$, $(A, B) \equiv (15, 0) \pmod{16}$, with $c \equiv 1 \pmod 8$ in both even cases. All three pairs are in the list of the rule for w_2, so $w_2 = -1$, $(2/c) = 1$, and $f = -w_2w_3(2/c) = +1$. □

Lemma E (the closed form). Write $m = V^2 - U^2$, $N = N_n$, σ_F the sign of F, $\delta = 1$ if $N \equiv 7 \pmod 8$ and 0 otherwise (so $\delta = [F > 0]$ for n odd and $\delta = [F < 0]$ for n even), and $\chi_4(x) = (-1)^{(x-1)/2}$ for odd x. As $(V/N) = 1$ and $(2/N) = 1$, $(ab/N) = (U/N)((V - U)/N)((V + U)/N)$. For $X = V \mp U$ (odd, prime to 3) reciprocity gives $(X/N) = (\sigma_X/N)(\sigma_F/|X|)(-6/|X|)\chi_4(|X|)^\delta$, since $F \equiv -54V^6$ modulo the primes of X; for $U = \pm 2^iU'$, $(U/N) = (\sigma_U/N)(\sigma_F/U')\chi_4(U')^{1+\delta}$, since $F \equiv -V^6$ modulo the primes of U. Also $(-1/N) = (-1)^\delta$. *n odd:* the exponent of $\chi_4(U')$ is $[F < 0] + 1 + \delta = 2$, the exponent of $\chi_4(|m|)$ is $[F < 0] + \delta = 1$, and $(-6/|m|)\chi_4(|m|) = (6/|m|)$, so $(ab/N) = (-1)^{\delta \cdot \#\{\text{negative among } U, V - U, V + U\}}(6/|m|)$; since $t_n \leqslant 1$ and $\delta = 1$ only for $t < -\rho$, where two of the three are negative, the sign is +1 and $(ab/N) = (6/|m|) = (m/3)$, using $m \equiv 1 \pmod 8$. *n even:* the exponent of $\chi_4(U')$ is 1 and that of $\chi_4(|m|)$ is even, so $(ab/N) = (-1)^{\delta \cdot \#\text{neg}}\chi_4(|U|)(-6/|m|)$; with $\chi_4(|U|) = \operatorname{sgn}(U)\chi_4(U)$ and $(-6/|m|) = \operatorname{sgn}(m)\chi_{-24}(m)$, a check over the intervals $t < -\rho$, $(-\rho, -1)$, $(-1, 0)$, $(0, 1)$ shows that all real signs cancel, so $(ab/N) = \chi_4(U)\chi_{-24}(m) = -(m/3)$, using $U \equiv 3 \pmod 4$ and $m \equiv 7 \pmod 8$. By Lemmas A and B, $m \equiv 1 \pmod 8$ for n odd, $m \equiv 7 \pmod 8$ for n even, and $m \equiv 2 \pmod 3$ if $3 \mid n$, $m \equiv 1 \pmod 3$ otherwise. So $(ab/N) = -1$ if n is odd and $3 \mid n$, +1 if n is odd and $3 \nmid n$, +1 if n is even and $3 \mid n$, −1 if n is even and $3 \nmid n$, which is the stated pattern (+1 for $n \equiv 0, 1, 5$ and −1 for $n \equiv 2, 3, 4$ modulo 6). □ The lemmas were checked numerically for $2 \leqslant n \leqslant 200$, and the case distinctions agree with the values computed there. The link to $W(E_{t_n})$ still requires F''_n squarefree (see above); the sequence $nP + T$ is not covered.

The prime divisors of F_n do not form a divisibility sequence (Computed; negative result). A Magma run (complete factorizations for $n \leqslant 7$, trial division to 10^4 up to $n = 18$) agrees with the table of this appendix: for $n = 2, \ldots, 7$ the number of prime factors is 2, 2, 2, 3, 4, 4, the Liouville values are +1, +1, +1, −1, +1, +1, no prime is repeated, and $F_n > 0$ only for $n = 7, 10, 17$ among $n \leqslant 18$. It also showed that $F_m \nmid F_n$ for all $m \mid n$ tested and that $\gcd(F_m, F_n) = 1$ whenever $\gcd(m, n) = 1$ for $2 \leqslant m < n \leqslant 18$. The second pattern does not persist. For all pairs $2 \leqslant m < n \leqslant 70$ (exact integers, prime-to-6 parts) 132 pairs have a common prime, and 28 of them have $\gcd(m, n) = 1$: for example 307 divides F_3, F_{23}, F_{37} and F_{53}, 2851 divides F_5 and F_{48}, 631 divides F_6 and F_{67}, 547 divides F_{10} and F_{41}. The explanation is that $p \mid F_n$ exactly when $nP \equiv \pm R_i \pmod p$ for the points R_i of $J_2(\mathbb{F}_p)$ over the roots of f,

so the set of such n is a union of up to six classes ±a_i modulo the order m_p of P, with no relation between different indices: for p = 23, m_p = 10 and n ≡ ±2 (mod 10), so only even indices share 23, which produced the apparent pattern; for p = 83, m_p = 39 and n ≡ ±12, ±16; for p = 307, m_p = 163 and n ≡ ±3, ±23, ±37, ±53. Hence F_n is not a divisibility sequence and the Liouville values λ(F″_n) at different n have no multiplicative or divisibility structure to exploit.

F″_n is not squarefree in general (Proved by exact modular computation). Computing ψ_k(P) modulo p³ by the division-polynomial recursion (checked against the exact values for p ∈ {23, 83, 307} and 2 ⩽ n ⩽ 40, 117 of 117) gives $v_{23}(F_{32}) = 2$, $v_{83}(F_{612}) = 2$ and $v_{307}(F_{3859}) = 2$ exactly, so $23^2 \mid F_{32}$, $83^2 \mid F_{612}$ and $307^2 \mid F_{3859}$. A search over all n ⩽ 400 and primes p < 6000 finds p² | F_n only at (n, p) = (32, 23), (198, 23) and (262, 23); primes above 6000 were not searched. The root number is nevertheless given by the exact formula W(E_{t_n}) = f(t_n)(−1)^{ω(R)}(ab/R), R the radical of F″_n, since a prime p ⩾ 5 dividing F gives multiplicative reduction of type I_k with local sign w_p = −(ab/p) whatever k is. Comparing with ε_nλ(F″_n) = f(ab/F″)(−1)^{Ω(F″)} one finds W(E_{t_n}) = ε_nλ(F″_n)·Π_{p : v_p(F″_n) even} w_p, so the identity W = ε_nλ(F″_n) holds exactly when every prime divides F″_n to an odd power. At the three examples $w_{23} = +1$ (n = 32), $w_{83} = -1$ (n = 612) and $w_{307} = +1$ (n = 3859); F_{32} has not been factored, so ω(R) is not known there. The period-six law for ε_n is unaffected, since its proof does not use squarefreeness; what changes is that the statement "W = +1 if and only if λ(F″_n) = ε_n" must be replaced, at indices where some prime has an even exponent, by "W = +1 if and only if f(−1)^{ω(R)}(ab/R) = +1", and Problem H, which asks for infinitely many n with this sign +1, is unchanged in nature.

Test cases for the non-squarefree branch of the root-number formula (to be checked independently). Among the 190 values t = u/v with |u|, v ⩽ 60 for which F(u, v) is divisible by the square of a prime p ⩾ 5, in 94 the exact formula W = f(t)(−1)^{ω(R)}(ab/R) and the naive expression ε·λ(F″) give different signs, so an independent root-number computation decides between them. The simplest is t = −4, where (a, b, c) = (−15, −8, 17), F = 4a³ + 27b²c = 15876 = 2²·3⁴·7², and the integral model is Y² = X³ − 294780X + 90870848; the naive expression gives −1 and the exact formula +1. Others (prime with even exponent in brackets, naive, exact): t = 5/11 (7², naive +1, exact −1), −3/13 (7², +1, −1), 9/10 (7², −1, +1), 7/13 (19², +1, −1), −5/22 (23², +1, −1), 17/8 (7², −1, +1). A control where both agree: t = 4 (7², −1, −1). Since the earlier checks of the formula against Magma used the radical form of the theorem, these curves test exactly the case in which the squarefree shortcut fails.

Proposition (infinitely many non-squarefree values; proved). For every integer k ⩾ 0, $23^2 \mid F_{32+230k}$, $83^2 \mid F_{612+3237k}$ and $307^2 \mid F_{3859+50041k}$. In particular F″_n = |F_n| is not squarefree for infinitely many n. *Proof.* J_2 has good reduction at every odd prime (discriminant -2^8). Let p be odd and r_p the order of P modulo p (10, 39, 163 for p = 23, 83, 307). The kernel of $J_2(\mathbb{Z}/p^2\mathbb{Z}) \to J_2(\mathbb{F}_p)$ has exponent p, so p·r_p·P ≡ O (mod p²) and the

class of nP modulo p^2, hence the class of t_n modulo p^2, is periodic in n with period dividing $p \cdot r_p$. For $p \mid F_n$ the denominator V_n is a unit modulo p and $F_n = V_n^6 f(t_n)$, so the condition $p^2 \mid F_n$ depends only on n modulo $p \cdot r_p$. The values n = 32, 612, 3859 were computed exactly modulo p^3 above ($v_p = 2$). □

The mechanism: one lift per class (proved when $r_pP \not\equiv O \bmod p^2$; verified for 55 primes). If $p \mid F_{n_0}$ then t_{n_0} reduces to a simple root r of f modulo p (f is separable modulo p, and r is not a branch value 1, ±i of the degree-2 map $t: J_2 \to \mathbb{P}^1$, because $f(1) = -54$ and $f(\pm i) = -8$). For $n = n_0 + j \cdot r_p$ one has $nP = n_0P + jD$ with $D = r_pP$ in the kernel of reduction, $z(jD) = jz(D) + O(p^2)$ in the formal group, so $t(nP) \equiv t_{n_0} + j \cdot p \cdot u \pmod{p^2}$ with u a unit when $D \not\equiv O \pmod{p^2}$; thus $f(t_n) \equiv p(a + j f'(r)u) \pmod{p^2}$ has exactly one solution j modulo p. Hence modulo $p \cdot r_p$ there is exactly one n with $p^2 \mid F_n$ for each class of n modulo r_p with $p \mid F_n$ (classes come in pairs ±a, since $t_{-n} = t_n$). The computation confirms this for the three primes (2, 4, 8 classes and 2, 4, 8 lifts modulo 230, 3237, 50041) and for all 55 primes $p < 3200$ that divide some F_n (23, 83, 149, 197, 307, 467, 541, 547, …): in every case the number of lifts equals the number of classes, so $r_pP \not\equiv O \pmod{p^2}$ always occurred. Summing (number of classes)/$(p \cdot r_p)$ over these primes gives a density of about 0.0107 of indices n for which F_n is divisible by the square of one of them (0.0087 from p = 23 alone), which predicts about 4 such $n \leqslant 400$ against the 3 found. Squarefree values are therefore the majority, but not all, and the local root-number factor of the primes with even exponent must be kept in the exact formula for W above.

Squarefree kernel and the even-valuation correction (Proved identities; densities heuristic). Write $|F_n| = d_n s_n^2$ with d_n the product of the primes p with $v_p(F_n)$ odd, $\chi_p(n) = (g(t_n)/p)$ with $g(t) = 2t(1 - t^2)$, and $A_n = (-1)^{\omega(d_n)} = \lambda(F''_n)$ (the Liouville value depends only on the primes of odd valuation). Since $W(E_{t_n}) = f(t_n) \Pi_{p \mid F_n,\, p \geqslant 5}(-\chi_p(n))$ for every valuation, one has exactly $W = \varepsilon_n \cdot A_n \cdot$ DON'T PANIC$_n$ with DON'T PANIC$_n = \Pi_{p \mid F_n,\, v_p \text{ even}}(-\chi_p(n))$ and $\varepsilon_n = \Pi_{p \mid d_n}\chi_p(n) = (ab/d_n)$, the periodic sign of the Theorem above; so A_n is the Liouville core of Problem H, not a new quantity. Restrictions on d_n: since s_n is prime to 6, $s_n^2 \equiv 1 \pmod{24}$ and $\sigma_F d_n \equiv F_n \pmod{24}$, and F_n mod 24 depends only on n mod 6 (checked for $2 \leqslant n \leqslant 200$: 1 if $n \equiv 0$, 17 if $n \equiv 2, 4, 7$ if $n \equiv 3$, 23 if $n \equiv 1, 5$), together with the Jacobi condition $(ab/d_n) = \varepsilon_n$. The correction DON'T PANIC$_n$ is +1 whenever F_n is squarefree. For a prime p the sign $\chi_p(n)$ depends only on the class of n modulo r_p, hence is constant for the primes with a single pair of classes ±a (p = 23: $\chi = -1$ always, local factor +1; p = 83, 149, 197: $\chi = +1$, local factor −1); only primes with four or more classes (307, 631, 823, …) have both values. Summing over the 55 primes $p < 3200$ that divide some F_n, the density of indices with $p^2 \mid F_n$ and local factor −1 is about 0.0017 and with local factor +1 about 0.0090 (mostly from p = 23). So DON'T PANIC$_n = +1$ for about 99% of indices, but this is a heuristic statement: the density of n with F_n squareful is at least 0.0087 (proved, from p = 23 alone), whether DON'T PANIC$_n = -1$ occurs infinitely often cannot be proved without control of the other primes of even valuation, and in any case a change of the target sign on a set of density ≈ 0.2% does not touch the Liouville obstruction in A_n.

Consecutive values: no square-class mechanism (Proved in the function field; supported at the points). Let $h(Q) = f(1 - x(Q))$ on J_2 and $Q \mapsto Q + P$ the translation. With $x(Q \pm P) = (x^2 - x + 2 \mp 2y)/(x - 1)^2$ one finds $t(Q \pm P) = (\pm 2y - x - 1)/(x - 1)^2$, and the product N_+N_- of the numerators of $f(t(Q + P))$ and $f(t(Q - P))$ is a polynomial in x, equal to $(x - 1)^{12}H(x)$ with H of degree 12, irreducible over $\mathbb{Q}$ and coprime to $F(x) = f(1 - x)$. Since the conjugation of $\mathbb{Q}(J_2)/\mathbb{Q}(x)$ sends Q to $-Q$, $N(h(Q)h(Q + P)) = h(Q)^2 \cdot h(Q + P)h(Q - P) = F(x)^2H(x)/(x - 1)^{12}$, which is not a square in $\mathbb{Q}(x)$. For any $r \in \mathbb{Q}(x)^\times$ the norm of $r \cdot h(Q)h(Q + P)$ is $r^2 \cdot N(h(Q)h(Q + P))$, again a nonsquare, so $h(Q)h(Q + P) \notin \mathbb{Q}(x)^\times \cdot \mathbb{Q}(J_2)^{\times 2}$ (constants included). This excludes the simplest function-field mechanism for the signs at consecutive indices; it does not exclude identities with functions involving y, and the relevant identity for the torsion translation (Appendix D) was not a function-field identity but a point-specific one at the roots of f. The point-specific test for the translation by P gives the same negative answer: for the 4151 points R of $J_2(\mathbb{F}_p)$ over the roots of f, $p < 40000$, the character $(g(t(R + P))/p)$ is not determined by p modulo 24 and by $\chi(R) = (g(t(R))/p)$ (the best predictor is right for 52.5% of the points, 50% being no information), the same holds for $(c(t(R + P))/p)$, and the product of the two characters takes the values $+1$ and -1 equally often (2056 and 2095). So among the translations tested only that by the 2-torsion point T produces a constant square-class identity at the roots of f; the doubling (negative norm, Appendix D) and the translation by P do not. A square-class identity would in any case constrain quadratic characters, not the Liouville values $\lambda(F''_n)$.

x(nP) is a rational square for every n (Proved; explains the 2-isogeny experiment). The point $P' = (2, 4)$ lies on $J_2' : Y^2 = X^3 + 4X^2 - 4X$ and the dual isogeny $J_2' \to J_2$ has $x(\hat{\phi}(Q)) = (Y/(2X))^2$, so $x(\hat{\phi}(P')) = 16/16 = 1 = x(P)$; hence $P = \pm\hat{\phi}(P')$ (this is the statement that the class $x(P) = 1$ of P in $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ is trivial) and $nP = \hat{\phi}(nP')$, so $x(nP) = (Y_n/(2X_n))^2$ is the square of a rational number for every n (checked for $1 \leqslant n \leqslant 200$, 200 of 200, while $x(nP + T)$ is a square in none of the 40 cases tested). Consequences. (a) With $x(nP) = m_n^2/D_n^2$, $V_n = D_n^2$ and $V_n - U_n = m_n^2$, so $t_n = (D_n - m_n)(D_n + m_n)/D_n^2$ and $U_n = (D_n - m_n)(D_n + m_n)$ (58 of 58 cases). (b) For an odd prime p the image $\hat{\phi}(J_2'(\mathbb{F}_p))$ has index 2 in $J_2(\mathbb{F}_p)$, because $\ker\hat{\phi}$ has order 2 and $\#J_2 = \#J_2'$, and it contains P; therefore the order of P modulo p divides $\#J_2(\mathbb{F}_p)/2$ and the index $[J_2(\mathbb{F}_p) : \langle P\rangle]$ is even for every odd prime p. This is the pattern seen for all 668 odd primes below 5000 in the experiment (even index 668, odd index 0, P in the dual-isogeny image 668 of 668) and is not a separate phenomenon; in particular P is a primitive root for no prime. (c) A prime p can divide F_n only at a root r of f modulo p for which $1 - r$ is a square modulo p (and, as before, $(1 - r)(1 + r^2)$ is a square): this holds for all 158 classes found among the primes $p < 3200$, while only 212 of the 450 roots of f modulo these primes satisfy it, so it cuts the candidate pairs (p, r) roughly in half. (d) $F_n = F(\xi_{n^2})$ with $\xi_n = m_n/D_n$, but $f(1 - \xi^2)$ is irreducible of degree 12 over $\mathbb{Q}$, so F_n does not factor into two rational values $A(\xi_n)A(-\xi_n)$. None of this controls the Liouville values $\lambda(F''_n)$.

Coprimality of the two sextic values along C_2 (Proved by a finite computation). Let J_2 : $y^2 = z^3 - 2z^2 + 2z$, $P = (1, 1)$, $t_n = 1 - z(nP) = a_n/b_n$ in lowest terms, $F(a, b) = a^6 - 30a^4b^2 - 24a^2b^4 - b^6$ and $G_2(a, b) = F(a + b, a - b)$, and let $N(m)$ be the part of $|m|$ prime to 6. Then $\gcd(N(F(a_n, b_n)), N(G_2(a_n, b_n))) = 1$ for every $n \geqslant 1$ for which t_n is finite. Proof: $\gcd(a_n + b_n, a_n - b_n)$ divides 2, so for odd p a common prime factor p must divide $\mathrm{Res}(F(t, 1), G_2(t, 1)) = 2^{17}\cdot 7\cdot 103\cdot 519521^2$ and $t_n \bmod p$ must be a common root of F and G_2 in $\mathbb{F}_p$ (the point at infinity is not a root of F). The common roots are $t = 4$ for $p = 7$, $t = 66$ for $p = 103$ and $t = 134939, 482030$ for $p = 519521$. J_2 has good reduction at odd primes (its discriminant is -2^8), the reduction of P has order 6, 54 and 698 modulo 7, 103 and 519521, and listing $z(nP)$ over a full period shows that $t(nP) = 1 - z(nP)$ is never a common root (for $p = 7$ and $p = 103$ the common root is moreover not the t-coordinate of any $\mathbb{F}_p$-point of the cover $w^2 = (1 - t)(1 + t^2)$, the value of the right-hand side being a non-residue; for $p = 519521$ both roots do have points above them, and the exclusion comes from the orbit). The statement was also checked directly on the members $n = 2, \ldots, 11$. By contrast the analogous statement for the reflection pairs of C_{−11/5} is false (the prime 3532913 divides both values at $n = 20062$, see above). Since the Liouville function is completely multiplicative, coprimality is not needed for $\lambda(A_nB_n) = \lambda(A_n)\lambda(B_n)$; it matters only for the independence heuristics, because a prime dividing both values would contribute a local factor to both signs.

At all four members the point $(2, y)$ is non-torsion (no multiple up to 12 is the point at infinity). For $8P$, $F(t) = 2^2\cdot 23\cdot 541\cdot p_{26}\cdot p_{43}$ with a 26-digit and a 43-digit prime: my own elliptic-curve factorization (B_1 up to 250000, 156 curves) missed the 26-digit factor, and Magma's Factorization found it in a few seconds. I verified that $4\cdot 23\cdot 541\cdot p_{26}\cdot p_{43} = |F|$ and that all four factors are probable primes, and recomputed the Legendre symbols independently. The members $9P$ and $9P + T$ (F of 91 digits after removing 2 and 3) were also factored completely, with an elliptic-curve program of my own in a few minutes each (the 63-digit prime factor at $9P + T$ and the 42-digit one at $9P$ are probable primes), and give $W = +1$ at both; the next members $10P$ and $10P + T$ have F of 113 digits.

Why it is hard. (i) $s(t)$ is the parity of the number of prime factors (with a character twist) of the values of an irreducible sextic form, so H is a Liouville-type statement; the unconditional results of Helfgott and Desjardins require all irreducible factors of the polynomial of multiplicative reduction to have degree $\leqslant 3$, and here the degree is 6 and remains divisible by 6 after any rational base change (§6). (ii) The values are taken along the thin sequence nP, with heights growing like e^{n²}. (iii) Making F equal to a constant times a square infinitely often is impossible (Faltings, genus 2).

Structure available to a solver. A prime $p \geqslant 5$ divides $F(t(R))$ exactly when R reduces modulo p to one of the 12 points of E' lying over the six roots of $D(t) = t^6 - 30t^4 - 24t^2 - 1$ (two points over each root, $E' \to \mathbb{P}^1_t$ having degree 2). Hence for fixed p the condition on n is a congruence modulo the order of P in $E'(\mathbb{F}_p)$ (a rank-of-apparition condition, as for

elliptic divisibility sequences), and the primes dividing F(t_n) are the primitive divisors of a family of such sequences. A proof of H could use this, or could show the equivalent statement for the parity of the number of such primes along n in an arithmetic progression.

Exact local periods (3-adic part Proved, 2-adic part Observed). Along t_n the 2-adic factor is $w_2 = -1$ and the 3-adic factor $w_3 = +1$ for all $n \leqslant 534$, so f(t_n) = +1. Along t′_n the factor w_2 alternates +, −, +, − (period 2) and w_3 has period 9 with the pattern + − + + + + − + + for n = 1, …, 9, hence f has period 18. For p = 3 this is proved: E′ has good reduction at 3 with $|E'(\mathbb{F}_3)| = 6$, w_3 depends on (A, B) mod 9, hence on the point modulo 9, hence on the image of the point in the finite group $E'(\mathbb{Z}/9)$ of order 18; so the orbit $n \mapsto nP + jT$ is periodic with period dividing 18, and the values for $n \leqslant 18$ were computed exactly. For p = 2 the periods 1 and 2 are established by computation up to n = 534; a proof needs the analogous finite quotient of $E'(\mathbb{Q}_2)$, which I did not verify. Along nP, Problem H is therefore exactly the statement that s(t_n) is even for infinitely many n, and along nP + T the statement that (−1)^{s(t′_n)} = f(n mod 18) infinitely often.

Arithmetic form of s(t). Since $ab \equiv 2t(1 - t^2)$ modulo squares, a prime $p \geqslant 5$ dividing F(t) with $t \equiv r \pmod{\text{DON'T PANIC}}$ for a root r of D in $\mathbb{F}_p$ is of split type exactly when $2r(1 - r^2)$ is a square modulo p. Thus s(t) counts the primes of degree one of the sextic field $\mathbb{Q}(r)$ (r a root of D) that divide t − r and split in $\mathbb{Q}(r)(\sqrt{2r(1 - r^2)})$; H is a statement about the parity of this count along t = t_n.

The twelve points (Observed). Let $Q_j \in E'(\bar{\mathbb{Q}})$ be the twelve points over the six roots of D; the two points over a root are ±Q_i. The polynomial $R(w) = w^{12} - 66w^{10} - 29922w^8 + 73214w^6 - 62559w^4 + 22032w^2 - 3456$, whose roots are their w-coordinates, is irreducible, so they form one Galois orbit (the norm of $(1 - r)(1 + r^2)$ is $-54 \cdot 64 = -3456$, not a square). For the 3241 primes $7 \leqslant p < 30000$ other than 23, the number of roots of R modulo p follows the distribution of fixed points of the group $C_2^6 \rtimes S_4$ of order 1536 acting on the 12 points: the frequencies of 0, 2, 4, 6, 8, 10, 12 roots are 0.687, 0.189, 0.096, 0.013, 0.0105, 0.0034, 0.0003 observed against 0.678, 0.191, 0.104, 0.013, 0.0098, 0.0039, 0.0007 predicted. So the Galois group appears to be the full wreath product, the largest group compatible with the block structure ±Q_i. The function D(t) has divisor $\Sigma_j Q_j - 12 \cdot O$ on E′, so the relation $\Sigma_j Q_j = O$ is the only one it forces, and it is trivial because Q_j and −Q_j occur together. A prime $p \geqslant 5$ divides F(t_n) exactly when $nP \equiv \pm Q_i$ modulo a prime above p for some i, a rank-of-apparition condition.

Consequence. If H holds, there are infinitely many t with rank $E_t(\mathbb{Q}) \geqslant 2$ or Sha(E_t) infinite (§6), hence rank ⩾ 2 if Sha(E_t)[2^∞] is finite; similar statements hold for the other covers C_c of §6.

A test-bed. The following Python script (with `sympy`) computes t_n, the primes ⩾ 5 of F(t_n) with their split type, the finite part f and the sign W for n ⩽ 6. For larger n replace `factorint` by an external factorization of F (for example PARI or ECM).

```
from fractions import Fraction as Fr
from math import gcd
from sympy import factorint, jacobi_symbol

a2, a4 = -2, 2   # E': v^2 = z^3 - 2 z^2 + 2 z

def add(P, Q):
    if P is None: return Q
    if Q is None: return P
    x1, y1 = P; x2, y2 = Q
    if x1 == x2:
        if y1 == -y2: return None
        lam = (3*x1*x1 + 2*a2*x1 + a4) / (2*y1)
    else:
        lam = (y2 - y1) / (x2 - x1)
    x3 = lam*lam - a2 - x1 - x2
    return (x3, lam*(x1 - x3) - y1)

def trip(u, v):
    d = gcd(gcd(abs(v*v - u*u), abs(2*u*v)), v*v + u*u)
    return (v*v - u*u)//d, (2*u*v)//d, (v*v + u*u)//d

def w2(A, B):
    if A % 2 == 1:
        if A % 4 == 3:
            return 1 if B % 8 == 4 else -1
        s = (-3*A + 27*B) % 16
        if B % 8 == 4:
            return 1 if s in (9, 13) else -1
        return 1 if s in (5, 9) else -1
    return 1 if B % 4 == 3 else -1

def w3(A, B):
    if B % 3 == 0 or A % 9 in (0, 6): return 1
    return 1 if (4*B) % 9 in (4, 7, 8) else -1

P, T = (Fr(1), Fr(1)), (Fr(0), Fr(0))
R = None
for n in range(1, 7):
    R = add(R, P)
    for lab, Q in ((str(n) + "P", R), (str(n) + "P+T", add(R, T))):
        t = 1 - Q[0]
```

```
        a, b, c = trip(t.numerator, t.denominator)
        Fv = 4*a**3 + 27*b*b*c
        if Fv == 0: continue
        s = sum(1 for p in factorint(abs(Fv)) if p >= 5
                and jacobi_symbol((a*b) % p, p) == 1)
        A16, B16 = a*pow(c, -1, 16) % 16, b*pow(c, -1, 16) % 16
        A9, B9 = a*pow(c, -1, 9) % 9, b*pow(c, -1, 9) % 9
        f = -w2(A16, B16)*w3(A9, B9)
        print(lab, t, "s =", s, "f =", f, "W =", f*(-1)**s)
```

The same problem on the cover C_4'. Let C_4' : $v^2 = w^3 + 8$, with R = (1, 3) of infinite order and the 2-torsion point T = (−2, 0), and let $t(R') = (w(R') - 2)/(w(R') + 2)$; the members are t(nR) = −1/3, −15, 191/675, 25665/36481, −17822401/8782563, … and t(nR + T) = 0, 11/12, −104/121, −28369/8112, 3723720/6651241, …. (Translation by T acts on the t-line by $t \mapsto -(1 + 3t)/(3(1 - t))$.) Here c is not a perfect square, so the factor (2/c) enters the finite part $f(t) = -w_2(t)w_3(t)(2/c)$. Along nR, $f(t(nR)) = (-1)^n$ (period 2), and along nR + T, f = +1 (period 1), both observed for n ⩽ 260. Hence W(E_{t(nR)}) = +1 exactly when s(t(nR)) ≡ n (mod 2), and W(E_{t(nR+T)}) = +1 exactly when s(t(nR + T)) is even. The data (primes ⩾ 5 of F, S = split, N = non-split; exponent in brackets when > 1):

member	t	primes ⩾ 5 of F	s	f	W
R	−1/3	1471 S	1	−1	+1
2R	−15	859 N, 5743 S	1	+1	−1
3R	191/675	7 S [2], 19 S, 173 N, 233 S, (10 digits) N	3	−1	+1
4R	25665/36481	103 N, (27 digits) S	1	+1	−1
5R	−17822401/8782563	(9 digits) N, (36 digits) N	0	−1	−1
2R + T	11/12	1901 N, 65587 N	0	+1	+1
3R + T	−104/121	7 N [2], 557 N, 1231 S, 120049 S	2	+1	+1
4R + T	−28369/8112	19 S, 3457 S, 2852387 N, (16 digits) N	2	+1	+1
5R + T	3723720/6651241	171553 N, (11 digits) N, (26 digits) S	1	+1	−1

The signs along nR are +, −, +, −, − and along nR + T they are +, +, +, − (for n = 2, …, 5), so no alternating pattern persists. As for C_2, the polynomial whose roots are the w-coordinates of the twelve points over the roots of the discriminant sextic $27w^6 - 108w^4 + 256w^3 - 432w^2 + 1728$ is irreducible of degree 12, and the number of its roots modulo the 3241 primes 7 ⩽ p < 30000 follows the distribution of $C_2^6 \rtimes S_4$ (frequencies 0.686, 0.189, 0.102, 0.012, 0.006, 0.004, 0.001 against 0.678, 0.191, 0.104, 0.013, 0.010, 0.004, 0.001). So the global root-number formula reduces the question on C_4' to the same parity statement as on C_2 and does not imply infinitely many positive root numbers.

Averaging the sign over the cover parameter (Heuristic; data Observed). (i) *Balance over all t.* For t = a/b with |a| ⩽ H and 1 ⩽ b ⩽ H the formula of §4 gives (total, W = +1, W = −1) = (39, 22, 17), (127, 67, 60), (511, 248, 263), (1111, 526, 585) and (3095, 1531, 1564) for H = 5, 10, 20, 30, 50, identical to the counts of an independent Magma computation of the root numbers (average sign −0.011 and positive fraction 0.495 at H = 50), and W(E_t) = W(E_{−t}) in 785 of 1547 pairs (0.507), so there is no visible pairing between t and −t. (ii) *The extra parameter does not enter the sign.* The pairs (t, s) with a rational point of E_t of abscissa x_s(t), or of abscissa c for the covers C_c, are exactly the pairs (t, P) with P ∈ E_t(ℚ), that is the rational points of the elliptic surface S. The sign W(E_t) depends only on t, through the same Liouville-type factor of the same sextic form (§6 and Appendix B), so averaging over s changes the measure on t but not the arithmetic problem. (iii) *With s unrestricted the target is false, not just unproved.* Then every t with a rational point is covered. Under the parity conjecture these t include all those with W = −1, about half of all t, together with those of even rank ⩾ 2; the proportion of W = +1 among them is the density of rank ⩾ 2 among positive-rank t, which is 0 by the minimalist heuristics for a family of generic rank 0. A positive limiting proportion (liminf > 0) is therefore expected only for a thin set forced to carry a point, such as the members of one cover C_c or C^{(s)} (where rank 1 and rank 2 are each expected with probability about 1/2). (iv) *With s bounded the sets are thin* (about (log H)^{1/2} members up to height H for each cover) and the question returns to the sign along a single sequence. (v) *Degrees.* After any parametrization t = P(u, v)/Q(u, v), every irreducible factor of the pulled-back discriminant has degree divisible by 6 (§6), so the unconditional averaged results for forms of degree ⩽ 3 do not apply.

Local state against global parity (Observed). Write W(E_t) = f(t)·Π(t) with the local factor f(t) = $-w_2(t)w_3(t)(2/c)$ and the global factor Π(t) = Π_{p | F, p ⩾ 5}(−(ab/p)) = (−1)^{s(t)}. For the 3095 rational t = a/b with |a|, b ⩽ 50, f = +1 for 1973 of them (63.7%) and f = −1 for 1122, so the local factor is biased, while W is balanced (positive fraction 0.495). The fraction of Π = +1 is 0.495 among the t with f = +1 and 0.505 among those with f = −1, and within each of the eight classes of (w_2, w_3, (2/c)) it lies between 0.41 and 0.57 (the extreme values come from the classes with 39 and 44 curves; the classes with at least 455 curves lie between 0.49 and 0.52). So the local state and the global parity appear independent, and the balance of the sign comes entirely from Π. Along the sequences nP and nR of the covers the local state is not a mixing process but a deterministic periodic orbit in a finite quotient of the group of points (period 1 or 2 for f along nP, nR and 18 along nP + T), so the content of the problem lies in Π alone.

Pairs (t, −t): the Liouville factor cancels (Proved identity, Observed accuracy of the factor-free form). Since F(−t) = F(t) and c(−t) = c(t), E_t and E_{−t} have the same primes of multiplicative reduction and the same hypotenuse. For p ⩾ 5 the local factors of W(E_t)W(E_{−t}) are χ_p(t) = w_p(t)w_p(−t) = (−(ab/p))(−(−ab/p)) = (−1/p) if p | F, χ_p(t) = 1 if p | c, and the archimedean signs cancel, so W(E_t)W(E_{−t}) =

$w_2(t)w_2(-t)w_3(t)w_3(-t)\cdot\Pi(-1/p)$, the product over the primes $p \geqslant 5$ dividing F. The first factor depends only on (A, B) modulo 16 and modulo 9, and the second is (−1)^k with k the number of primes $p \equiv 3 \pmod 4$ dividing F. This is the root number of E_t over $\mathbb{Q}(i)$, because E_{-t} is the twist of E_t by −1, and it involves no Liouville function of the values of F. For squarefree $F'' = |F|/(2^{\alpha}3^{\beta})$ one has $\Pi(-1/p) = (-1/F'') = (-1)^{(F''-1)/2}$, so the root number of E_t over $\mathbb{Q}(i)$ is then a periodic function of t. On the 979 pairs $t = a/b$, $1 \leqslant a, b \leqslant 40$, the identity holds in 979 of 979 cases, and the form with $(-1/F'')$ in place of $\Pi(-1/p)$ holds in 938 of 979 (95.8%; it fails when a prime $\equiv 3 \bmod 4$ divides F to an even power). Along C_2 the products W(t)W(−t) are −, −, −, +, −, −, +, +, −, −, − for t = −1, 3/4, −7, −40/9, 31/49, −561/400, 161/961, 34440/34969, −69409/529, −2047837/2965284, −917447/5013121, with no constant pattern. This does not give the sign of a single E_t, because $W(E_t) = f(t)(-1)^{s(t)}$ still contains the Liouville-type factor, but it shows that $W(E_t)$ and $W(E_{-t})$ are determined by one common unknown bit. It would give a rank statement for a pair (t, −t) both carrying a known point: if the product is −1, one of the two curves has W = +1 and a point, hence rank ⩾ 2 (or Sha infinite). The covers considered here do not contain such pairs: $C_c(-t)$ is the twist by −1 of $C_c(t)$, and the only odd twists $d(-t) = -d(t)$ with sections, $d = 2t(t^2 - 1)$ and $-t(1 + t^2)$, have covers of rank 0. For the involutions of the covers that come from translation by a 2-torsion point ($\tau(t) = (t + 1)/(t - 1)$ on C_2 and $\tau(t) = -(1 + 3t)/(3(1 - t))$ on C_4') the singular fibres are not permuted (the numerator of $D(\tau(t))$ is coprime to D), so the primes of F(t) and $F(\tau(t))$ are unrelated and the Liouville factors do not cancel.

Pairs (t, M(t)) for the torsion involution M (Proved for the primes 2, 5, 13, 17; Observed on 1110 pairs). Let $M(t) = (10 - 11t)/(10t + 11)$, the involution induced by translation by the 2-torsion point of the rank-2 Jacobian of $C_{-11/5}$ (M(M(t)) = t). On the circle $t = \tan(\theta/2)$ it is the reflection $\theta \mapsto 2\beta - \theta$ with $\tan\beta = 10/11$, that is $(A, B) \mapsto ((21A + 220B)/221, (220A - 21B)/221)$, and the hypotenuse is multiplied by $221 = 13\cdot17$; it is the composition of $t \mapsto -t$ with the rotation by the angle (21, 220, 221), and it reproduces the large primes 891481, 16931, 300916397, 2003 and 1114721 of the partners M(0), M(−7), M(−4), M(−2) and M(2) in an independent report. Writing $\chi_p(t) = W_p(E_t)W_p(E_{M(t)})$, over the 1110 pairs with $t = u/v$, $|u|, v \leqslant 30$ one finds $\chi_2 = -1$, $\chi_5 = +1$, $\chi_{13} = -1$ and $\chi_{17} = +1$ in all 1110 cases, while $\chi_3 = -1$ in 183 cases and +1 in 927. These four values hold for every rational t: (i) at p = 2, the local sign w_2 depends only on (A, B) modulo 16; $221 \equiv 13$ is a unit with inverse 5, so $(A, B) \mapsto (9A + 12B, 12A + 7B)$ modulo 16, and a check of the 16 attainable classes shows $w_2(M(t)) = -w_2(t)$ for all of them; (ii) at p = 5, the reflection is (A, −B) modulo 5, so $E_{M(t)}$ agrees locally with the twist of E_t by −1, which is a square in $\mathbb{Q}_5$, and the local signs agree; (iii) at p = 13 and 17 the hypotenuse of M(t) acquires one more factor p than that of t, giving the local sign (2/13) = −1 and (2/17) = +1. (An earlier version of this paragraph used the rotation by the same angle instead of M and reported χ_2 as non-constant; that was the wrong map.) Since $\chi_2\chi_{13} = +1$, these constants do not affect the global sign. The remaining factor $\Pi_{p \notin \{2, 5, 13, 17\}} \chi_p(t)$ is not a character sum without Liouville factors: M does not permute

the six singular fibres (the numerator of D(M(t)) is coprime to D; the resultant of F(t, 1) and G(t, 1), where G(a, b) = F(10b − 11a, 10a + 11b), is $2^{20}5^6\cdot7\cdot239^2\cdot6133^2\cdot20873^2\cdot3532913\cdot585807239^2$), so the primes dividing F(t) and F(M(t)) are unrelated, and the product is the product of two independent copies of the Liouville-type factor Π of §6.

Common prime factors along the reflection pairs of C_{−11/5} (Proved; Observed for n ⩽ 20). Let P be a generator of the rank-2 group of C_{−11/5} and M(t) = (10 − 11t)/(10t + 11) the translation by 2-torsion. A computation with exact integers shows gcd(F″(t(nP)), F″(t(nP + T))) = 1 for 0 ⩽ n ⩽ 20 (the values reach 8000 digits). A prime p ⩾ 5 dividing both F(a, b) and G(a, b) = F(10b − 11a, 10a + 11b) for primitive (a, b) must divide $\mathrm{Res}(F(t,1), G(t,1)) = 2^{20}5^6\cdot7\cdot239^2\cdot6133^2\cdot20873^2\cdot3532913\cdot585807239^2$, and in fact must have a common root of F and G in the projective line over $\mathbb{F}_p$. There is none for p = 5, 13, 17, 239, 6133, 20873. For p = 7 (common root t = 4) and p = 585807239 (two common roots) the fibre of C_{−11/5} over the root has no $\mathbb{F}_p$-point ($w^2 = (1 + t^2)q_c(t)$ is a non-residue), so no rational point of the cover reduces to it, and these primes never divide both. For p = 3532913 the common root t = 3198986 has $\mathbb{F}_p$-points, M fixes it modulo p, and the point B of the cover over t = −201/43 (w < 0, which has the heights of a generator of the lattice) has order 882828 modulo p with t(nB) ≡ 3198986 and t(nB + T) ≡ 3198986, both roots of F, for n ≡ 20062 and n ≡ 461476 (mod 882828) (checked by modular arithmetic). So 3532913 divides both F″ values at those n, the first being n = 20062, and the coprimality observed for n ⩽ 20 does not hold for all n; for every generator of the lattice, residue classes of n with this property exist whenever the relevant points lie in the cyclic subgroup it generates modulo p.

Appendix C. A method: reducing statements about all multiples of a point to finite computations

This appendix records a method that was used several times in the paper (the periods of the local signs, the common prime factors of two sextic values, the fixed-prime lemma for the reflection pairs) in a form that can be reused.

Setting. Let J be an elliptic curve over $\mathbb{Q}$, $P \in J(\mathbb{Q})$ a point of infinite order, and $t : J \to \mathbb{P}^1$ a rational function defined over $\mathbb{Q}$ (here a degree-2 function, t = 1 − z on $y^2 = z^3 - 2z^2 + 2z$, or the corresponding function on the Jacobian of C_{−11/5}). For n with t(nP) finite write t(nP) = a_n/b_n in lowest terms. Let F and G be forms with integer coefficients (here $F(a, b) = a^6 - 30a^4b^2 - 24a^2b^4 - b^6$ and G the form F composed with the Möbius involution induced by translation by a 2-torsion point).

Lemma (reduction to finitely many primes and a finite orbit). Let p be a prime at which J has good reduction, and let m_p be the order of P modulo p. If p divides both F(a_n, b_n) and G(a_n, b_n), then (i) p divides the homogeneous resultant Res(F, G), and (ii) t(nP) mod p is a common root of F and G in $\mathbb{P}^1(\mathbb{F}_p)$. Both conditions depend only on n modulo

m_p, because (n + m_p)P ≡ nP (mod p). Hence "gcd(F(a_n, b_n), G(a_n, b_n)) has no prime factor outside a given finite set for every n" is decided by finitely many orbit computations.

Proof. If p divides F(a, b) and G(a, b) with (a, b) primitive, the point (a : b) of $\mathbb{P}^1(\mathbb{F}_p)$ is a common zero of F and G, so the two forms have a common factor modulo p and p divides their resultant. The orbit of the reduction of P is periodic with period m_p ⩽ p + 1 + 2√p, and t(nP) mod p depends only on nP mod p. □

Algorithm.

1. Compute and factor R = Res(F(X, 1), G(X, 1)) (and treat the points at infinity separately: here F(1, 0) = 1, so ∞ is not a root of F). The primes dividing R, together with the primes of bad reduction of J and the primes dividing the determinant of the Möbius map, form the finite exceptional set.
2. For each exceptional prime p, factor F and G modulo p and list the common roots r in $\mathbb{F}_p$.
3. (Cheap exclusion.) If the fibre of the cover over t = r has no $\mathbb{F}_p$-point (the quadratic character of the right-hand side of the cover equation at t = r is −1), no rational point of the cover reduces to r and p is excluded for all n.
4. (Orbit test.) Otherwise compute the orbit nP mod p for n = 1, …, m_p − 1 and record the residue classes of n with t(nP) ≡ r.
5. If no class is recorded for any exceptional prime, the two values are coprime for every n; otherwise the recorded classes (with their modulus m_p) describe exactly the n at which the prime divides both.

Results. On C_2 (J_2 : $y^2 = z^3 - 2z^2 + 2z$, P = (1, 1)) the exceptional primes are 7, 103 and 519521 with orders of P equal to 6, 54 and 698 and no hits, so the two sextic values are coprime for every n. On C_{−11/5} the exceptional prime 3532913 has orbit length 882828 for the point B over t = −201/43 and hits the common root t = 3198986 at n ≡ 20062 and 461476 (mod 882828), so the analogous statement is false there.

Variant for translates: the pair test (step 4′). To decide whether a prime p can divide the values of F at nP and at (n + k)P, collect the finite set S of points of J($\mathbb{F}_p$) lying over the roots of F in $\mathbb{F}_p$ (at most 12 points). A pair $R_1, R_2 \in S$ with $R_2 = R_1 + kP$ is necessary. This needs no orbit, so it works for primes of any size: at p = 2325076196570957, where the orbit of P has length of order 10^{15} and cannot be walked, the test takes 0.0 s. It is necessary but not sufficient: if a pair exists one must still decide whether R_1 lies in the subgroup generated by P (by walking the orbit when p is small, or by a discrete logarithm), as was done for 47, 18637 and 36997 in Proposition D3 (Appendix D), where it does not. The code below is the pair test for k = 2 on C_2.

```
from sympy import sqrt_mod
from sympy.polys.galoistools import gf_factor, gf_from_int_poly, gf_degree
from sympy.polys.domains import ZZ

def roots_f(p):                       # roots in F_p of t^6 - 30 t^4 - 24 t^2
- 1
    g = gf_from_int_poly([1, 0, -30, 0, -24, 0, -1], p)
    lc, facs = gf_factor(g, p, ZZ)
    return sorted((-gg[1]) % p for gg, m in facs if gf_degree(gg) == 1)

def add(P, Q, p):                     # group law on y^2 = x^3 - 2 x^2 + 2 x
over F_p
    if P is None: return Q
    if Q is None: return P
    (x1, y1), (x2, y2) = P, Q
    if x1 == x2:
        if (y1 + y2) % p == 0: return None
        lam = (3*x1*x1 - 4*x1 + 2) * pow(2*y1, -1, p) % p
    else:
        lam = (y2 - y1) * pow(x2 - x1, -1, p) % p
    x3 = (lam*lam + 2 - x1 - x2) % p
    return (x3, (lam*(x1 - x3) - y1) % p)

def pair_test(p):                     # pairs R1, R2 = R1 + 2P over roots of
f (t = 1 - x)
    S = set()
    for r in roots_f(p):
        x = (1 - r) % p
        v = (x**3 - 2*x*x + 2*x) % p
        if v == 0: S.add((x, 0)); continue
        s = sqrt_mod(v, p)
        if s is not None: S |= {(x, s), (x, (-s) % p)}
    two_P = (pow(4, -1, p), (-5 * pow(8, -1, p)) % p)
    return [(R, add(R, two_P, p)) for R in S if add(R, two_P, p) in S]
```

Other uses in this paper. The same idea, that a local quantity along nP depends only on nP modulo a prime power and is therefore periodic, underlies the periodicity of the finite part f(t_n) of the root number (proved for p = 3, period 18 for nP + T; verified for p = 2 up to n = 534, not proved by this method), and the finite residue checks behind the constancy of the characters χ_p of the involution M at p = 2, 5, 13, 17. Those checks were done by different scripts than the one below, and this appendix does not reprove them.

Limits. The method decides statements about finitely many primes. It says nothing about the number of prime factors of the values, so it does not touch the Liouville-parity problem H. The orbit test costs the orbit length, which can be as large as the prime itself

(about $3.5\cdot10^6$ steps was easy, about $6\cdot10^8$ would need compiled code); the pair test of step 4′ avoids this cost and decides most primes at once and it needs good reduction at the primes tested and care with primitive normalizations of the coordinates (here the common divisors of the reflected coordinates divide 2, 13 and 17 and are handled separately).

Code (Python with sympy). The script below reproduces the C_2 computation and can be adapted to another cover by changing the curve equation, the point and the map t.

```
from sympy import symbols, Poly, expand, resultant, factorint,
legendre_symbol
from sympy.polys.galoistools import gf_factor, gf_from_int_poly, gf_degree
from sympy.polys.domains import ZZ

a, b, t = symbols('a b t')
F = lambda x, y: x**6 - 30*x**4*y**2 - 24*x**2*y**4 - y**6
G2 = expand(F(a + b, a - b))              # the involution (a, b) -> (a + b,
a - b) of C2

Fc = [int(c) for c in Poly(F(t, 1), t).all_coeffs()]
Gc = [int(c) for c in Poly(G2.subs({a: t, b: 1}), t).all_coeffs()]
Res = resultant(Poly(F(t, 1), t).as_expr(), Poly(G2.subs({a: t, b: 1}),
t).as_expr(), t)
print("Res =", factorint(int(Res)))

def linear_roots(co, p):
    f = gf_from_int_poly(co, p)
    lc, facs = gf_factor(f, p, ZZ)
    return sorted((-g[1]) % p for g, m in facs if gf_degree(g) == 1)

def orbit_z(p):
    # y^2 = z^3 - 2 z^2 + 2 z over F_p with P = (1, 1): returns z(nP) for n =
1, ..., m_p - 1
    def add(P, Q):
        if P is None: return Q
        if Q is None: return P
        x1, y1 = P; x2, y2 = Q
        if x1 == x2:
            if (y1 + y2) % p == 0: return None
            lam = (3*x1*x1 - 4*x1 + 2) * pow(2*y1, -1, p) % p
        else:
            lam = (y2 - y1) * pow(x2 - x1, -1, p) % p
        x3 = (lam*lam + 2 - x1 - x2) % p
        return (x3, (lam*(x1 - x3) - y1) % p)
    P = (1, 1); Q = P; zs = [1]
    while True:
```

```
        Q = add(Q, P)
        if Q is None: break
        zs.append(Q[0])
    return zs

for p in [q for q in factorint(int(Res)) if q >= 5]:
    common = sorted(set(linear_roots(Fc, p)) & set(linear_roots(Gc, p)))
    zs = orbit_z(p)
    order = len(zs) + 1
    hits = [n + 1 for n, z in enumerate(zs) if (1 - z) % p in common]
    for r in common:
        val = ((1 - r) * (1 + r*r)) % p            # cover C2: w^2 = (1 - t)
(1 + t^2)
        print("p =", p, "root", r, "cover has points above it:",
              val == 0 or legendre_symbol(val, p) == 1)
    print("p =", p, "order of P =", order, "common roots =", common, "hits
=", hits or "none")
```

Appendix D. Reflection, translation by 2P and doubling: sorted register of imported results

This appendix sorts the results of two research files dated 5 November 2026 (a differential update on reflection and doubling, and an addendum on the sextic character fields) against the rest of the paper. Every item was re-derived in exact arithmetic unless marked otherwise. Notation: $f(t) = t^6 - 30t^4 - 24t^2 - 1$, $K = \mathbb{Q}(\alpha)$ with $f(\alpha) = 0$, $M(t) = (t + 1)/(t - 1)$, $c(t) = -6t(1 + t^2)$, $s(t) = (-2t^5 + 60t^3 + 46t - 4)/3$, $h = 3s$. For $t = a/b$ in lowest terms put $u = M(t) = c/d$ in lowest terms (so $(c, d) = (a + b, a - b)/g$ with $g \in \{1, 2\}$), $C_t = -6ab(a^2 + b^2)$, $C_u = -6cd(c^2 + d^2)$, and let R_t, R_u be the products of the primes $p \geq 5$ dividing $F(a, b)$ and $F(c, d)$.

D.1 Proved (re-derived here)

#	Statement	Basis
1	$-c(M(\alpha)) = s(\alpha)^2$ in K, i.e. $s(t)^2(t - 1)^3 \equiv 12(t + 1)(t^2 + 1) \bmod f$, and $\mathrm{Res}(f, h) = -186624 = -2^8 3^6$.	remainder 0, resultant recomputed
2	For every prime $p \geq 5$ and every root r of f in $\mathbb{F}_p$, $(c(M(r))/p) = (-1/p)$.	from 1: $s(r) \neq 0 \bmod p$ because $\mathrm{Res}(f, h)$ is a unit; also 150 of 150 roots for $5 \leq p < 1000$ (the source says 148; the difference is only which primes were counted)

#	Statement	Basis
3	$C_tC_u = 2ab(a^2 - b^2)\cdot(6(a^2 + b^2)/g^2)^2$.	identity of polynomials
4	**Theorem D1 (reflected Jacobi identity, all t).** $(C_t/R_t)(C_u/R_u) = (-2ab(a^2 - b^2)/(R_tR_u))$.	proof below; 4405 of 4405 cases with
5	$2P = (1/4, -5/8)$ and $t(Q + 2P) = (4(3x - 7)(4x + 1) - 80y)/(4(4x - 1)^2)$ for $Q = (x, y)$ on $y^2 = x^3 - 2x^2 + 2x$ with $4x \neq 1$.	checked for $Q = nP$, $n = 1, 3, \ldots, 12$
6	$\mathrm{Res}_x(f(1 - x), N(x)) = 2^{186}3^{27}23^{12}47^2947^24241^{12}18637\cdot36997^2\cdot232507619657 0957^2$ with deg N = 24, and $\mathrm{Res}_x(f(1 - x), 4x - 1) = -97543 = -23\cdot4241$.	exponents and the trivial cofactor recomputed
7	The doubling map is 𓀀(t) = $(t^4 - 2t^2 + 8t - 3)/(4(t - 1)(t^2 + 1))$, $t_{2n} = 𓀀(t_n)$ $(n = 1, \ldots, 8)$; $H = U^6 - 30U^4V^2 - 24U^2V^4 - V^6$ has degree 24, is irreducible over $\mathbb{Q}$, has constant term −97543, $\gcd(f, H) = 1$, and $\mathrm{Res}(f, H) = 2^{36}\cdot19\cdot229\cdot797\cdot7351\cdot12113\cdot363773\cdot609613\cdot9624457993 \cdot1386618529213$.	recomputed
8	$N_{K/\mathbb{Q}}(\gamma) = -14878279856690158253 1/2^{33}$ for $\gamma = c(𓀀(\alpha))$; hence γ and −γ are non-squares in K, $[K(\sqrt{\gamma}) : K] = 2$ and $\gamma \notin \mathbb{Q}^\times K^{\times 2}$.	recomputed; the negative sign is the whole argument, since $N(ds^2) = d^6N(s)^2 > 0$
9	$c_1 = c(\alpha)$, $c_2 = c(M(\alpha)) = -s^2$, $c_3 = c_1c_2$ are non-squares in K; $[c_2] = [-1]$, $K(\sqrt{c_2}) = K(i)$, $[L : K] = 4$ and $[L : \mathbb{Q}] = 24$ for $L = K(\sqrt{c_1}, i)$; $\sigma(\alpha) = -\alpha$ sends c_1 to $-c_1$, so $\sigma[c_1] = [c_3]$ and $K(\sqrt{c_1}) \cong K(\sqrt{c_3})$ over $\mathbb{Q}$.	$N(c_1) = (-6)^6\cdot N(\alpha)\cdot N(1 + \alpha^2) = 46656\cdot(-1)\cdot64 < 0$ shows c_1 and $-c_1$ are non-squares; −1 is not a square because f has real roots

Proof of Theorem D1. (i) For $p \geqslant 5$ dividing $F(a, b)$: $F \equiv -b^6$ if $p \mid a$, $F \equiv a^6$ if $p \mid b$, $F \equiv -8b^6$ if $p \mid a^2 + b^2$, and $F(1, 1) = -54$ excludes $p \mid a - b$, so $p \nmid ab(a^2 + b^2)(a - b)$ and $(C_t/p) \neq 0$. (ii) Since $C_u = d^4c(M(t))$ up to the factor g^4, $(C_u/p) = (c(M(r))/p) = (-1/p)$ for $r = a/b \bmod p$, by item 2. Exchanging the roles of t and u (M is an involution) gives $(C_t/p) = (-1/p)$ for $p \mid R_u$. (iii) For Jacobi symbols $(n/mk) = (n/m)(n/k)$ holds always, so $(C_tC_u/(R_tR_u)) = (C_t/R_t)(C_u/R_u)(C_t/R_u)(C_u/R_t) = (C_t/R_t)(C_u/R_u)(-1/R_tR_u)$; no coprimality of R_t and R_u is used. (iv) By item 3, C_tC_u is $2ab(a^2 - b^2)$ times a square coprime to R_tR_u, because $p \nmid a^2 + b^2$ for $p \mid R_t$ and $p \nmid c^2 + d^2$ for $p \mid R_u$. Combining gives the identity. □

The hypotheses that the source files list as outstanding (coprimality, denominators, local reduction) are therefore not needed for Theorem D1. What they leave open is the 2- and 3-adic part, which is the finite factor f(t)f(M(t)) already controlled in §4.

Corollary D2 (pair of root numbers). At a prime p ⩾ 5 dividing F(a, b) one has (ab/p) = (C_t/p) (2240 of 2240 pairs on |u|, v ⩽ 30, and by 27b²c ≡ −4a³ together with c ≡ −4a³/(27b²) in the triple of §4), so W(E_t) = f(t)(−1)^{ω(R_t)}(C_t/R_t) and

W(E_t)W(E_{M(t)}) = f(t)f(M(t))·(−1)^{ω(R_t) + ω(R_u)}·(−2ab(a² − b²)/(R_tR_u)).

On C_2 the two sextic values are coprime (Appendix C), so ω(R_tR_u) = ω(R_t) + ω(R_u) and, if the product N(F·G_2) is squarefree, the pair product is f(t)f(M(t))·λ(R)·(−2ab(a² − b²)/R) with R = R_tR_u: one Liouville value of the degree-12 form F·G_2 and one explicit Jacobi symbol. This reorganizes the correlation Σ λ(N_n)λ(N_n′) of the main text but does not control λ(R); it is not a distribution result.

Proposition D3 (translation by 2P; Proved by a finite computation). For every n ⩾ 1, gcd(N(F(a_n, b_n)), N(F(a_{n+2}, b_{n+2}))) = 1, where t_n = a_n/b_n = 1 − x(nP) in lowest terms and N(m) is the part of |m| prime to 6. Proof. A prime p ⩾ 5 dividing both values has t_n and t_{n+2} among the roots of f modulo p. Then x = x(nP) satisfies f(1 − x) ≡ 0 together with N(x) ≡ 0 (item 6) or 4x ≡ 1 (the degenerate case Q = ±2P), so p divides Res(f(1 − x), N(x)) or Res(f(1 − x), 4x − 1) = −97543, that is p lies in {23, 47, 947, 4241, 18637, 36997, 2325076196570957}. The pair test of Appendix C (step 4′) excludes four of them outright: for p = 23, 4241 and 2325076196570957 there is no pair of points R_1, $R_2 = R_1 + 2P$ over roots of f, and for p = 947 the two roots of f have no point of J(𝔽_p) above them. For p = 47, 18637 and 36997 pairs exist (2, 1 and 2 of them), but the orbit of P has order 7, 9368 and 18336 modulo p and contains no n for which both nP and (n + 2)P lie over roots of f, so these primes never divide both values. □ Checks: a scan of all primes 5 ⩽ p < 20000 finds pairs only at 47 and 18637, as the candidate list predicts, and the exact gcd for 1 ⩽ n ⩽ 60 is 1 in every case.

D.2 Matches with the rest of the paper

The nine signs W(E_{t_n}) = (+, −, −, −, −, +, +, −, +) for n = 1, …, 9 agree with the table of Appendix B (t_1 = 0 is the torsion member with W = +1), and the nine products W(E_{t_n})W(E_{M(t_n)}) = (−, −, +, +, +, −, +, +, +) agree with the products of the two columns nP and nP + T of that table for n = 2, …, 9 (t = −1 gives −1 at n = 1). The doubling products D_n = ε_nε_{2n} = (−, +, −, +) for n = 1, …, 4 and the lag correlations C_1, …, C_5 = 0, −1/7, 0, −3/5, 0 were recomputed. All of these are observations on short samples: D_n = $(-1)^n$ for n ⩽ 4 and the C_m prove nothing about a limit.

D.3 Candidates and open items

- **Translation by 2P (item 6).** The primes 23, 47, 947, 4241, 18637, 36997 and 2325076196570957 were the candidates for common divisors of the discriminant values at Q and Q + 2P. All seven are now excluded (Proposition D3), so the two values are coprime for every n.
- **Doubling (items 7, 8).** The scope of item 8 is limited: it excludes only a class in $\mathbb{Q}^\times K^{\times 2}$. A relation of [γ] with classes such as [c_1] or other elements of K is not excluded and has not been tested. The characters of $K(\sqrt{\gamma})$ are not local factors of the root number of the doubled curve, whose bad primes come from H.
- **Automorphisms.** The reported fact $\mathrm{Aut}_{\mathbb{Q}}(K) = \{1, \sigma\}$ was not recomputed; the cubic resolvent $y^3 - 30y^2 - 24y - 1$ has discriminant $3^9 \cdot 23$, not a square, which is consistent with it. Only the existence of σ is used above.

D.4 Not established or rejected

- **$W_p(E_t) = -(-1/p)$ for $p \mid F$.** Not proved and not needed. The proved local sign at $p \mid F(t)$ is $-(ab/p) = -(C_t/p)$ (§4). The reported test (78 of 148 agreements) compared the character at the reflected parameter with the local sign of the other curve, at primes that need not divide that curve's discriminant, so its failure does not contradict anything in this paper.
- **A prime-modulo-4 cancellation rule** for the combined multiplicative contribution: matched 6 of 9 cases (failed at n = 2, 6, 8). Rejected.
- **A universal square-class identity for doubling** of the reflection type with a rational constant: false by item 8, but see its scope above.

Appendix E. The 2-isogeny and the Galois groups of the sextic and of the character extension: sorted register

This appendix sorts a research file on the 2-isogeny of J_2 and on the splitting fields of f and of $R(z) = \mathrm{Res}_t(f(t), z^2 - g(t))$, with $g(t) = 2t(1 - t^2)$. Everything marked *recomputed* was verified here in exact arithmetic. Notation: $F(x) = f(1 - x)$, N the splitting field of f, M the splitting field of R, a_1, a_2, a_3 one root from each pair $\pm a_i$ of roots of f.

E.1 Recomputed

1. $F(x) = x^6 - 6x^5 - 15x^4 + 100x^3 - 189x^2 + 162x - 54$; $G(x) = x^6F(2/x) = -54x^6 + 324x^5 - 756x^4 + 800x^3 - 240x^2 - 192x + 64$; $\gcd(F, G) = 1$. So the root divisor of F is not preserved by $x \mapsto 2/x$ and F does not descend.
2. The isogeny $J_2 \to J_2'$: $Y^2 = X^3 + 4X^2 - 4X$ is $X = y^2/x^2 = x + 2/x - 2$.

3. $F(x)F(2/x) = H_6(X)$ with $H_6 = -54X^6 + 1998X^4 - 4060X^3 + 2730X^2 - 624X + 64$, irreducible over $\mathbb{Q}$. Hence for a root $x = 1 - \alpha$ of F the value $\beta = X(x)$ has degree 6, $Q(\beta) = Q(\alpha)$, and the isogeny gives another generator of the same sextic field, not a smaller one.
4. $R(z) = z^{12} - 109188z^8 - 991440z^4 - 186624$, irreducible over $\mathbb{Q}$; $R(0) = -186624 = N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(g(\alpha)) = -2^8 3^6$, so $g(\alpha)$ is a unit away from 2 and 3.
5. **$\mathrm{Gal}(N/\mathbb{Q}) \cong S_4$, of order 24 (proved).** Write $f(t) = h(t^2)$ with $h(y) = y^3 - 30y^2 - 24y - 1$, irreducible with discriminant $3^9 \cdot 23$, not a square, so the Galois group of h is S_3. The roots satisfy $y_1 y_2 y_3 = 1$, hence $\sqrt{y_1}\sqrt{y_2}\sqrt{y_3} = \pm 1$ and the group of f lies in $\{\varepsilon \in C_2^3 : \Pi\varepsilon_i = 1\} \rtimes S_3 \cong V_4 \rtimes S_3 \cong S_4$. The subgroup $V_4 \cap \mathrm{Gal}$ is S_3-invariant, so it is 0 or V_4; it is not 0, because a Frobenius element of cycle type (2, 4) occurs (below) and S_3 has no element of order 4. So $\mathrm{Gal}(N/\mathbb{Q}) = S_4$ and $[N : \mathbb{Q}] = 24$. The cycle types of f modulo p for the 33857 primes $5 \leqslant p < 4 \cdot 10^5$ ($p \neq 23$) are (1,1,2,2): 0.3774, (3,3): 0.3334, (2,4): 0.2483, (1^6): 0.0409, against 9/24, 8/24, 6/24, 1/24 predicted for S_4 on the six roots, and no other type occurs.
6. The pairs $\pm a_i$: since g is odd, $g(-a) = -g(a)$, so $[g(-a)] = [-1][g(a)]$ and $(g(-a)/p) = (-1/p)(g(a)/p)$. The roots of R are $\pm\sqrt{g(a_i)}$ and $\pm i\sqrt{g(a_i)}$, so $M = N(i, \sqrt{g(a_1)}, \sqrt{g(a_2)}, \sqrt{g(a_3)})$ and $[M : N] \leqslant 16$, that is $|\mathrm{Gal}(M/\mathbb{Q})| \leqslant 384$ instead of the naive $24 \cdot 2^6 = 1536$. (The two displayed products $g(a_1)g(-a_1)g(a_2)g(-a_2)$ and $g(a_1)g(-a_1)g(a_3)g(-a_3)$ are squares for the trivial reason $g(a)g(-a) = -g(a)^2$; that they generate all relations is equivalent to $|\mathrm{Gal}(M/\mathbb{Q})| = 384$.)
7. $N \subset M$: $S(w) = \mathrm{Res}_t(f(t), w - g(t))$ is irreducible ($R(z) = S(z^2)$ is), so $\mathbb{Q}(g(\alpha)) = \mathbb{Q}(\alpha)$.

E.2 Independence of the square classes and the order 384 (proved; Chebotarev check)

The order $|\mathrm{Gal}(R/\mathbb{Q})| = 384$, that is $[M : N] = 16$, was reported from Magma; it is proved in the paragraph after the next, and the statistics below are a consistency check. The lower bound needs $i \notin N$, which holds because the only quadratic subfield of N is $\mathbb{Q}(\sqrt{69})$, and the independence of $[g(a_1)]$, $[g(a_2)]$, $[g(a_3)]$ modulo squares and -1. A Chebotarev check supports it: of the 1386 primes below $4 \cdot 10^5$ that split completely in N (fraction 0.0409 against 1/24), R splits completely at 89 (0.0642 against 1/16 = 0.0625), and for the 683 of them with $p \equiv 1 \pmod 4$ the number of residues among the three symbols $(g(a_i)/p)$ is distributed as 84, 258, 252, 89 against 85, 256, 256, 85 for independence.

Proposition E ($|\mathrm{Gal}(M/\mathbb{Q})| = 384$; proved). The classes $[g(a_1)]$, $[g(a_2)]$, $[g(a_3)]$, $[-1]$ are independent in $N^\times/N^{\times 2}$, hence $[M : N] = 16$ by Kummer theory and $|\mathrm{Gal}(M/\mathbb{Q})| = 24 \cdot 16 = 384$. *Proof.* Let $V = \mathbb{F}_2^4$ with basis $e_1, e_2, e_3, \varepsilon$ mapping to $[g(a_1)]$, $[g(a_2)]$, $[g(a_3)]$, $[-1]$, and let $K \subset V$ be the kernel of this map to $N^\times/N^{\times 2}$. The group $\mathrm{Gal}(N/\mathbb{Q}) = C_2^2 \rtimes S_3 \cong S_4$ acts on V: S_3 permutes the e_i, an even sign change flipping the roots a_i, $i \in S$, sends e_i to $e_i + \varepsilon$ for $i \in S$ (because $g(-a) = -g(a)$), and ε is fixed. K is stable under this action. A

computation (enumerating the subspaces of $\mathbb{F}_2^4$ closed under the generators) shows that V has exactly 7 submodules, and that the only one containing none of ε, σ = $e_1 + e_2 + e_3$ and σ + ε is 0. Now ε ∉ K because −1 is not a square in N, and σ, σ + ε ∉ K: the product of the six values of g is $\mathrm{Res}(f, g) = -2^8 3^6 = -(g(a_1)g(a_2)g(a_3))^2$ (as $g(a)g(-a) = -g(a)^2$), so $g(a_1)g(a_2)g(a_3) = \pm 432 = \pm 3 \cdot 12^2$, whose class is [±3], and $[-g(a_1)g(a_2)g(a_3)] = [\mp 3]$. Since $\mathrm{Gal}(N/\mathbb{Q}) = S_4$ has a unique subgroup of index 2, $\mathbb{Q}(\sqrt{69})$ is the only quadratic subfield of N, so none of −1, 3, −3 is a square in N. Hence K = 0. □ In particular M = N(i, $\sqrt{g(a_1)}$, $\sqrt{g(a_2)}$, $\sqrt{g(a_3)}$) has degree 16 over N (the roots of R are $\pm\sqrt{g(a_i)}$ and $\pm i\sqrt{g(a_i)}$), and the field identification N ⊂ M is item E.1.7 ($\mathbb{Q}(g(\alpha)) = \mathbb{Q}(\alpha)$ because S is irreducible), so no further field-identification step remains.

E.3 Relation to the rest of the paper

The expression $g(t) = 2t(1 - t^2)$ is the t-form of the product ab of the legs of the triple, so the symbols (g(a)/p) at the roots a of f modulo p are the local signs at the primes p | F, $w_p = -(g(a)/p)$ (§4). Moreover $-3(1 + \alpha^2)/(1 - \alpha^2) = c(\alpha)/g(\alpha)$, with $c(t) = -6t(1 + t^2)$ as in Appendix D, is a square in $\mathbb{Q}(\alpha)$: it is a square modulo p at all 5938 roots of f in $\mathbb{F}_p$ for p < 60000. Hence [g(α)] = [c(α)] and M contains the field L = K($\sqrt{c_1}$, i) of Appendix D (degree 24 over $\mathbb{Q}$). The question asked at the end of the research file, whether the prime-divisor character product in the root-number formula can be handled by reciprocity, is answered by the Hilbert-symbol reduction of Appendix B: along nP the product of the local signs at the primes dividing F_n reduces to local symbols at 2, 3 and ∞ and is periodic of period 6 (proved). In Chebotarev terms each individual sign w_p is equidistributed over $\mathrm{Gal}(M/\mathbb{Q})$, but along nP the root a(p, n) is fixed by n modulo the order of P, which is why no distribution statement follows.

E.4 What this does not give

None of this is a distribution theorem for the global root numbers, nor a step toward infinitely many specializations of rank at least 2: it constrains the characters at a fixed prime (the oddness of g and the evenness of f), and the remaining obstruction is the Liouville value λ(F″_n) (Problem H).

Appendix F. The parity problem along the orbit: sieve, Galois structure, searches and covers

Appendix F began as a study of the quadratic-residue sieve on the roots of f and became the main record of the parity programme and of the Picard-number computations. Each paragraph F.n is self-contained. The table gives its topic and its status (Proved, Observed,

Conditional, Open, or Remark for a comment that proves nothing); the paragraphs are grouped under five subheadings. Where a later paragraph corrects an earlier one, the title of the earlier one says so (F.12, F.29, F.33, F.35, F.43, F.44).

§	Topic	Status
F.1	Sieve on the roots of f: admissible roots, statistics for p up to 5000	Proved
F.2	Galois group of the sieve field L; the square identity $-3(1-\alpha^4) = s(\alpha)^2$	Proved
F.3	Density 175/1024 of the primes with an admissible root	Proved
F.4	The second obstruction: R lies in the group generated by P	Open
F.5	The compositum L*, of group order 24576	Proved
F.6	Independent confirmation of L*; corrected local sign formula	Proved
F.7	The fifth relation, from the cubic	Proved
F.8	Further classes; z + 2 is a square	Proved
F.9	Classes of v − k in the cubic field	Proved
F.10	The relations from ideal factorizations; class number 1	Proved
F.11	Relation to Verzobio's primitive-divisor theorems	Observed
F.12	Relation to Helfgott's thesis (the equivalence claimed here is withdrawn in F.24)	Conditional
F.13	Rational multisections: conic base change and lattice constraints	Observed
F.14	The rational trisection class: net, double plane, bitangents	Observed
F.15	No smooth rational $\mathbb{Q}$-curve of degree at most 21 over the t-line	Proved
F.16	Where the Liouville values come from; weaker variants	Observed

§	Topic	Status
F.17	Elliptic-net test: no information passes between different n	Observed
F.18	The squarefree part of F_n tends to infinity (Faltings)	Proved
F.19	No multiplication or translation identity for h(x(Q))	Proved
F.20	Valuations at 2 and 3 of F_n and G_n	Proved
F.21	The period-3 law along nP + T	Proved
F.22	Valuation laws $v_2(D_n)$, $v_3(D_n)$ from the formal group	Proved
F.23	$\lambda = \chi_4 \cdot (-1)^{\{\Omega_1\}}$: a reformulation that does not help	Remark
F.24	Precise statement of the remaining bit; correction of F.12	Proved
F.25	gcd(F_n, G_n) = 1	Proved
F.26	The quotient sextic of the involution $x \mapsto 2/x$	Proved
F.27	Literature status: primitive divisors versus parity	Remark
F.28	The exact root-number formula for every rational t	Proved
F.29	Status of the open items at the time of F.28	Remark
F.30	Why C_2 and not C_4 or C_4'	Observed
F.31	The one-bit test on the factored members	Observed
F.32	Result at 10P + T	Observed
F.33	What a period-4 law would imply	Remark
F.34	What is and is not excluded among algebraic routes	Proved
F.35	Conjecture D: $\delta_{2n} = \delta_n$ (refuted in F.38)	Remark
F.36	Independence of h and its doubling pullback modulo squares	Proved
F.37	Conjecture D on the cover C_4'	Observed
F.38	The period-4 law and Conjecture D refuted at n = 11 and n = 12	Observed
F.39	Status of N(10P); what a partial factorization gives	Remark

§	Topic	Status
F.40	Problem H in terms of the sums T_r(X)	Remark
F.41	Doubling and tripling laws; general squareclass independence	Proved
F.42	The even/odd asymmetry of divisibility is a one-prime effect	Observed
F.43	A second section on the twist of a constant-x cover: case analysis	Remark
F.44	The cover c = 13/9: ρ = 13	Conditional
F.45	The covers c = −11/5, 14/45, 1, 4; the sign rule; corrections	Conditional
F.46	Is ρ = 13 for every c? Scan of 24 further values	Observed

F.A The sieve, the Galois structure and the class relations (F.1–F.10)

F.1 The sieve (Proved). Since x(nP) is a rational square (Appendix B) and $y^2 = (1 - t)(1 + t^2)$ at every point, a prime $p \geq 5$ can divide F_n only if $t_n \equiv r \pmod p$ for a root r of f in $\mathbb{F}_p$ with both $1 - r$ and $1 + r^2$ nonzero squares in $\mathbb{F}_p$ ($f(1) = -54$ and $f(i) = -8$ rule out vanishing). Let $G_p(t) = \gcd(f, (1 - t)^{(p-1)/2} - 1, (1 + t^2)^{(p-1)/2} - 1)$ in $\mathbb{F}_p[t]$; its roots are exactly the admissible roots, all in $\mathbb{F}_p$. If $\deg G_p = 0$ then $p \nmid F_n$ for every n (also when $p \mid V_n$, since then $F_n \equiv U_{n^6} \not\equiv 0$). Recomputed for the 667 primes $5 \leqslant p \leqslant 5000$: 397 have no root of f, 59 have roots but no square $1 - r$, 93 have a square $1 - r$ but no admissible root, and 118 have an admissible root, that is 118 of the 270 primes with roots (0.437); of the 660 roots 328 pass the first condition and 173 pass both (0.262). The orbit of P modulo p meets the roots for 81 primes, 37 admissible primes are missed by the orbit, and no orbit hit violates the sieve.

F.2 The Galois group of the closure of $K = \mathbb{Q}(r, \sqrt{(1 - r)}, \sqrt{(1 + r^2)})$ (Proved). Let $L = N(\sqrt{(1 - \alpha)}, \sqrt{(1 + \alpha^2)} : f(\alpha) = 0)$. Write $A_i^+ = [1 - a_i]$, $A_i^- = [1 + a_i]$, $B_i = [1 + a_i^2]$ for the three pairs $\pm a_i$ in $N^\times/N^{\times 2}$ (nine classes). *Relation.* $-3(1 - \alpha^4) = s(\alpha)^2$ with $s(t) = (t^5 - 29t^3 - 26t)/3$ (verified exactly: $s^2 + 3(1 - t^4) \equiv 0 \bmod f$), so $A_i^+ A_i^- B_i = [1 - a_i^4] = [-3]$ for each i; two independent relations. This also proves the identity $[g(\alpha)] = [c(\alpha)]$ of Appendix E, because $c(t)/g(t) = -3(1 - t^4)/(1 - t^2)^2$, which had been checked only at 5938 roots. *Rank.* Evaluating the nine symbols at all labelings ($a_1 a_2 a_3 = 1$, 24 per prime) for the 714 primes below $2 \cdot 10^5$ that split completely in N gives 128 distinct vectors of rank 7, so at most two independent relations exist, and the two above are all of them. Hence the span has

dimension 7, $[L : N] = 2^7$, and $|\mathrm{Gal}(L/\mathbb{Q})| = 24 \cdot 128 = 3072$. Note that $\sqrt{-3} \in L$ but $\sqrt{-3} \notin N$ (the only quadratic subfield of N is $\mathbb{Q}(\sqrt{69})$), so the character $\kappa(p) = (-3/p)$ is independent of the Frobenius class in N.

F.3 Density (Proved by Chebotarev; checked numerically). Seven independent characters on Gal(L/N) are κ and, for each pair i, the symbols of $1 - a_i$ and $1 + a_i$, with the symbol of $1 + a_i^2$ equal to κ times their product. A Frobenius element of type (1^6) fixes all six roots; given κ the three pairs are independent and a pair is admissible (symbol of $1 + a^2 = +1$ and at least one of $1 \mp a$ a square) with probability 1/4 if $\kappa = +1$ and 1/2 if $\kappa = -1$, so at least one admissible root occurs with probability $\tfrac{1}{2}(1 - (3/4)^3) + \tfrac{1}{2}(1 - (1/2)^3) = 93/128$. A Frobenius of type (1,1,2,2) (nine of 24) fixes exactly one pair and gives probability $\tfrac{1}{2} \cdot \tfrac{1}{4} + \tfrac{1}{2} \cdot \tfrac{1}{2} = 3/8$; other types fix no root. Hence the set of primes p for which f has an admissible root has natural density

$$(1/24)(93/128) + (9/24)(3/8) = 175/1024 \approx 0.1709,$$

with density $181/1536 \approx 0.1178$ among $p \equiv 1 \pmod 3$ ($\kappa = +1$) and $43/192 \approx 0.2240$ among $p \equiv 2 \pmod 3$ ($\kappa = -1$). For the 25994 primes $5 \leqslant p < 3 \cdot 10^5$ ($p \neq 23$) the observed proportions are 1516/12970 = 0.1169, 2912/13024 = 0.2236 and 0.1703 overall; the model with nine free classes would give 0.1721 for both residue classes, so the asymmetry modulo 3 confirms the relation. So the primes that can divide some F_n have upper density at most 175/1024.

F.4 The second obstruction (heuristic). An admissible root r gives a point R of $J_2(\mathbb{F}_p)$ with $x(R) = 1 - r$ a nonzero square, i.e. R lies in the index-2 subgroup $\hat{\phi}(J_2'(\mathbb{F}_p))$ that contains $\langle P \rangle$. The remaining condition $R \in \langle P \rangle$ is of Artin type (it depends on the index of $\langle P \rangle$ in that subgroup) and is not a Chebotarev condition in a fixed number field; among the 118 admissible primes below 5000, 81 are hit by the orbit and 37 are not.

F.5 The compositum of the sieve field and the character field (Proved under the following reading of L*). Let $L^* = L \cdot M$, where L is the field of F.2 and $M = N(i, \sqrt{g(a_1)}, \sqrt{g(a_2)}, \sqrt{g(a_3)})$ is the field of Appendix E, that is $L^* = N(\sqrt{1 - a_i}, \sqrt{1 + a_i}, \sqrt{1 + a_i^2}, \sqrt{-1}, \sqrt{g(a_i)})$. Then $[N : \mathbb{Q}] = 24$, $[L : N] = 2^7$, $[L^* : N] = 2^{10}$ and $[L^* : L] = 8$, so $|\mathrm{Gal}(L^*/\mathbb{Q})| = 24 \cdot 2^{10} = 24576$. *Proof.* Over the 7-dimensional span of the nine classes of F.2 one adjoins the four classes $[-1]$, $[g(a_1)]$, $[g(a_2)]$, $[g(a_3)]$, an 11-element generating set. The only relation among them is $[g(a_1)g(a_2)g(a_3)] = [\pm 432] = [\pm 3] = [\pm 1] \cdot [-3]$, with $[-3]$ already in the span of F.2 (Appendix E: $g(a_1)g(a_2)g(a_3) = \pm 432$), so the rank is at most $7 + 4 - 1 = 10$. Evaluating the symbols of the nine classes together with $(-1/p)$ and $(g(a_i)/p)$ at all 24 labelings of the 714 primes below $2 \cdot 10^5$ that split completely in N gives vectors of rank 10, so the rank is exactly 10. (Equivalently $[g(a_i)] = [2][a_i][1 - a_i][1 + a_i]$ with $[2] = B_1B_2B_3$ in the span, so over L the new classes are $[-1]$ and $[a_i]$ with $a_1a_2a_3 = 1$; adjoining -1, 2, 3 alone raises the rank only to 8, and adjoining -1, 2, 3 and a_1, a_2, a_3 gives 10.) □ This is the field that carries both the sieve characters (the symbols of $1 - r$ and $1 + r^2$) and the local root-number characters (the symbols of $g(r)$). The new classes are independent of those of L, so the local sign −

(g(r)/p) at an admissible root is equidistributed given admissibility: among the 4458 admissible roots below $2\cdot10^5$, (g(r)/p) = +1 for 2265 (0.508) and = −1 for 2193, and the proportions are the same modulo 3 (1110 and 1106 for p ≡ 1, 1155 and 1087 for p ≡ 2). This is a statement about a fixed prime, not about the primes dividing F_n along nP.

F.6 The independent dossier on L*: confirmation, the extra relation, and a correction (Proved; one formula corrected). An independent dossier defines L = N(√A_i, √B_i, √C_i) and L* = L(√D_i, √E_i) with A_i = 1 − a_i, B_i = 1 + a_i, C_i = 1 + a_i², D_i = 2a_i, E_i = −2a_i, and reports [L* : N] = 2^{10}, [L* : L] = 8 from Magma (exact number-field square tests, 2047 products, one extra relation). This is the same field as the L* of F.5: D_iE_i = −4a_i² gives [E_i] = [−1][D_i], and g(a_i) = D_iA_iB_i with A_iB_i = [−3]C_i in the span of L, so L(√−1, √g(a_i)) = L(√D_i, √E_i). *Certificate.* With the consistent labelings ($a_1a_2a_3$ = +1, 24 per prime; the dossier's remark that independently sorted labels are not compatible across primes is correct, and the diagnostic rank 11 it mentions is not a lower bound) the symbol vectors of the nine old classes have rank 7 and those of all fifteen classes A, B, C, D, E have rank 10 (560 primes below $1.5\cdot10^5$ splitting completely in N; 128 and 1015 distinct vectors). *Upper bound, with the extra relation proved symbolically.* The product of the roots of g(Z) = $Z^3 - 30Z^2 - 24Z - 1$ is $z_1z_2z_3$ = +1 (not −1), so $a_1a_2a_3$ = ±1, and Π(1 + z_i) = −g(−1) = 8, so $C_1C_2C_3$ = 8. From $-3(1 - a^4) = s(a)^2$ one gets $A_iB_i = 1 - a_i^2 = -s(a_i)^2/(3C_i)$. Hence for a labeling with $a_1a_2a_3$ = +1, $C_1D_1A_2B_2D_2A_3B_3D_3 = C_1\cdot(8a_1a_2a_3)\cdot s(a_2)^2s(a_3)^2/(9C_2C_3) = 64s(a_2)^2s(a_3)^2/(9(C_2C_3)^2) = (8s(a_2)s(a_3)/(3C_2C_3))^2$, an explicit square in N, which is the relation reported by the dossier. The label-invariant form is Π_i $C_iD_i = 64a_1a_2a_3 = \pm64$, that is $[C_1C_2C_3D_1D_2D_3] = [a_1a_2a_3]$; for a labeling with $a_1a_2a_3$ = −1 one replaces one D_i by E_i. With the four binary relations (the three triple sums A_iB_iC_i and the three pair sums D_iE_i are each equal) the formal rank 15 drops to 11 and the extra relation gives 10. *Correction to the dossier's section 6.* The local sign at a root r of f modulo p is w_p = −(g(r)/p) with g(r) = 2r(1 − r²), and since (1 − r²)(1 + r²) ≡ −3 modulo squares, (g(r)/p) = (−3/p)(2r/p)((1 + r²)/p). The factor ((1 + r²)/p) cannot be dropped in general: (g(r)/p) = (−3/p)(2r/p) at only 2956 of the 5938 roots of f modulo p for p < 60000 (0.498), while the complete formula holds at all 5938. It may be dropped exactly when 1 + r² is a square modulo p, which is the case for every root met by the orbit nP ($y^2 = (1 - t)(1 + t^2)$ with 1 − t a square), so along nP one has w_p(E_{t_n}) = −(−3/p)(2t_n/p). As the dossier says, [L* : L] = 8 describes a finite character extension and does not by itself show variation of the global root numbers along the orbit or infinitely many specializations of rank at least 2.

F.7 The fifth relation, derived from the cubic (Proved; checked for all 48 labelings). Let z_i = a_i² be the roots of g(Z) = $Z^3 - 30Z^2 - 24Z - 1$, so e_1 = 30, e_2 = −24, e_3 = 1, and put ε = $a_1a_2a_3$. Then $\varepsilon^2 = e_3 = 1$, so ε = ±1, and Π(1 + z_i) = 1 + e_1 + e_2 + e_3 = 8, so $C_1C_2C_3$ = 8, while $D_1D_2D_3 = 8\varepsilon$. Hence the *direct relation* Π_i $C_iD_i = 64\varepsilon$, that is $[C_1C_2C_3D_1D_2D_3] = [\varepsilon]$, is a square exactly when ε = +1; it comes from the coefficients of the cubic alone and does not use the identity $s^2 = -3(1 - T^4)$. In the quotient basis of the dossier it is the product of this relation with two of the four binary relations: from $A_iB_iC_i = -s(a_i)^2/3$ one has A_iB_i =

$-s(a_i)^2/(3C_i)$, so $C_1D_1A_2B_2D_2A_3B_3D_3 = C_1D_1D_2D_3\cdot s(a_2)^2s(a_3)^2/(9C_2C_3) = \varepsilon\cdot(8s(a_2)s(a_3)/(3C_2C_3))^2$, using $C_1 = 8/(C_2C_3)$. Verified at 60 digits for all 48 labelings (24 with $\varepsilon = +1$ and 24 with $\varepsilon = -1$): the displayed product equals $\varepsilon\cdot\rho^2$, $\rho = 8s(a_2)s(a_3)/(3C_2C_3)$, and $\Pi C_iD_i = 64\varepsilon$. Consequences. (a) The relation is not in the span of the four binary relations (it contains no E_i), and it is invariant under $\mathrm{Gal}(N/\mathbb{Q})$ because $\varepsilon \in \mathbb{Q}$: the 24 relabelings induced by the Galois group are exactly the labelings with $\varepsilon = +1$, and for a labeling with $\varepsilon = -1$ the same product is $-\square$, equivalently one replaces one D_i by $E_i = -D_i$. (b) The same identities give a consistency check: $\Pi(1 - z_i) = 1 - e_1 + e_2 - e_3 = -54 = \Pi(-s(a_i)^2/(3C_i))$ forces $s(a_1)s(a_2)s(a_3) = \pm 108$, and indeed $s(a_1)s(a_2)s(a_3) = 108\varepsilon$, because $\Pi_i\, s(a_i) = \varepsilon\cdot\mathrm{Res}(g, Z^2 - 29Z - 26)/27 = 2916\varepsilon/27$.

F.8 Other families of classes, and a third kind of relation: z + 2 is a square (Proved exactly; the searches are certified lower bounds). *Pair classes.* For the twenty classes A_i, B_i, C_i, $[-1]$, a_i, $a_i + a_j$, $a_i - a_j$ $(i < j)$ and $[\Pi(z_i - z_j)/81] = [\pm V69]$, the symbol vectors at the 463 primes below $1.2\cdot10^5$ that split completely in N (all 24 labelings with $a_1a_2a_3 = 1$) have rank 16, and exactly four relations occur: $a_1a_2a_3 = 1$ (the cubic's constant term), the two relations $A_iB_iC_i = [-3]$ (the s-identity), and the tautology $\Pi(a_i \pm a_j) = \Pi(z_i - z_j)$; so the six classes $[a_i \pm a_j]$ are free (six new dimensions). For the classes $[z_i + z_j]$ and $[1 \pm a_ia_j]$ the only relations are consequences of $a_1a_2a_3 = 1$ and the s-identity (for instance $1 \pm a_1a_2 = (1 \pm a_3^{-1})$ gives $p_{12} \equiv B_3a_3$ and $q_{12} \equiv -A_3a_3$), and the three classes $[z_i + z_j]$ are free. *A relation of a third kind.* A search over all 117649 polynomials $P(a)$ of degree $\leqslant 5$ with coefficients in $[-3, 3]$ for which the Kummer class of $P(a)$ in $\mathbb{Q}(a)^\times/\mathbb{Q}(a)^{\times 2}$ is a rational class (tested at 542 degree-one primes below 4000) finds, apart from trivial squares, only the multiples of $1 - a^4$ (class $[-3]$) and of $a^2 + 2$ (class $[1]$). The second is a new identity: with $z = a^2$ a root of $g(Z) = Z^3 - 30Z^2 - 24Z - 1$,

$$z + 2 = \nu^2,\ \nu = (-z^2 + 32z + 14)/9 = 6 - 9/(z + 2),$$

exactly in $\mathbb{Q}(z)$ (the difference vanishes modulo g, and $N(z + 2) = -g(-2) = 81$). Equivalently $g(Y - 2) = Y^3 - 36Y^2 + 108Y - 81 = Y^3 - (6Y - 9)^2$, so with $Y = v^2$ the number $v = 6 - 9/Y$ is a root of $q(v) = v^3 - 6v^2 + 9$, $g(v^2 - 2) = -q(v)q(-v)$, and the cubic field of the sextic is $\mathbb{Q}(v)$ with $\mathrm{disc}\, q = 3^5\cdot23$ (totally real, $N(v - k) = -q(k)$). Hence $z_i = v_i^2 - 2$ and $f(t) = \Pi_i\,(t^2 + 2 - v_i^2)$; the substitution $t = (2 - u^2)/(2u)$ splits the degree-12 polynomial $(2u)^6 f((2 - u^2)/(2u))$ into the two sextics $u^6 \mp 12u^5 + 6u^4 \pm 24u^3 + 12u^2 \mp 48u + 8$ over $\mathbb{Q}$, and $N = \mathbb{Q}(v_1, v_2, v_3, \sqrt{(v_i^2 - 2)})$. This relation is of neither earlier kind: it is not a consequence of the s-identity or of the cubic's constants, and its class is trivial (so the counts of F.2, F.5 and F.6 are unchanged). The shape $Y^3 = (6Y - 9)^2$ is that of a cuspidal cubic, which is where the singular fibres E_t of the family with $\Delta = 0$ are expected to put it; the classes of the elements $v_i - k$ of the totally real cubic field (norms $-q(k)$) are the natural next candidates for further relations.

F.9 The classes of v − k in the cubic field (Proved: five explicit squares; rank 10 certified). Let v be a root of $q(v) = v^3 - 6v^2 + 9$ and $K_3 = \mathbb{Q}(v)$ (totally real, real roots ≈ −1.124, 1.399, 5.725). For the fifteen classes [−1], [2], [3], [7], [23] and [v − k], k = −3, …, 6, in $K_3^\times/K_3^{\times 2}$, the symbols at the 25906 degree-one primes of K_3 below $3\cdot10^5$ have rank 10, so there are at most five relations, and there are exactly five, each an explicit square in $\mathbb{Q}(v)$ (the difference with the square vanishes modulo q):

$-(v-6) = (v^2/3 - 2v)^2$, $23(v+2) = (-v^2+2v+8)^2$, $(v^2-9)(v-5) = (v^2-5v)^2$, $(v^2-1)(v-3)(v-5) = (3 - v^2/3)^2$, $v(v-1) = (v^2/3 - v)^2$.

The first follows at once from $v^2(v-6) = -9$, and the norms explain the others: N(v + 2) = 23, N(v − 5) = 16, N(v) = −9, N(v − 1) = −4, N(v + 1) = −2, N(v − 3) = 18 (N(v − k) = −q(k)). In particular [v_i + 2] = [23] is the same class for the three conjugates, and [C_i] = [v_i² − 1] = [(v_i − 3)(v_i − 5)], so the classes C_i of the sieve are expressed through the v-classes. Among the old classes together with a_i, v_i and v_i − k (46 classes, all labelings) the symbol vectors have rank 17 at the 1386 primes below $4\cdot10^5$ that split completely in N; most of the many short relations found there are consequences of the five relations above and of their Galois conjugates, and the relations that hold in N but not in K_3 (for instance [v_i − 5] = 1, where v − 5 is negative at two real embeddings) come from the non-real elements of N and have not been derived here.

F.10 The five relations from ideal factorizations, and a reading of the Kummer ranks 7 and 10 (Proved, except where marked). The field $K_3 = \mathbb{Q}(v)$, $q(v) = v^3 - 6v^2 + 9$, has discriminant $621 = 3^3\cdot23$ ($\mathbb{Z}[v]$ has index 3). Modulo 2, $q = (v+1)(v^2+v+1)$; modulo 3, $q \equiv v^3$ (and 3 is totally ramified); modulo 7, $q = (v-2)(v^2+3v-1)$; modulo 23, $q = (v+2)(v-4)^2$. So $2 = \mathfrak{p}_2\mathfrak{p}_2'$ (norms 2 and 4), $3 = \mathfrak{p}_3^3$, $7 = \mathfrak{p}_7\mathfrak{p}_{49}$, $23 = \mathfrak{P}\mathfrak{P}^2$ with $\mathfrak{P} = (23, v+2)$ and $\mathfrak{P}' = (23, v-4)$ ramified. From N(v − k) = −q(k) the principal ideals are $(v+1) = \mathfrak{p}_2$, $(v-1) = \mathfrak{p}_2^2$, $(v-5) = \mathfrak{p}_2^4$, $(v-3) = \mathfrak{p}_2\mathfrak{p}_3^2$, $(v+3) = \mathfrak{p}_2^3\mathfrak{p}_3^2$, $(v) = (v-6) = \mathfrak{p}_3^2$, $(v+2) = \mathfrak{P}$, $(v-4) = \mathfrak{P}'$, $(v-2) = \mathfrak{p}_7$. The element $v^2/3$ is integral (characteristic polynomial $x^3 - 12x^2 + 12x - 3$, norm 3) and generates $\mathfrak{p}_3$, and $2/(v+1)$ is integral of norm −4 and generates $\mathfrak{p}_2'$. By Minkowski's bound $(6/27)\sqrt{621} \approx 5.5$ only the primes of norm 2, 3, 4 matter (5 is inert), all principal, so the class number is 1. *Derivation of the relations.* The map from the 15 classes [−1], [2], [3], [7], [23], [v − k] to the parities of ideal exponents at $\mathfrak{p}_2, \mathfrak{p}_2', \mathfrak{p}_3, \mathfrak{P}, \mathfrak{P}', \mathfrak{p}_7, \mathfrak{p}_{49}$ has rank 7 (the images of [v + 1], [2], [3], [23], [v − 4], [v − 2], [7] are independent). Its kernel has dimension 8 and is spanned by the classes with square ideal: [−1], [v], [v − 1], [v − 5], [v − 6], [(v − 3)(v + 1)], [(v + 3)(v + 1)], [23(v + 2)]. Since the class number is 1 these are classes of units, and the unit group of a totally real cubic field modulo squares has dimension 3 (units of rank 2 and −1), so at least 8 − 3 = 5 relations hold; five are found, hence the rank is exactly 7 + 3 = 10, and the five explicit squares of F.9 are the unit relations. Units occurring: $(v-1)/(v+1)^2$, $9/v^3 = v/(v^2/3)^2$, $(v-5)/(v+1)^4$ and $z = a^2 = v^2 - 2$ (norm 1: z is itself a unit, and $1 - z = 3 - v^2$ has norm −54). *Relations involving z.* Adding [z] and [1 − z] to the fifteen classes does not raise the rank (still 10, 25906 primes), and gives the further relations [v − 5][z] = 1, [(v + 3)

$(v - 3)][z] = 1$, $[(v^2 - 1)(v - 3)][z] = 1$ and $[3][v - 3][v - 6][3 - v^2] = 1$. In N one has $[z] = [a^2] = 1$, so $[v_i - 5] = 1$, $[(v_i + 3)(v_i - 3)] = 1$ and $[C_i] = [v_i - 3]$ hold there, which explains the relation $[v - 5] = 1$ that held in N but not in K_3 (F.9); with the s-identity $A_iB_iC_i = [-3]$ one also gets $[A_iB_i] = [-3][v_i - 3]$. *Reading the ranks 7 and 10.* The seven dimensions of the old classes decompose as 3 (the classes $[C_i] = [v_i - 3]$, one for each conjugate of v) + 1 (the class $[-3] = A_iB_iC_i$) + 3 (the quotients $[A_i/B_i] = [(1 - a_i)/(1 + a_i)]$, which are not elements of $\mathbb{Q}(v)$), and the extension of F.5 adds 1 ($[-1]$) + 2 ($[a_1]$, $[a_2]$, with $a_1a_2a_3 = 1$) for the total of 10. What this reading does not yet give is a proof of the independence of these classes by valuations (the lower bounds remain the symbol-vector certificates), in particular for the three quotients $[A_i/B_i]$.

F.B Relations to the literature and the search for a rational multisection (F.11–F.15)

F.11 Relation to Verzobio's primitive-divisor theorems (Observed; framework only). M. Verzobio, *Primitive divisors of sequences associated to elliptic curves*, J. Number Theory 209 (2020) 378–390, studies the denominators D_n of $x(nP + Q)$ for P of infinite order. Theorem 1.3: if Q is a torsion point of prime order, D_n has a primitive divisor for all $n \geqslant C$ (ineffective, through Siegel's theorem); Theorem 1.5: for E over $\mathbb{Q}$, P of infinite order and $Q \in E(\mathbb{Q})$, the number of primes $p \leqslant x$ with $Q \in \langle P \rangle$ modulo p is $\gg \sqrt{(\log x)}$; the link to the elliptic Artin (Lang–Trotter) conjecture is that $Q \in \langle P \rangle$ modulo p exactly when p divides some D_n. *Application to J_2.* The translation by $T = (0, 0)$ is $x \mapsto 2/x$, so $x(nP + T) = 2/x(nP) = 2D_n^2/m_n^2$ (checked for $1 \leqslant n \leqslant 90$), and the denominator sequence of $\{nP + T\}$ is, up to the factor 2, m_n^2, the squared numerator root of $x(nP) = m_n^2/D_n^2$. Theorem 1.3 with $Q = T$ (prime order 2) therefore gives a primitive prime divisor of m_n for all large n; computationally every m_n with $3 \leqslant n \leqslant 90$ has a prime dividing no earlier m_k (the only exception is $n = 2$, where $m_2 = 1$). The same argument as in the paper gives the rank-of-apparition structure $p \mid m_n \Longrightarrow r_p \mid 2n$ and $r_p \nmid n$ (r_p the order of P modulo p; no violation among the 40 prime-index pairs with $p \leqslant 10^5$ and $n \leqslant 25$), so $\{m_n\}$ behaves like an elliptic divisibility sequence. *Limits.* The points R_i over the roots of f that govern the primes of F_n are non-torsion points over a field of degree 12, so neither Theorem 1.3 (torsion Q) nor Theorem 1.5 ($Q \in E(\mathbb{Q})$) applies to F_n; the step $rnP \equiv O$ in the proof uses that Q is torsion; the condition $R_i \in \langle P \rangle$ modulo p of F.4 is exactly the set $N_{P,Q}(x)$, for which only a $\sqrt{(\log x)}$ lower bound is known unconditionally; and primitive divisors give new primes, not the parity of the number of prime factors or of their exponents, so they do not bear on Problem H.

F.12 Relation to Helfgott's thesis (Conditional framework; read from the abstract and Chapter 1 of the thesis only; the equivalence stated in bold below is withdrawn in F.24). H. A. Helfgott, *Root numbers and the parity problem*, Princeton Ph.D. thesis (2003), arXiv:math/0305435, studies the average of $W(\text{DON'T PANIC}(t))$ over $t \in \mathbb{Z}$ or $t \in \mathbb{Q}$ (lattice points (x, y) in sectors) for a one-parameter family DON'T PANIC over $\mathbb{Q}(t)$. Let $M_{\text{DON'T PANIC}}(x, y)$ be the product of the

homogenized polynomials of the places of multiplicative reduction and B′_DP that of the quite bad places. Theorem 1.7.3 shows that if M_DP is not constant then W averages to zero over ℚ, conditionally on the square-free sieve for B′_DP (conjecture DP$_2$) and on Chowla's conjecture DP$_2$ for M_DP (the average of λ(M_DP(x, y)) over $\mathbb{Z}^2$ is zero); Propositions 1.7.9 and 1.7.10 show that, given DP$_2$, the two statements are equivalent. Unconditional cases need deg_irr(B′) ⩽ 6 and deg M_DP ⩽ 3 (Theorem 1.3′), and DP$_2$ is proved for homogeneous forms of degree at most 3 (the square-free sieve DP$_2$ is known for deg_irr ⩽ 6 by Greaves). *Our family.* The six I_1 fibres form one irreducible place, so M_DP = F(x, y) is the irreducible binary sextic and j is non-constant; B′ has irreducible factors of degree at most 6, so DP$_2$ holds by Greaves, and Theorem 1.3′ does not apply because deg M_DP = 6 > 3. Hence, if I read the hypotheses correctly, **the average of W(E_t) over t = u/v in a box is zero if and only if Chowla's conjecture DP$_2$ holds for F(u, v)**, and the family is non-degenerate (W takes both signs, as the formula and the data show). This is the same parity barrier as in Problem H, now in its thick form. *Agreement with our decomposition.* Helfgott's §2.1 writes W as a pliable function times λ(M_DP(x, y)) times a correction for non-squarefree values handled by the square-free sieve, and his Lemma 2.3.27 uses the Hilbert-symbol reciprocity Π(a/DP)^{v_DP(b)} = (pliable function)·Π(b/DP)^{v_DP(a)}; our reciprocity reduction and the periodic factor ε_n of the Theorem above are the analogue along the orbit, and the factor Π w_p over primes of even exponent is his square-free correction. *Limits.* The thesis averages over thick sets of parameters; the orbit t_n = 1 − x(nP) is a set of exponentially growing height to which none of its theorems applies, and nothing there gives infinitely many n with W(E_{t_n}) = +1. *A conditional route it suggests.* If the K3 surface S contained a curve R ≅ $\mathbb{P}^1$ over ℚ that is a multisection of the t-fibration, the base change by R → $\mathbb{P}^1$_t would give a family over ℚ(s) with a tautological section, hence rank ⩾ 1 for all but finitely many s (specialization), and Helfgott's theorem would give infinitely many s with W = +1 conditionally on DP$_2$ for the pulled-back discriminant (a form of degree 6d) and DP$_2$ if its degree exceeds 6, hence rank ⩾ 2 if Sha[2^∞] is finite. No such rational curve has been found: the searches of §8 found only genus-1 and higher covers, and every irreducible factor of the pulled-back discriminant has degree divisible by 6, so an unconditional version is excluded.

F.13 The search for a rational multisection: the conic base change and lattice constraints (negative results; one finite target). *The conic.* The map t = (s² − 1)/(2s) is the double cover of the t-line by the conic w² = 1 + t² (the Pythagorean parametrization), and rank E(ℚ(s)) = rank E(ℚ(t)) + rank E^{(1+t²)}(ℚ(t)). The twist is y² = x³ + 4(1 − t⁴)x − 16t(1 + t²)², with discriminant proportional to (t² + 1)³f(t) of degree 12: fibres 2·III + 6·I_1, a rational elliptic surface whose Mordell–Weil group has rank 8 − 2 = 6 over $\bar{\mathbb{Q}}$(t). Counting points on the surface modulo p (the counts satisfy N ≡ 1 + p² mod p at every prime tested) gives the Frobenius trace on the rank-6 lattice, whose average over the 376 primes 5 ⩽ p < 2600 is −0.016, so the rank over ℚ(t) is 0. The same traces give the multiplicity of every quadratic character χ_c in the Galois representation on the lattice, that is the rank of E^{c(1+t²)}

over $\mathbb{Q}(t)$ (the rank of E over the cover $w^2 = c(1 + t^2)$): for all squarefree c with $|c| \leqslant 120$ the multiplicity is within noise of 0 (the largest is 0.12). So no double cover of the t-line branched at $t = \pm i$ gives positive rank. *Lattice constraints.* Since the generic Mordell–Weil group is trivial, $NS(S) = U \oplus E_7 \oplus E_7$, and Galois ($i \mapsto -i$) swaps the two E_7 summands, so a $\mathbb{Q}$-rational curve R has class $D = dO + bF - (\lambda, \lambda)$ with $d = D \cdot F$ the degree over the t-line, λ a dominant weight of E_7 in the root lattice ($n_2 + n_5 + n_7$ even) taken in both summands, $D^2 = 2\delta - 2$ with δ the number of nodes, $d \cdot b = d^2 + \delta - 1 + |\lambda|^2$. The necessary conditions for an irreducible curve are $D \cdot O = b - 2d \geqslant 0$, $D \cdot \Theta_j = n_j \geqslant 0$ and $D \cdot \Theta_0 = d - \Sigma m_j n_j \geqslant 0$ at each III* fibre (multiplicities $m_j = 2, 2, 3, 4, 3, 2, 1$). Enumerating: there is **no smooth rational $\mathbb{Q}$-curve ($\delta = 0$) of degree $d \leqslant 4$ over the t-line**; the smallest is $d = 5$, with one class ($\lambda = \omega_2 + 3\omega_7$, $b = 10$, $D \cdot O = 0$), then $d = 9$ and $d = 11$. A bisection ($d = 2$) must be nodal with $D^2 = 0$, so it is a member of a genus-1 pencil: there are two classes, the x-fibration ($\lambda = \omega_6$, $b = 4$, $D \cdot O = 0$) and $\lambda = 2\omega_7$ ($b = 5$, $D \cdot O = 1$); the singular members of the x-fibration $\{x = c\}$ are at the roots of $(c^2)^3 - 16c^2 - 64 = 0$, which has no rational root, so no rational bisection of that pencil exists over $\mathbb{Q}$ (as found in §8). The smallest class of rational trisections is $d = 3$, $\delta = 2$, $\lambda = \omega_2 + \omega_7$, $b = 6$, $D^2 = 2$, $D \cdot O = D \cdot \Theta_0 = 0$: the net $|D|$ gives a double plane $S \to \mathbb{P}^2$, its rational members are the preimages of the bitangents of the branch sextic, and a $\mathbb{Q}$-rational bitangent would give a rational base change of degree 3 with a section, disjoint from O. Constructing this net explicitly and testing for rational bitangents is a finite computation that has not been done. *Consequence.* The conditional route of F.12 requires such a curve; none exists in degree $\leqslant 2$ over $\mathbb{Q}$, and in degree 3 only this nodal class remains.

F.14 The rational trisection class: the net, the double plane and its bitangents (Computed; negative result). *The net $|D|$.* Let $D = 3O + 6F - \Sigma_{t = \pm i}\Sigma_k c_k\Theta_k$ with $c = C^{-1}(e_2 + e_7) = (3, 5, 6, 9, 7, 5, 3)$ (Bourbaki labels), the class with $\lambda = \omega_2 + \omega_7$ of F.13. In the integral model $Y^2 = X^3 + 4ac^3X - 8bc^5$ ($a = 1 - t^2$, $b = 2t$, $c = 1 + t^2$) the singular point of the III* fibre at $t = \pm i$ is the E_7 point $y^2 = x^3 + (\text{unit})\tau^3 x + O(\tau^5)$ with $\tau = t \mp i$. Putting it in normal form by $\xi = X - \rho(\tau)$, where ρ is the root of $X^3 + a_4X + a_6$ of valuation 2 (so that $Y^2 = \xi(\xi^2 + 3\rho\xi + A + 3\rho^2)$ with no constant term), the seven exceptional curves are the crepant monomial valuations $(w_\tau, w_\xi, w_Y) = (2,2,3), (2,3,5), (3,4,6), (4,6,9), (3,5,7), (2,4,5), (1,3,3)$ for $\Theta_1, \ldots, \Theta_7$ (found as the seven solutions of $w_\tau + w_\xi + w_Y = \nu(f) + 1$ with at least two monomials of minimal weight, and matched to the Dynkin diagram through midpoints: $(3,4,6) = \tfrac{1}{2}((2,2,3) + (4,6,9))$, $(3,5,7) = \tfrac{1}{2}((4,6,9) + (2,4,5))$, $(2,4,5) = \tfrac{1}{2}((3,5,7) + (1,3,3))$). For $f = p_0 + p_2X + p_3Y$ with $\deg p_0 \leqslant 6$, $\deg p_2 \leqslant 2$, p_3 constant (dimension 11), the conditions $\nu_k(f) \geqslant c_k$ are $\text{ord}(p_0 + p_2\rho) \geqslant 3$, $\text{ord}(p_2) \geqslant 1$ and none on p_3, that is 8 real linear conditions, leaving exactly dimension 3 as Riemann–Roch requires ($D^2 = 2$, $h^0 = 3$). The solutions are $f = \lambda c^3 + \kappa cX + \mu Y = c^3(\lambda + \kappa x + \mu y)$, so $|D|$ is the pullback of the lines $\lambda + \kappa x + \mu y = 0$ of the (x, y)-plane. *The double plane.* $S \to \mathbb{P}^2_{(x,y)}$, $(t, x, y) \mapsto (x, y)$, has degree 2: eliminating $B = (x^3 + 4Ax - y^2)/8$ from $A^2 + B^2 = 1$ gives a quadratic in A with discriminant $-F/16$ for the **branch sextic $F = (y^2 - x^3)^2 - 16x^2 - 64$**. F is irreducible over $\mathbb{Q}$, its affine part

is smooth (the Gröbner basis of (F, F_x, F_y) is {1}), its only singular point is (0 : 1 : 0) at infinity, and with x = r − 1/r its normalization is the hyperelliptic curve $Y^2 = r(r^6 + r^4 + 7r^2 - 1)$ of genus 3. A line gives a rational member of |D| exactly when it is a bitangent (two nodes on the preimage). *Bitangents.* For y = mx + q the polynomial F(x, mx + q) is monic of degree 6; imposing $F(x, mx + q) = (x^2 + ax + b)^2(x^2 + cx + d)$ gives six equations in (a, b, c, d, m, q); eliminating c and d leaves four equations whose grevlex Gröbner basis has 35 elements and is zero-dimensional; converting to lex order (FGLM) the eliminant in q has degree 60 and is **irreducible over** $\mathbb{Q}$. Hence there is no bitangent with rational slope and intercept; vertical lines x = c cannot be bitangent, since $F(c, y) = (y^2 - c^3 - \kappa)(y^2 - c^3 + \kappa)$ with $\kappa = 4\sqrt{(c^2 + 4)} \neq 0$ would need both factors to have a double root at y = 0. So **no $\mathbb{Q}$-rational rational trisection of this class exists**, and with F.13 there is no $\mathbb{Q}$-rational rational multisection of degree $\leqslant 3$ (smooth ones are excluded up to degree 4 by the enumeration, and nodal ones of degree 2 and 3 by the pencil and this bitangent computation). *The degree-5 class.* The unique candidate smooth class of degree 5 ($\lambda = \omega_2 + 3\omega_7$ at both fibres, b = 10) is automatically Galois-invariant, so if irreducible it would be a $\mathbb{Q}$-conic; but over $\bar{\mathbb{Q}}$ there are numerically possible (−2)-classes of degree 2, 2O + 4F − ($2\omega_7$ at one fibre, ω_6 at the other) and its conjugate, with intersection number −1 with it, and rational bisections over $\bar{\mathbb{Q}}$ exist (the twist E^{(1+t²)} has Mordell–Weil rank 6 over $\bar{\mathbb{Q}}(t)$); if these classes are realised by curves the degree-5 class is reducible. This is settled in F.15 (the degree-5 class is not represented by an irreducible curve).

F.15 The degree-5 class and all smooth invariant classes up to degree 21 are not curves (Proved; the argument for the degree-5 class is due to the reviewer, the extension is computed). *The class R.* Let R = 2O + 4F − $2\omega_7$ (at the fibre t = i) − ω_6 (at the fibre t = −i), and $\bar{R}$ its Galois conjugate. Both $2\omega_7$ and ω_6 lie in the root lattice of E_7 ($n_2 + n_5 + n_7$ is even), so R is an integral class generated by O, F and fibre components; $|2\omega_7|^2 = 6$, $|\omega_6|^2 = 4$, $(2O + 4F)^2 = 8$, hence $R^2 = 8 - 6 - 4 = -2$ and R·F = 2. By Riemann–Roch on a K3 surface ($\chi = 2 + R^2/2 = 1$) R or −R is effective, and −R is not because F is nef and (−R)·F = −2; so R is effective (it need not be nef or irreducible). *The degree-5 class.* $D_5 = 5O + 10F - (\omega_2 + 3\omega_7)$ at each fibre has $D_5^2 = -2$ and $D_5 \cdot R = -20 + 20 + 20 - [(\omega_2 + 3\omega_7, 2\omega_7) + (\omega_2 + 3\omega_7, \omega_6)] = 20 - (12 + 9) = -1$, using $(\omega_2, \omega_7) = 3/2$, $(\omega_7, \omega_7) = 3/2$, $(\omega_2, \omega_6) = 3$, $(\omega_7, \omega_6) = 2$. If an irreducible curve C had class D_5, then C·R < 0 would force C to be a component of every effective divisor of the class R; but C·F = 5 > 2 = R·F and every component has non-negative degree over the t-line, a contradiction. So **no irreducible curve has the class D_5.** *Extension.* The same argument shows that an irreducible curve C of degree d > 2 satisfies $C \cdot R' \geqslant 0$ for every effective class R′ of degree < d, in particular for every integral class with $R'^2 = -2$ and $0 < R' \cdot F < d$ (effective by Riemann–Roch, whatever its other intersections). For all smooth $\mathbb{Q}$-invariant candidate classes (δ = 0) of odd degree $d \leqslant 21$ (one class for d = 5, 9, 11; two for d = 13; one for d = 15; four for d = 17; three for d = 19; four for d = 21) a search over the integral (−2)-classes of lower degree finds in every case one of degree 2 with D·R′ between −1 and −5 (the values −1, −2, −2, −2 and −3, −3, between −4 and −2, between −4

and −3, between −5 and −3), so none is represented by an irreducible curve: **there is no smooth rational $\mathbb{Q}$-curve of degree ⩽ 21 over the t-line**. For a nodal class ($D^2 = 2\delta - 2 \geqslant 0$) the same inequality gives a lower bound on the number of nodes: with $R = 2O + 4F - 2\omega_7 - \omega_6$, $D\cdot R = 2b - 2(\lambda, \omega_7) - (\lambda, \omega_6) \geqslant 0$ where $b = (d^2 + \delta - 1 + |\lambda|^2)/d$, so a rational multisection of degree d needs δ large enough. The case d = 3 is the class of F.14 (δ = 2), for which no $\mathbb{Q}$-curve exists.

F.C The parity problem: what is proved and what is not (F.16–F.29)

F.16 Where the Liouville values come from, and the weaker variants (Computed; one provable variant). *p-adic structure.* For a prime p dividing F_n and n in a class a modulo r_p, $nP - R$ lies in the formal group over $\mathbb{Z}_p$, and the logarithm gives $v_p(F_n) = 1 + v_p((n - N_p)/r_p)$ for a p-adic integer N_p attached to the class: the set of n with $p^e \mid F_n$ is one class modulo $r_p p^{e-1}$ for each class modulo r_p. This is the mechanism behind F.15′s one lift per class and was checked at the levels e = 1, 2, 3 for p = 23 (2 classes modulo 10, 46 residues modulo $10\cdot 23^e$ at every level) and e = 1, 2 for p = 83 (4 classes modulo 39, 332 residues modulo $39\cdot 83^e$). So each prime contributes a completely explicit p-adic 'centre', and $\lambda(F_n) = \Pi_p(-1)^{v_p(F_n)}$ is a product of such local factors. *The rough part.* The primes below $3\cdot 10^4$ that divide some F_n (358 of them) explain only a negligible part of F_n: for n = 10, 20, 40, 60, 80, 100, 120 the digits explained are 6.1, 0.0, 9.3, 0.0, 3.3, 10.9, 4.2 out of 113, 450, 1802, 4055, 7210, 11266, 16223 digits. The part of F_n supported on primes p with $r_p > n$ occurs for at most one class and carries more than 99.9% of $\log F_n$ for n ⩾ 40; its parity is untouched by any sieve (this is the parity problem), and heuristically $\omega(F_n)$ is Poisson with mean about $\log\log F_n \approx 2\log n$, so $\lambda(F_n)$ equidistributes while the number of n with F_n having at most k prime factors is finite. *Weaker variants.* (i) *Quadratic twists (provable).* For $q \nmid N_E$ prime, $W(E^{(q)}) = W(E)\chi_q(-N_E)$; since E_{t_n} has a non-torsion point, $s_2(E_{t_n}/\mathbb{Q}) \geqslant 1$, and if $W(E^{(q)}) = -1$ then $s_2(E^{(q)}/\mathbb{Q})$ is odd, hence ⩾ 1, and the 2^∞-Selmer corank over $\mathbb{Q}(\sqrt{q})$ is $s_2(E/\mathbb{Q}) + s_2(E^{(q)}/\mathbb{Q}) \geqslant 2$. Both values of $\chi_q(-N_E)$ occur for infinitely many q, so for every n there are infinitely many quadratic fields over which E_{t_n} has Selmer corank ⩾ 2 (rank ⩾ 2 if Sha[2^∞] is finite); this does not give rank ⩾ 2 over $\mathbb{Q}$. (ii) The pair version ($W(E_{t_n}) = +1$ or $W(E_{t_n'}) = +1$ infinitely often) is slightly weaker, since its negation needs the Liouville values of both sequences locked to periodic patterns, but it is not accessible either. (iii) Statements about averages, logarithmic densities or subsequences in progressions are equivalent in difficulty because they all contain the parity of the rough part.

F.17 Test of the elliptic-net idea: no flow of information between different n (Computed; negative). The values F_n are norms of values at nP of the Miller-type function $f_R(Q) = x(Q) - x(R) = -\sigma(Q + R)\sigma(Q - R)/(\sigma(Q)^2\sigma(R)^2)$, that is slices of a rank-2 elliptic net W(n, m) for the points P and R. The net relation $W(p+q+s)W(p-q)W(r+s)W(r) + W(q+r+s)W(q-r)W(p+s)W(p) + W(r+p+s)W(r-p)W(q+s)W(q) = 0$ with p, q, r on the P-

axis and s = (0, 1) expresses a relation among the terms W(·, 1) and the EDS terms ψ_n of P; it is a sum of three products, not a single product, so it gives no multiplicative relation between the values λ(F_n) of the Liouville function at different n (λ is not additive). The only way a relation between λ(F_n) and λ(F_m) could arise is therefore through shared primes. Measured for 2 ⩽ m ⩽ 64, the part of F_n supported on primes that also divide some other F_m has 0.0, 3.9, 3.0, 1.8, 0.0, 2.7, 4.8, 0.0 digits for n = 4, 8, 12, 16, 24, 32, 48, 64, out of 17.8, 71.9, 162.9, 287.2, 647.8, 1154.1, 2595.2, 4613.9 digits, that is at most 5.4% for n = 8 and at most 0.3% for n ⩾ 32; these shared primes are the small primes of the explicit classes of F.15–F.16. So the prime sets of different F_n are essentially disjoint, λ(F_n) and λ(F_m) are products over essentially disjoint sets of primes, and no information on the parity of the rough part passes from one index to another. The complete factorizations available (n = 2, …, 9) give λ(F_n) = +1, +1, +1, −1, +1, +1, +1, −1 with 2, 2, 2, 3, 4, 4, 4, 5 prime factors, with no pattern in n modulo 6 (+1 at n = 3 and −1 at n = 9), but eight values prove nothing.

F.18 The squarefree part of F_n tends to infinity (Proved, by Faltings' theorem; ineffective). Write x(nP) = A_n/B_n² in lowest terms and C_n = B_n¹²h(A_n/B_n²) with h(x) = f(1 − x) = x⁶ − 6x⁵ − 15x⁴ + 100x³ − 189x² + 162x − 54; since t_n = (B_n² − A_n)/B_n², C_n = F(U_n, V_n) = F_n. The function h(x(Q)) on J_2 has the twelve distinct simple zeros of F.16 and a pole of order 12 (even) at O, so for any fixed non-zero d ∈ ℚ the double cover X_d : w² = h(x)/d of J_2 is geometrically irreducible, branched at exactly twelve points, and by Riemann–Hurwitz 2g − 2 = 2(2·1 − 2) + 12 gives g = 7. *Proposition.* For every fixed squarefree integer d (of either sign) there are only finitely many n with C_n = d·u², u ∈ ℤ. *Proof.* If C_n = du² then h(x(nP))/d = (u/B_n⁶)², so nP lifts to a point of X_d(ℚ); the points nP are pairwise distinct because P has infinite order, so infinitely many such n would give infinitely many rational points on a curve of genus 7, contradicting Faltings' theorem. □ *Consequences.* The set of squarefree integers supported on a finite set of primes S is finite (at most 2^{|S|+1} signed classes), so for every finite S all sufficiently large n have a prime p ∉ S dividing F_n to an odd power; equivalently sqf(|F_n|) → ∞, and in particular F_n is a perfect square for only finitely many n. The constants are not effective. *What it does not give.* The squarefree part may tend to infinity through a single large prime, so the proposition does not give ω(sqf(|F_n|)) → ∞, let alone both parities: λ(|F_n|) = (−1)^{ω(sqf(|F_n|))} may still be constant for all large n. The same argument applies to the translate sequence nP + T and to any sequence nP + Q with Q ∈ J_2(ℚ). (The relevant statement on ordinary signs of elliptic divisibility sequences is that of Silverman and Stephens, not about Liouville signs.)

F.19 No multiplication or translation identity for φ = h(x(Q)) (Proved). Let D be the reduced divisor of the twelve zeros of φ on J_2, so div φ = D − 12(O). *Lemma.* Let α : J_2 → J_2 be an étale morphism of degree δ (an isogeny followed by a translation) and suppose φ∘α = cφψ² with c ∈ $\bar{ℚ}$^× and ψ ∈ $\bar{ℚ}$(J_2)^×. Then α^{−1}(D) = D as sets, so δ = 1. *Proof.* div(φ∘α) = α^*D − 12α^(O) and the pole divisor has even coefficients, so the identity forces α^*D −

D ≡ 0 (mod 2) as divisors; α^*D is reduced (α étale) with 12δ points and D is reduced, so every point of one set lying outside the other has odd coefficient, hence the two sets coincide, and 12δ = 12. □ *Proposition.* For every integer m with |m| ⩾ 2 there is no identity φ([m]Q) = cφ(Q)ψ(Q)² over $\bar{\mathbb{Q}}$, a fortiori over $\mathbb{Q}$: the ratio φ([m]Q)/φ(Q) is not in $\bar{\mathbb{Q}}^\times\bar{\mathbb{Q}}(J_2)^{\times 2}$ (here δ = m² ⩾ 4, and [m]^*D has 12m² points). For m = 2 this agrees with gcd(h, H_2) = 1 for the degree-48 divisor [2]^*D (Appendix D). *Translations.* By the Lemma an identity φ(Q + R) = cφ(Q)ψ² requires R + D = D, so R would permute the twelve points; for the three 2-torsion points $T_j = (e_j, 0)$, $e_j \in \{0, 1 + i, 1 - i\}$, the numerator of h(x(Q + T_j)) has gcd of degree 0 with h over $\mathbb{Q}(i)$, so the supports are disjoint and no identity holds (for T = (0, 0) this is gcd(F, G) = 1 with $G(x) = x^6F(2/x)$, Appendix E.1); torsion translations of other orders were not checked. *Scope.* This excludes identities between values of φ at Q and at α(Q) that come from rational functions on J_2. It says nothing about identities that hold only at the points nP, and it does not contradict the point-specific identity c(M(α)) = −s(α)² of Appendix D, which concerns the values of a different function (the character) at the twelve zeros of φ, not φ itself.

F.20 The valuations at 2 and 3 of the values of the sextic along nP and nP + T (Proved; checked exactly for 1 ⩽ n ⩽ 200). Write x(nP) = a_n/d_n in lowest terms ($a_n = m_n^2$, $d_n = D_n^2$ are squares, F.18) and $F_n = d_n^6h(a_n/d_n)$, $G_n = d_n^6h_T(a_n/d_n)$ with $h_T(X) = X^6h(2/X) = -54X^6 + 324X^5 - 756X^4 + 800X^3 - 240X^2 - 192X + 64$. Since x(nP + T) = 2/x(nP) = $2d_n/a_n$ with a_n a square, $G_n = a_n^6h(2d_n/a_n)$ is the value C'_n of F.18 for the sequence nP + T. *Theorem.* For every n ⩾ 1: $v_2(F_n) = v_3(F_n) = 0$, $v_2(G_n) = 1$, and $v_3(G_n) = 4$ if n ≡ 4, 5 (mod 9) and 3 otherwise. *Proof.* (a) At 2 the curve reduces to y² = x³ and P = (1, 1) to a nonsingular point of order 2, so a_n is odd for every n (d_n is odd for n odd and even for n even); F ≡ a⁶ + a⁴d² + a²d⁴ ≡ 1 (mod 2), and G ≡ −54a⁶ ≡ 2 (mod 4) because all other coefficients of h_T are divisible by 4. (b) The discriminant −256 gives good reduction at 3 and P has order 3 modulo 3 (3P = (49/9, −287/27)). If 3 | n then 9 | d_n and 3 ∤ a_n, so F ≡ a⁶ ≡ 1 (mod 3), and in G the term −54a⁶ has valuation 3 while every other term has valuation at least 6; if 3 ∤ n then x(nP) ≡ 1 (mod 3) and h(1) = −1 ≢ 0, so 3 ∤ F_n. (c) For 3 ∤ n write x(nP) = 1 + 3z; the identity $h_T(1 + 3z) = -54(729z^6 - 81z^4 + 32z^3 - 9z^2 + 1) = -54Q(z)$ holds, Q(z) ≡ 2z + 1 (mod 3) vanishes mod 3 only for z ≡ 1 (mod 3), and for z = 1 + 3w, Q ≡ 6 (mod 9), so $v_3(Q) = 1$ exactly when x(nP) ≡ 4 (mod 9) and 0 otherwise; d_n is a 3-adic unit. (d) The formal parameter of 3P is −x/y = 147/287 of valuation 1, so 9P = 3 lies in the second kernel of reduction, 9P ≡ O (mod 9), and nP mod 9 depends only on n mod 9. From x(P) = 1, x(2P) = 1/4 ≡ 7, x(4P) = 961/400 ≡ 4 and x(−Q) = x(Q) one gets x(nP) mod 9 = ∞, 1, 7, ∞, 4, 4, ∞, 7, 1 for n ≡ 0, …, 8, so x(nP) ≡ 4 (mod 9) exactly for n ≡ 4, 5 (mod 9). □ *Consequences.* With $\lambda_{2,3}(N) = (-1)^{v_2(N)+v_3(N)}$: $\lambda_{2,3}(F_n) = +1$ for all n and $\lambda_{2,3}(G_n) = -1$ if n ≡ 4, 5 (mod 9), +1 otherwise (the same for F_nG_n). Hence λ(|F_n|) is the Liouville value of the part prime to 6 of F_n, and the full Liouville value of the translate's C-value differs from that of its part prime to 6 by the explicit period-9 sign −1 for n ≡ 4, 5 (mod 9); the

lemma gives the 3-adic structure behind the period 9 (period 18 with the 2-adic factor) of the finite factor of the root number of E_{t′_n} but does not by itself determine the local root numbers w_2, w_3, (2/c), and it says nothing about the primes p ⩾ 5.

F.21 The period-3 law along nP + T (Theorem; proof complete, with three finite local evaluations done by exact computation; checked directly for 2 ⩽ n ⩽ 200). Let t′_n = 1 − x(nP + T) = 1 − 2/x(nP) = U′_n/V′_n and ε′_n = ε(t′_n) with the exact definition of Appendix B. *Theorem.* For every n ⩾ 2, ε′_n = +1 if and only if 3 | n. *Proof.* (1) With x(nP) = m²/D², t′_n = (m² − 2D²)/m², so V′ = m² is an odd square (a_n = m² is odd by F.20), gcd(U′, V′) = 1 and the common factor of the Pythagorean triple of t′ is 2; moreover t′ < 1 and t′ ∉ {0, ±1}. (2) Because V′ is a square, the reciprocity formula of Appendix B applies verbatim: ε(t′) = f(t′)·Π_{v ∈ {2,3,∞}} (2t(1−t²), f(t))_v (−1, t)_v (−6, 1−t²)_v, where the primes p ⩾ 5 dividing U′V′(V′² − U′²) cause no trouble since F ≡ −V⁶, U⁶, −54V⁶ modulo them; the 2- and 3-parts of the value G_n (F.20) enter only through the symbols at 2 and 3. (3) Real place: the real factor is +1 for every t < 1, t ≠ 0 (checked on 2313 random points; by cases t < −1, −1 < t < 0, 0 < t < 1). (4) Hence ε′_n = Ψ_2(t′_n)Ψ_3(t′_n) with $\Psi_2 = -w_2\cdot(2/c)\cdot h_2$ and $\Psi_3 = w_3\cdot h_3$; each depends only on t′_n as an element of $\mathbb{Q}_2$ resp. $\mathbb{Q}_3$, since w_2, w_3 are read from A = (1−t²)/(1+t²), B = 2t/(1+t²) modulo 16 resp. 9 (continuous in t, 1 + t² ≠ 0) and (2/c) = +1 here (c ≡ 1 mod 8: V′² ≡ 1 mod 16 and (1 + t′²)/2 ≡ 1 mod 8 for t′ ≡ ±1 mod 8). (5) Classes of t′_n, from the valuation laws of Appendix B and F.20: 2-adically, n odd: $v_2(t' + 1) = 1 + v_2(n^2 - 1) \geqslant 4$; n ≡ 2 (mod 4): t′ ≡ −7 (mod 64) (x′ = 8D′²/m² with D′ odd); n ≡ 0 (mod 4): $v_2(x') = 1 + 2v_2(n) \geqslant 5$, so t′ ≡ 1 (mod 32). 3-adically, 3 | n: $v_3(t' - 1) = 2v_3(n) \geqslant 2$; 3 ∤ n: t′ ≡ 8, 2, 5 (mod 9) for n ≡ ±1, ±2, ±4 (mod 9) (from the table of F.20 via x′ = 2/x). (6) Local constancy: on each of these classes Ψ_2 resp. Ψ_3 is constant. Near the singular points t = ±1 the symbols containing 1 − t² = ε·(unit) cancel (the residue vanishes because −6f(±1) = 324 = 18² is a square) and f(t)/f(±1) is a square on the stated balls (relative perturbation ≡ 0 mod 8 resp. mod 3); on the classes t ≡ −7 (mod 64) and t ≡ 2, 5 (mod 9) all unit square classes of t, 1 − t², f(t) are fixed and w_2, w_3 are determined by t modulo 64 resp. 9. This was confirmed on 300 random rational points u/m² per class (all constant). (7) The constants, by exact evaluation at n = 2, 3, 4, 8: $\Psi_2 = -1$ on all three 2-adic classes; $\Psi_3 = -1$ for 3 | n, and $\Psi_3 = +1$ on the three classes t′ ≡ 8, 2, 5 (mod 9). Therefore ε′_n = (−1)(−1) = +1 for 3 | n and (−1)(+1) = −1 for 3 ∤ n. □ *By-products (proved above, checked for n ⩽ 200).* G_n ≡ 10 (mod 16) for every n, so G_n/2 ≡ 5 (mod 8) and the part N′_n of |G_n| prime to 6 satisfies (2/N′_n) = −1 exactly when n ≡ 4, 5 (mod 9). *Consequence.* The period-6 theorem for nP (Appendix B) and the period-3 theorem for nP + T are now both proved; the sign ε of the root number is completely determined along both orbits, and what remains of H is exactly the parity of the number of prime factors p ⩾ 5 (with multiplicity) of F_n resp. G_n, i.e. the Chowla problem of F.12.

F.22 The valuation laws $v_2(D_n) = v_2(n)$ and $v_3(D_n) = v_3(n)$ from the formal group (Theorem; proved; checked for 1 ⩽ n ⩽ 300; closes open problem 2). Write x(nP) = m_n²/D_n² in lowest terms, D_n > 0 (so V_n = D_n²). *Theorem.* For every n ⩾ 1, v_2(D_n) =

$v_2(n)$ and $v_3(D_n) = v_3(n)$. *Proof.* Let $z = -x/y$ be the formal parameter and $E_r = \{Q : v(z(Q)) \geqslant r\}$. Three facts. (a) If $v_p(x(Q)) = -2r < 0$ then $v_p(z(Q)) = r$, because on $y^2 = x^3 - 2x^2 + 2x$ the term x^3 has strictly smaller valuation than $2x^2$ and $2x$; so "depth r" and "$v(x) = -2r$" are the same statement, and $v_p(D) = r$. (b) For a unit $k \in \mathbb{Z}_p$, $k = kz + z^2 g(z)$ with $g \in \mathbb{Z}_p[[z]]$, hence $v([k]z) = v(z)$ whenever $v(z) \geqslant 1$. (c) Doubling: $x(2Q) = (x^2 - 2)^2/(4x(x^2 - 2x + 2)) = (x/4)\cdot(1 - 2x^{-2})^2/(1 - 2x^{-1} + 2x^{-2})$; if $v_2(x) = -2r$ with $r \geqslant 1$ the last factor is a 2-adic unit, so $v_2(x(2Q)) = v_2(x) - 2$: doubling raises the 2-adic depth by exactly 1 (checked on 59 points). *$p = 2$.* Modulo 2 the curve is $y^2 = x^3$ with singular point $(0, 0)$; the nonsingular points are O and $(1, 1)$, and $\bar{P} = (1, 1)$ has order 2 since $2P = (1/4, -5/8)$ reduces to O. So for n odd, $n\bar{P} = \bar{P}$ and $x(nP) \in \mathbb{Z}_2$, D_n is odd; for n even $nP \in E_1$, and $2P$ has $v_2(x) = -2$, i.e. depth 1 ($z(2P) = 2/5$). For $n = 2^s k$, $s \geqslant 1$, k odd, $nP = k$: by (c) iterated $s - 1$ times the point 2^{s-1} has depth exactly s, and by (b) multiplication by the odd k keeps it. By (a), $v_2(D_n) = s = v_2(n)$. *$p = 3$.* The discriminant -256 gives good reduction at 3; $2P = (1/4, -5/8)$ is 3-integral and $3P = (49/9, -287/27)$ reduces to O, so $\bar{P}$ has order exactly 3. If $3 \nmid n$ then $n\bar{P} \neq O$ and $3 \nmid D_n$. If $3 \mid n$ then $3P \in E_1$ with $z(3P) = 21/41$ of valuation exactly 1. For $p = 3$ the condition $v(z) > v(p)/(p - 1) = 1/2$ holds on E_1, so $v_3(z([3]Q)) = 1 + v_3(z(Q))$ there (Silverman, AEC, IV Prop. 6.4). Writing $n = 3^s k$ with $3 \nmid k$, $nP = k$ has depth exactly $1 + (s - 1) = s$, and (a) gives $v_3(D_n) = s = v_3(n)$. □ *Remark.* The same argument gives, for every prime p of good reduction at which $\bar{P}$ has order r_p (the rank of apparition) and p odd, $v_p(D_n) = 0$ if $r_p \nmid n$ and $v_p(D_n) = v_p(D_{r_p}) + v_p(n/r_p)$ if $r_p \mid n$; this is the standard elliptic-divisibility-sequence law and explains why different F_n share almost no primes (F.18). Together with Appendix B and F.20 it removes the last unproved input behind the period-6 and period-3 theorems.

F.23 A reformulation that does not help (Remark). For odd $N > 0$ one has $N \equiv (-1)^{\Omega_3(N)} \pmod 4$, where Ω_3 counts prime factors $\equiv 3 \pmod 4$ with multiplicity, hence $\lambda(N) = \chi_4(N)\cdot(-1)^{\Omega_1(N)}$, and along our sequences $\chi_4(|F_n|) = (-1)^n \cdot \mathrm{sgn}(F_n)$ ($F_n \equiv 1 \pmod 8$ for n even, $7 \pmod 8$ for n odd). This is an identity for every odd integer; it replaces the parity of Ω by the parity of Ω_1 with a known explicit sign, so the two problems are equivalent, and $\mathrm{sgn}(F_n)$ is not periodic (nP equidistributes on the real component). It does not reduce the Chowla problem of F.12.

F.24 Precise status of the remaining parity bit (Correction and Proposition). *Correction of F.12.* The statement that H is "equivalent" to Chowla's conjecture for $F(u, v)$ is withdrawn. Helfgott's theorem concerns the average of $W(E_t)$ over the full two-parameter family $t = u/v$; the orbit t_n is a density-zero subset of any such family. Chowla for $F(u, v)$ in two variables does not imply anything along the thin sequence (U_n, V_n), and H, which only asks for infinitely many n with $W = +1$, does not imply Chowla (it asks for no cancellation). What is true is the weaker statement of the barrier in §6 (the paragraph "A barrier for the sign"): the unconditional equidistribution methods do not apply to a degree-6 form, and a sign statement needs a parity-type input for values of F along the orbit. Neither implication with Chowla is established. *The bit.* Since $v_2(F_n) = v_3(F_n) = 0$

(F.20), the part of |F_n| prime to 6 is |F_n| itself, and the unknown is δ_n = λ(|F_n|) = (−1)^{ω(rad)}·(−1)^{Σ(e_p − 1)} (the second factor accounts exactly for repeated primes). *Proposition.* Assuming the exact root-number formula W = f(t_n)·(−1)^{ω(R)}·(ab/R) with R the radical of F″_n, and with ε_n = f(t_n)·(ab/F″_n), one has W(E_{t_n}) = ε_n·δ_n·κ_n, where κ_n = Π_{p : e_p even} (−(ab/p)) (so κ_n = 1 when every exponent is odd). Consequently, H fails if and only if W_n = −1 for all n ⩾ n_0, and for those n with F_n squarefree this says δ_n = −ε_n: the Liouville function along the orbit would be eventually equal to the explicit period-6 pattern −ε_n. *Remark on targets.* (i) Cancellation of δ_n alone, or a bound |Σ_{n⩽N} δ_n| ⩽ (1 − c)N, does not suffice for H: the pattern δ_n = −ε_n has bounded partial sums (ε is periodic with ε_n = +1 exactly for n ≡ 0, 1, 5 (mod 6), three plus signs and three minus signs per period, so its mean is 0) and still gives W ≡ −1. The correct target is a bound for the sums of W_n = ε_nδ_nκ_n, i.e. non-correlation of δ_n with the periodic function ε_n, for instance |Σ_{n⩽N} W_n| ⩽ (1 − c)N, which forces both signs of W with positive lower density. (ii) Squarefreeness of F_n along a positive proportion of n is not known (F.18 gives sqf(|F_n|) → ∞ and infinitely many non-squarefree F_n), so Route A (restricting to squarefree indices) is itself open, and even on such indices the parity barrier remains.

F.25 Universal coprimality of the sequences along nP and nP + T (Theorem; proved; verified independently by exact computation). With x(nP) = A_n/B_n², F_n = B_n¹²h(A_n/B_n²) and G_n = B_n¹²h_T(A_n/B_n²) (F.20; G_n is the value of the sextic along nP + T), *Theorem.* gcd(|F_n|, |G_n|) = 1 for every n ⩾ 1. *Proof.* F_n = F_h(A_n, B_n²) and G_n = F_h(2B_n², A_n) are values of two binary sextics at a coprime pair, so a common prime p divides their resultant, |Res(h, h_T)| = 2¹⁷·7·103·519521² (recomputed). The prime 2 is excluded because F_n is odd (F.20: v_2(F_n) = 0). If p | B_n then F_n ≡ A_n⁶ ≢ 0 (mod p) (h is monic), so p ∤ F_n. Otherwise x(nP) reduces to a common root r ∈ 𝔽_p of h and h_T, and (r, ·) must be an 𝔽_p-point of J lying in ⟨P̄⟩. p = 7: the only common root is r = 4, and r³ − 2r² + 2r is a non-square mod 7, so r is the x-coordinate of no $\mathbb{F}_7$-point. p = 103: the only common root is r = 38, again the x-coordinate of no 𝔽_{103}-point. p = 519521: the common roots are r = 37492 and 384583; both are x-coordinates of 𝔽_p-points Q, but neither is the x-coordinate of a point of ⟨P̄⟩ (P̄ has order 698; equivalently 698Q ≠ O for both). So no prime divides both F_n and G_n. □ (Brute-force check: gcd = 1 for 1 ⩽ n ⩽ 150.) *Consequences.* (a) The parts of F_n and G_n prime to 6 are coprime, so the root numbers of E_{t_n} and E_{t′_n} involve disjoint sets of primes at the bad places coming from the sextic. (b) From Appendix B and F.21, ε_nε′_n = (−1)^n for all n ⩾ 2 (ε_n = +1 iff n ≡ 0, 1, 5 (mod 6), ε′_n = +1 iff 3 | n). Hence, assuming the exact root-number formula of F.24, W(E_{t_n})·W(E_{t′_n}) = (−1)^n·λ(|F_nG_n|)·s_n·κ_nκ′_n with s_n = −1 exactly when n ≡ 4, 5 (mod 9) (F.20): the joint sign of the pair is the Liouville sign of the degree-12 value F_nG_n up to explicit periodic and squareful corrections. *What this does not give.* The multiplicativity λ(|F_nG_n|) = λ(|F_n|)λ(|G_n|) holds without coprimality; coprimality only excludes shared primes as a source of dependence and does not make the two Liouville signs

independent. An algebraic relation forced by the involution x ↦ 2/x is excluded by F.19 (the ratio h(2/x)/h(x) is not in $\bar{\mathbb{Q}}^\times \cdot \bar{\mathbb{Q}}(J)^{\times 2}$, so no squareclass identity relates the two sequences). The joint parity λ(|F_nG_n|) is again a parity problem for a binary form (of degree 12) along the orbit, and the barrier of F.24 remains.

F.26 The quotient sextic of the involution x ↦ 2/x (Proposition; verified by exact computation). Let υ = x + 2/x be the coordinate on the quotient J/⟨T⟩ = J′ (x′ = υ − 2, φ(P) = (1, −1) on J′: $Y^2 = X^3 + 4X^2 - 4X$) and $R(U) = -54U^6 + 648U^5 - 1242U^4 - 11404U^3 + 62082U^2 - 113832U + 73224$. *(a) Identities.* h(X)h_T(X) = $X^6 R(X + 2/X)$, and R(U) = Res_Z($Z^2 - UZ + 2$, h(Z)) is the quadratic norm of h; its roots are υ_i = x_i + 2/x_i for the six roots x_i of h. *(b) Arithmetic of R.* R is irreducible over $\mathbb{Q}$, as is $R(X + 2) = -54X^6 + 1998X^4 - 4060X^3 + 2730X^2 - 624X + 64$; disc(R) = $2^{20} \cdot 3^{18} \cdot 23^2 \cdot 519521^2$ and disc(h) = $2^6 \cdot 3^{18} \cdot 23^2$ are both squares, with ratio $2^{14} \cdot 519521^2$; R stays irreducible over $\mathbb{Q}(\sqrt{d})$ for d ∈ {−1, ±2, ±3, ±5, ±6, ±7, ±23, ±69}; its Galois group has order 24 (the group S_4 of Appendix E, acting on the six roots inside A_6). Since R is irreducible of the same degree 6 as h, $\mathbb{Q}(\upsilon_i) = \mathbb{Q}(x_i)$: R defines the same sextic field K = $\mathbb{Q}(\alpha)$ as h (it is a Tschirnhaus transform, υ = α + 2/α). *(c) The product as one value on J′.* For x(nP) = A_n/B_n^2, put $p_n = A_n^2 + 2B_n^4$ and $q_n = A_n B_n^2$ (coprime). Then F_nG_n = R_h(p_n, q_n), where R_h is the binary sextic of R, and p_n/q_n − 2 is the x-coordinate of nφ(P) on J′ (checked for n ⩽ 40): the product of the two sequences of F.25 is a degree-6 binary form along one orbit of the quotient curve. *(d) Where the exceptional primes of F.25 come from.* Res(h, $X^2 - 2$) = $2^3 \cdot 7 \cdot 103$: a root of h with $x^2 = 2$ is fixed by the involution and is automatically a common root of h and h_T, which explains 2, 7, 103 (the common roots 4 mod 7 and 38 mod 103 are the square roots of 2). The prime 519521 is a genuine collision: h has the six roots 37492, 134940, 162299, 357224, 384583, 482031 modulo 519521 and two of them satisfy 37492·384583 ≡ 2, i.e. are swapped by the involution, so R has a double root and 519521^2 | disc(R)/disc(h). *(e) What it does not give.* The quotient does not lower the degree (it stays 6, irreducible, Galois group S_4), so the unconditional equidistribution results (irreducible factors of degree ⩽ 3) remain inapplicable, F.19 excludes any squareclass identity between the two sequences, and the joint parity λ(|F_nG_n|) = λ(|R_h(p_n, q_n)|) is again a parity problem for a sextic form along an orbit, now on J′. (Computations done in sympy, reproducing every assertion of the supplied Magma script.)

F.27 Literature status: primitive divisors versus parity (Remark). Two kinds of results must be kept apart. Primitive-divisor theorems say that new primes appear; Chowla-type theorems control the parity of the number of prime factors, which is what the root-number strategy needs. For the first kind, Verzobio (J. Number Theory 209 (2020) 378–390) proves that for {nP + Q} with Q a torsion point of prime order the denominator of x(nP + Q) has a primitive divisor for all large n, and relates the problem to the Lang–Trotter conjecture; his CM paper (arXiv:2010.10175) requires P non-torsion and endomorphisms f, g ≠ 0 with f(P) = g(Q), i.e. Q commensurable with P. A prime 𝔓 divides F_n exactly when nP ≡ ±R_i (mod 𝔓) for a point R_i with h(x(R_i)) = 0; these points are non-torsion, defined over

a field of degree 12 and not commensurable with P, so neither hypothesis holds, and his comparison of old and new primes (which uses rQ = O) does not carry over: old primes of F_n lie in gcd(F_n, B_{n±k}) for k < n, and an unconditional gcd bound of Siegel/Vojta type would be needed (not checked to exist in this generality). Even a full primitive-divisor theorem for F_n would give one prime not dividing any earlier F_k; this is weaker in content than sqf(|F_n|) → ∞ (F.18) combined with F.25 and the valuation laws, and gives no information on the parity of the number of the other primes. For the second kind, Helfgott proves Chowla for reducible and for irreducible cubic forms (arXiv:math/0408142 and its companion) and describes the parity problem as the obstruction beyond degree 3; these statements concern forms over pairs (x, y), not thin sequences. Two literature searches found no Chowla-type result along an elliptic divisibility orbit (this is a statement about those searches, not a proof that none exists). Conclusion: extending Verzobio's method to the divisor h(x) = 0 is a legitimate study of the prime structure of F_n but is not a route to H.

F.28 The exact root-number formula holds for every rational t (Theorem; proved; confirmed by an independent analytic computation; closes open problem 5). For t = U/V in lowest terms put $c_0 = U^2 + V^2$, $a = V^2 - U^2$, $b = 2UV$ (reduced to the primitive triple). *Theorem.* For every t, W(E_t) = f(t)·Π_{p ⩾ 5, p | F(U,V)} (−(ab/p)), the product running over the distinct primes, with f(t) as in Appendix B; in particular the exponents of F need not be odd. *Proof.* On the model $y^2 = x^3 + a_4x + a_6$, $a_4 = 4(V^2 - U^2)c_0^3$, $a_6 = -16UVc_0^5$, one has the identity $\Delta = 2^{12}c_0^9F(U, V)$. Let p ⩾ 5 divide F. If $p \mid V^2 - U^2$ then $F \equiv -54V^6 \not\equiv 0$, and if $p \mid c_0$ then $F \equiv -8V^6 \not\equiv 0 \pmod p$; hence $p \nmid c_0$ and $p \nmid c_4 = -48a_4$. So the reduction at p is multiplicative of type I_e with e = v_p(F), whatever e is. The local root number of a multiplicative fibre is −1 if split and +1 if non-split, independently of e. It is split exactly when $-c_6$ is a square mod p, where $-c_6 \equiv -6\cdot UV\cdot c_0$ up to squares; since F ≡ 0, the proved square identity $3(\alpha^4 - 1) = s(\alpha)^2$ (F.8) gives $(-3c_0(V^2 - U^2)/p) = 1$, hence $(-c_6/p) = (ab/p)$ (checked in 618 cases with no exception). So w_p = −(ab/p), and the primes not dividing F contribute through f(t). □ *Consequence.* The identity W = ε_n·λ(F″_n) holds exactly when every prime of F″_n has odd exponent; in general W = ε_n·δ_n·κ_n (F.24), and the assumption "exact root-number formula" made there is now a theorem. *Independent verification.* The sign was computed analytically from the functional equation (method validated on 11a, 14a, 389a with sign +1 and 37a with sign −1): for t = −4 ($F = -3^4\cdot 7^2$, the case where the naive formula fails) the unique conductor consistent with the functional equation is $2^5\cdot 3^4\cdot 7\cdot 17^2 = 5243616$ and the sign is +1, equal to the exact formula and different from the naive value −1; for t = 4 the sign is −1, as predicted. No other t with a square prime factor ⩾ 5 and |F| ⩽ $2\cdot 10^6$ exists.

F.29 Status of the open items at the time of F.28 (superseded by §9 and F.46). *Closed:* (1) the squareclass obstruction for the translate by T: h·h_T is squarefree of degree 12 and coprime to $x(x^2 - 2x + 2)$, so neither h·h_T nor $h\cdot h_T\cdot x(x^2 - 2x + 2)$ is a square in $\bar{\mathbb{Q}}(x)$, hence h·h_T is not in $\bar{\mathbb{Q}}^\times\cdot\bar{\mathbb{Q}}(J)^{\times 2}$ (this completes F.19 for the translation by T); (2) the

exact root-number formula at squareful F_n (F.28); (3) the period-3 law along nP + T (F.21); (4) the valuation laws at 2 and 3 (F.22); (5) the primitive-divisor route to H (F.27: not a route); (6) coprimality of the sequences along nP and nP + T (F.25). *Open but not used by any theorem:* the reducibility over $\bar{\mathbb{Q}}$ of the degree-5 invariant class (over $\mathbb{Q}$ it has no irreducible representative); the factorization of the 107-digit cofactor of F_{10}, which would add one data point to the table of λ(|F_n|) and has no theoretical content. *Blocked:* Problem H itself. By F.24 it is equivalent (given the proved formulas) to the statement that W_n = −1 does not hold for all large n, i.e. that the Liouville sign along the orbit does not eventually equal the explicit period-6 pattern −ε_n on squarefree indices; every route examined in this paper (sieve methods, function-field algebra, the geometric multisection search F.13–F.15, Faltings F.18, primitive divisors F.27, the quotient F.26, coprimality F.25, local and global congruences (Appendix B, F.20, F.21)) leaves exactly this parity statement, which requires a Chowla-type input along a thin orbit.

F.D Data, conjectures tested and refuted (F.30–F.42)

F.30 Why C_2 and not C_4 or C_4' (Remark; data computed exactly). *Size of the members.* On C_2 (z = x(nP) on J, t_n = 1 − z) the numerators and denominators of t_n have 1, 1, 2, 3, 5, 7, 10, 12, 16, 19, 23, 27 digits for n = 1, …, 12; on C_4' ($v^2 = w^3 + 8$, R = (1, 3), t = (w − 2)/(w + 2)) they have 1, 2, 3, 5, 8, 11, 15, 19, 24, 29, 35, 41 digits, i.e. the canonical height of the generator is about 1.5 times larger, and the members are correspondingly harder to compute with. *Number of rational points of height at most B* (all t = u/v with |u|, v ⩽ 3000): C_2: 5, 5, 7, 7, 9, 9 for B = 10, 30, 100, 300, 1000, 3000; C_4': 3, 5, 5, 6, 7, 7; the constant-x curve c = 4 ($w^2 = (1 + t^2)(3t^2 − t + 5)$): 1, 2, 2, 2, 2, 2, namely only t = 4/3 and t = −11/2 (the two rank-2 fibres with W = +1), so its next members are astronomically large and no further data can be obtained from it; c = 1: 2, 2, 2, 2, 3, 4. *Structure.* Only C_2 is an elliptic curve in the form (z, w) with t = 1 − z a coordinate function whose denominators are squares (x-coordinates of rational points on J have square denominators), so V_n = D_n² and (2/c′) = 1; this is what makes Hilbert reciprocity eliminate every symbol except those at 2, 3, ∞ and allows the exact period-6 and period-3 theorems of Appendix B and F.21, the valuation laws of F.22, the coprimality of F.25 and the quotient description of F.26 (all of which use the rational 2-torsion point T = (0, 0) of J and the involution x ↦ 2/x). On C_4' the denominator V is not a square, the factor (2/c) enters (observed period 2 along nR and period 1 along nR + T) and no corresponding theorem has been proved; for c = 4 there is no such parametrization at all. *What is the same.* All covers carry the same discriminant sextic D(t); the pulled-back divisor has irreducible factors of degree divisible by 6, the global formula reduces the sign question on C_4' to the same parity statement as on C_2, and the unknown is the same bit δ_n. So no cover is better for H itself; C_2 is the best for data and for proofs. (Not computed: the canonical height of the generator of the c = 4 Jacobian, for lack of PARI/Sage here; the point counts above are the proxy.)

F.31 The one-bit test on the completely factored members (Observation; data recomputed from scratch). For n = 2, …, 9 along nP and along nP + T the part N of |F| prime to 6 has been factored completely (the product of the primes was checked to equal N; every factor passed a strong probable-prime test, which has no known counterexample but is not a proof; all exponents are 1, so κ = 1). From each factorization I computed s = the number of split primes ((ab/p) = +1), f from the 2-adic and 3-adic tables, W = f·(−1)^s, ε = f·(ab/N) from the exact definition of Appendix B, and δ = (−1)^ω. Results: **W = ε·δ in 16 of 16 rows, and ε equals the proved law (period 6 along nP: +1 iff n ≡ 0, 1, 5 mod 6; period 3 along nP + T: +1 iff 3 | n) in 16 of 16 rows**, so the only data-dependent quantity is the single bit δ; for nP + T this also checks F.21 on members of size up to 71 digits. Table (kind, n, digits of N, ω, f, s, W, ε, δ): nP: (2, 5, 2, +, 1, −, −, +), (3, 10, 2, +, 1, −, −, +), (4, 18, 2, +, 1, −, −, +), (5, 29, 3, +, 1, −, +, −), (6, 41, 4, +, 2, +, +, +), (7, 55, 4, +, 2, +, +, +), (8, 73, 4, +, 1, −, −, +), (9, 91, 5, +, 0, +, −, −); nP + T: (2, 3, 1, −, 1, +, −, −), (3, 10, 3, −, 0, −, +, −), (4, 16, 2, +, 1, −, −, +), (5, 27, 2, −, 2, −, −, +), (6, 39, 3, +, 1, −, +, −), (7, 56, 5, +, 2, +, −, −), (8, 71, 4, +, 1, −, −, +), (9, 91, 4, −, 3, +, +, +). W = +1 occurs at n = 6, 7, 9 along nP and n = 2, 7, 9 along nP + T, i.e. at 6 of these 16 members. *A pattern, with a warning.* The bit δ equals +, +, +, −, +, +, +, − along nP and −, −, +, +, −, −, +, + along nP + T for n = 2, …, 9: both are exactly periodic of period 4 on these 8 terms. For independent fair signs the chance of a given period-4 pattern on 8 terms is 1/16 per sequence (1/256 for both), and about 8.6% per sequence (0.7% for both) that some period ⩽ 4 fits, but this was noticed after looking at the data and the null model is doubtful: ω(N) grows roughly like 2 log n plus a constant (observed 2, 2, 2, 3, 4, 4, 4, 5 along nP), so for such small n the parity of ω changes at fairly regular steps, which is a transient effect and cannot persist, since the gaps between increments of ω grow with n. It is not a theorem and carries no weight for H; it is recorded because it is testable at once. *Pre-registered predictions at n = 10* (period 4 would give δ = + along nP and δ = − along nP + T). N(10P) = 547·2557·12323894684537·C_{94} with C_{94} = 21495883243311689523812343835828307461517398133538256874117249503974813071583928916255842633 99 (94 digits, composite, no prime factor below 10^6, not a perfect power): period 4 predicts that C_{94} has an odd number of prime factors (three), whence ω even, δ = +, W(10P) = −1 (here ε_{10} = −1); if C_{94} is a product of two primes then δ = −1 and W(10P) = +1 (rank ⩾ 2 under the parity conjecture). N(10P + T) = 1009·3580555487435623·22085981153898979·C_{78} with C_{78} = 47747769652087619462246775233103229702383327826553830077216652720597472450014 9 (78 digits, composite, no prime factor below 10^6, not a perfect power): period 4 predicts that C_{78} has an even number of prime factors (two), whence ω odd, δ = −, and since ε'_{10} = −1, W(10P + T) = +1; if C_{78} has three prime factors the pattern fails and W(10P + T) = −1. Neither cofactor could be factored with the tools available here (one core, no sieve software); a quadratic sieve or NFS implementation (Magma's Factorization, PARI's factor, msieve) should do the 78-digit number first.

F.32 Result of the pre-registered test at 10P + T (Observation; factorization supplied by the user with PARI/GP, product and primality rechecked). The 78-digit cofactor of F.31 factors as C_{78} = 928615468896957507550954470402067379 · 514182363436225319787643322205506400496631 (36 and 42 digits, both probable primes, product equal to C_{78}). So N(10P + T) = 1009 · 3580555487435623 · 22085981153898979 · p_{36} · p_{42} has ω = 5, δ = −1 (as period 4 predicted), the Legendre symbols (ab/p) are +, +, +, −, + (s = 4), f = +1, hence W(E_{t(10P+T)}) = f·(−1)^s = **+1**, equal to ε'_{10}·δ = (−1)(−1) as in F.31 (and ε'_{10} = f·(ab/N) = −1 as F.21 requires). At t = 562339668795541727 3/5979893522075890561 the point (2, y) lies on E_t and is non-torsion (no multiple up to 12 is the point at infinity), so under the parity conjecture E_{t(10P+T)} has rank ⩾ 2 (or Sha is infinite). Including n = 10 the sign W = +1 now occurs at 7 of the 17 members known along nP (n ⩾ 2) and nP + T (n ⩾ 2). *How much this confirms.* The prediction that C_{78} has an even number of prime factors was the a priori more likely outcome (a 78-digit composite with no prime factor below 10^6 most likely has two prime factors), so the confirmation is weak evidence for a period-4 pattern; the sharper test is the 94-digit cofactor C_{94} of F.31 along nP, for which period 4 predicts an odd number (three) of prime factors, the less likely outcome a priori.

F.33 What a period-4 law for δ would imply (Remark; hypothetical; the pattern is refuted in F.38). Suppose δ_n = λ(|F_n|) were +1 for n ≢ 1 (mod 4) and −1 for n ≡ 1 (mod 4) along nP, and δ′_n = −1 for n ≡ 2, 3 and +1 for n ≡ 0, 1 (mod 4) along nP + T, as observed for 2 ⩽ n ⩽ 9 and (nP + T) n = 10. Together with the proved periodic laws for ε_n and ε′_n this would give W = ε·δ·κ with κ = 1 on squarefree indices and hence a period-12 sign: along nP, W = +1 exactly for n ≡ 0, 6, 7, 9, 11 (mod 12) (density 5/12); along nP + T, W = +1 exactly for n ≡ 0, 2, 7, 9, 10, 11 (mod 12) (density 1/2). These patterns reproduce all 17 signs computed so far. Then H would hold along both orbits, with positive density, and under the parity conjecture infinitely many members would have rank ⩾ 2. The law differs from the H-false pattern δ = −ε exactly at the indices where W = +1 (n = 6, 7, 9, 11, 12, … along nP). The first untested predictions are W(11P) = W(12P) = W(11P + T) = W(12P + T) = +1, but these members have N of about 135 to 160 digits and cannot be factored; the only feasible test left is the 94-digit cofactor of F.31 at n = 10. *Why this is not a route to a proof.* The law would say that the Liouville sign along the orbit is a deterministic function of n mod 4, which contradicts the Chowla-type heuristic that predicts independent signs; no mechanism is known (a prime p divides F_n for n in at most 12 residue classes ±k modulo the order of F modulo a prime above p, but the parity of the number of such primes up to size e^{cn²} is not controlled by this, and squareclass identities of three kinds are proved impossible (F.19, F.28, F.34) while identities of other kinds have not been examined (F.34)). Since the size of N grows like e^{n²}, the law can never be tested for large n by factorization, so it would remain a conjecture even if every feasible check agreed.

F.34 What is and is not excluded among algebraic routes, and the elliptic-net relation (Proposition; verified by computation). *Correction.* The statement in F.33 that the algebraic routes are all excluded was too strong. *Proved excluded:* (i) squareclass relations between h(x(Q)) and h(x([m]Q)) for |m| ⩾ 2 and between h(x(Q)) and h(x(Q + T)) for the 2-torsion points T (F.19, F.28), as identities in $\bar{\mathbb{Q}}(J)$; (ii) F_n a square for infinitely many n (F.18); (iii) degree reduction by the quotient J → J′ (F.26) and by rational base change (barrier of §4); (iv) shared primes between the two sequences outside an explicit finite set (F.25). *New (the three-term relation).* For P, Q ∈ J with u = x(P), v = x(Q) and α a root of h one has the exact identity $(u - v)^2(x(P + Q) - \alpha)(x(P - Q) - \alpha) = N_\alpha(u, v) := (uv - 2)^2 - 2\alpha[(u + v)(uv + 2) - 4uv] + \alpha^2(u - v)^2$ (checked on exact points), so $F(P + Q)F(P - Q) = \mathrm{Norm}_{K/\mathbb{Q}}(N_\alpha(u, v))/(u - v)^{12}$. The polynomial N_α is irreducible over $\mathbb{Q}(\alpha)$ as a polynomial in (u, v, α), squarefree in u, and coprime to u − α, v − α, ρ(u), ρ(v) and u − v (ρ(x) = x(x² − 2x + 2)); at u = α it equals $-4\rho(\alpha)v + (\alpha^2 - 2)^2$, a nonzero polynomial for every root α of h. Since the square classes in $\bar{\mathbb{Q}}(J \times J)$ of elements of $\bar{\mathbb{Q}}(u, v)$ are those modulo squares and the factors ρ(u), ρ(v), no product of the form N_α·(u − α)^i(v − α)^j·ρ(u)^kρ(v)^l with exponents in {0, 1} is a square. Hence there is no squareclass identity relating F(P + Q)F(P − Q) to F(P) and F(Q); the elliptic-net relation does not give a parity relation among the F_n. *Bounded searches, not proofs:* second sections of twists (low height and degree), smooth rational curves of degree ⩽ 4 and invariant classes of degree ⩽ 21 (F.13–F.15), polynomial twists with coefficients ⩽ 4 (§8). *Not examined:* arithmetic identities that hold only at the points nP and not as identities of functions (for example relations in the class group or unit group of the degree-6 field K or of the degree-12 field L between the ideals $(A_n - \alpha B_n^2)$); descent or Heegner-type constructions of a second point on E_{t_n} not coming from the parity theorem; and relations involving three or more terms F_{n_1}, F_{n_2}, F_{n_3} beyond the three-term relation above. For ideals of K one fact is clear: every prime ideal dividing $A_n - \alpha B_n^2$ (outside the primes dividing the index and the discriminant) has residue degree 1, because α reduces into $\mathbb{F}_p$, so Ω(F_n) counts only degree-1 primes, and the evenness of the Frobenius classes (disc h is a square, Gal ⊂ A_6) constrains all primes above p together, not these alone.

F.35 Conjecture D: δ_{2n} = δ_n (Hypothetical; refuted in F.38). *Statement.* For all n ⩾ 1, λ(|F_{2n}|) = λ(|F_n|) along the orbit nP. *Data.* With $\delta_1 = +1$ ($F_1 = -1$) and $\delta_2, \delta_4, \delta_6, \delta_8 = +, +, +, +$ the four testable cases $\delta_2 = \delta_1$, $\delta_4 = \delta_2$, $\delta_6 = \delta_3$, $\delta_8 = \delta_4$ hold (probability 1/16 for independent signs). The analogue along nP + T fails ($\delta'_4 = + \neq \delta'_2 = -$, and $\delta'_{10} = - \neq \delta'_5 = +$), so D, if true, is not a symmetric property of the two orbits. *Consequence for H.* From Appendix B, ε_{2n}ε_n = (−1)^n for all n (checked for n < 200 and clear from the table of ε modulo 6), hence W_nW_{2n} = (−1)^n·δ_nδ_{2n}·κ_nκ_{2n}. Under D this is W_nW_{2n} = (−1)^nκ_nκ_{2n}: for odd n with κ_nκ_{2n} = +1 (for instance when F_n and F_{2n} are squarefree) exactly one of the members n, 2n has W = +1. So D would imply H as soon as infinitely many odd n have κ_nκ_{2n} = +1; this squarefreeness-type condition is not known to hold for infinitely many n. *Rival law and the experiment.* The period-4 law of F.33

violates D at n = 5 (it predicts $\delta_{10} = +$, D predicts $\delta_{10} = \delta_5 = -$). The 94-digit cofactor C_{94} of F.31 decides between them: a product of two primes gives $\omega(N(10P)) = 5$, $\delta_{10} = -1$ (D holds, period 4 fails, $W(10P) = +1$); a product of three primes gives $\delta_{10} = +1$ (period 4 holds, D fails, $W(10P) = -1$). Two primes is the a priori more likely outcome, so a confirmation of D would again be weak evidence. *Why no algebraic mechanism exists.* By the doubling formula $F_{2n} = \mathrm{Res}_\alpha(h, q_\alpha(A_n, B_n^2)) = \mathrm{Norm}_{K/\mathbb{Q}}(q_\alpha(A_n, B_n^2))$ with $q_\alpha(a, b) = (a^2 - 2b^2)^2 - 4\alpha ab(a^2 - 2ab + 2b^2)$ (checked exactly for $n \leqslant 4$, with ratio 1). D would follow at once from a squareclass identity $\varphi([2]Q)\cdot\varphi(Q) \in \bar{\mathbb{Q}}^\times\cdot\bar{\mathbb{Q}}(J)^{\times 2}$, which makes $F_{2n}F_n$ a square up to a bounded factor; F.19 proves that this identity is impossible. In the cases computed F_n and F_{2n} are coprime, so D would be a deterministic correlation between the Liouville signs of two coprime numbers, in contradiction with the Chowla heuristic exactly as the period-4 law is.

F.36 Independence of h and its doubling pullback modulo squares (Proposition; proved by exact computation; elementary form of the case m = 2 of F.19). Let $\delta(X) = (X^2 - 2)^2/(4X(X^2 - 2X + 2)) = x(2Q)$ as a function of $X = x(Q)$, and $\mathrm{Num}(X) = \mathrm{Res}_\alpha(h(\alpha), (X^2 - 2)^2 - 4\alpha X(X^2 - 2X + 2))$, so that $h(\delta(X)) = \mathrm{Num}(X)/(4X(X^2 - 2X + 2))^6$ (checked exactly). *Proposition.* The classes of h and of $h(\delta(X))$ in $\mathbb{Q}(X)^\times/\mathbb{Q}^\times\cdot(\mathbb{Q}(X)^\times)^2$ are independent: none of h, $h(\delta(X))$, $h\cdot h(\delta(X))$ is a constant times a square in $\mathbb{Q}(X)$. *Proof.* The denominator $(4X(X^2 - 2X + 2))^6$ is a square, so $h(\delta(X)) \equiv \mathrm{Num}(X)$. Here Num has degree 24 and is irreducible over $\mathbb{Q}$ (computed), h is irreducible of degree 6, and $\gcd(h, \mathrm{Num}) = 1$ (computed). Then h, Num and $h\cdot\mathrm{Num}$ each have an irreducible factor of multiplicity one, hence are not constants times squares. □ *Remarks.* (i) The facts used are the irreducibility of Num (degree 24) and $\gcd(h, \mathrm{Num}) = 1$, not the irreducibility and coprimality results for h, h_T and R (F.25, F.26), which concern the involution $x \mapsto 2/x$ and not doubling; those facts were established only in the computation above. (ii) Over $\bar{\mathbb{Q}}$ the same holds: $h\cdot\mathrm{Num}$ is squarefree of degree 30, and coprime to $\rho(X) = X(X^2 - 2X + 2)$ ($\mathrm{Num}(0) = 4^6$, $\mathrm{Num}(1 \pm i) \neq 0$), so $h\cdot\mathrm{Num}\cdot\rho^k$ is not a square in $\bar{\mathbb{Q}}(X)$ either, which is the case m = 2 of F.19 in $\bar{\mathbb{Q}}(J)$. (iii) The proposition excludes only identities of rational functions (F.34): it shows that Conjecture D (F.35) cannot be obtained from $F_{2n}F_n = c\cdot(\text{square})$, and does not exclude arithmetic coincidences at the points nP.

F.37 Conjecture D on the cover C_4', and why a proof of D is not to be expected (Observation). On C_4' ($v^2 = w^3 + 8$, $R = (1, 3)$, $t(nR) = (w(nR) - 2)/(w(nR) + 2)$) the part N of $|F(U, V)|$ prime to 6 was factored completely for $n = 1, \ldots, 7$ (products checked, all factors probable primes); the number of prime factors counted with multiplicity Ω and $\delta = (-1)^\Omega$ are: n = 1: $N = 1471$, $\Omega = 1$, $\delta = -$; n = 2: $859\cdot 5743$, $\Omega = 2$, $\delta = +$; n = 3: $7^2\cdot 19\cdot 173\cdot 233\cdot 3923578273$ (18 digits, squareful), $\Omega = 6$, $\delta = +$; n = 4: $103\cdot p_{27}$, $\Omega = 2$, $\delta = +$; n = 5: $647264489\cdot p_{36}$, $\Omega = 2$, $\delta = +$; n = 6: eight distinct primes, $\Omega = 8$, $\delta = +$; n = 7: a prime of 18 digits times a prime of 69 digits, $\Omega = 2$, $\delta = +$ (n = 8 has 110 digits and was not factored). The analogue of Conjecture D ($\delta_{2n} = \delta_n$) therefore fails at n = 1 ($\delta_2 = + \neq \delta_1 = -$) and holds at n = 2 and n = 3. Counting all three orbits, the analogue of D holds in 8 of the 11 tests available (4 of 4 along

nP on C_2, 2 of 4 along nP + T, 2 of 3 on C_4'), consistent with chance. *Why a proof is out of reach.* (i) By F.36, D cannot come from an identity of rational functions. (ii) F_n and F_{2n} have no common prime in the cases computed (more generally p | F_n and p | F_{2n} force $nP^- \equiv \pm R^-$ and $2nP^- \equiv \pm R''$ modulo p, which restricts p to a thin set), so D would assert that the Liouville signs of two essentially disjoint sets of primes are equal for every n. (iii) Even a statement at the level of ideals of K would not suffice: every prime ideal dividing $A_n - \alpha B_n^2$ has residue degree 1 (F.34), so the sign is (−1)^{number of prime ideals counted with multiplicity}, and an identity forcing that number to agree for two coprime ideals would have to hold with no square-class mechanism to explain it. (iv) If the signs behave like independent fair coins, as the Chowla heuristic predicts, D fails for infinitely many n with probability one; a proof of D would disprove that heuristic along the orbit. The sensible use of D is therefore as a test, not as a target: it predicts $\delta_{10} = \delta_5 = -1$, equivalently that C_{94} (F.31) is a product of two primes, which is the likely outcome a priori and therefore weak evidence if it occurs.

F.38 The period-4 law and Conjecture D are refuted at n = 11 and n = 12 (Observation; complete factorizations, all factors probable primes, products checked). *Along nP + T, n = 11* (135 digits): N = 491 · 37157059 · 181611589 · 3389375653 · 17257011513996899591 · 350857089532834962879832747183405516817663446786100985829827058984675985527307279704636 7, six distinct primes, so ω = 6 and δ'_{11} = **+1**; the Legendre symbols give s = 3, f = +1, W = −1, ε'_{11} = −1 (as F.21 requires, 3 ∤ 11) and W = ε′δ′. The period-4 law of F.31 predicted δ'_{11} = −1: **refuted.** *Along nP, n = 12* (163 digits): N = 23 · 83 · 18397 · 7851197 · 172604961553 · 3022285517285783321 · 6344487564223421983068803760187024008273285782888349481767151634472736124778907875032940285904419080000374027575756680 3, seven distinct primes, so ω = 7 and δ_{12} = **−1**; s = 3, f = +1, W = −1, ε_{12} = +1 (as F.17 requires, 12 ≡ 0 mod 6) and W = εδ. The period-4 law predicted δ_{12} = +1, and Conjecture D (F.35) predicted $\delta_{12} = \delta_6$ = +1: **both refuted.** These are also two further checks of the proved laws for ε (Appendix B at n = 12, F.21 at n = 11) on numbers of 163 and 135 digits, and of W = ε·δ. With these data W = +1 occurs at 7 of the 19 members known (nP and nP + T, n ⩾ 2). Not factored: N(11P), a 136-digit composite with no prime factor below $2 \cdot 10^6$ and none found by ECM with moderate bounds, 4013443238136047027943207311329615913049700315566516636507517490310778274740116167953214179614151497454294097047330487257135673478856081; N(10P) cofactor C_{94} (F.31); N(12P + T) (a computation was still running). *Conclusion.* The period-4 pattern of δ was the transient effect anticipated in F.31: ω(N) grows roughly like 2 log n, so the parity of ω changes at regular steps for small n and then drifts. The data are again compatible with independent signs, and no pattern of δ is supported.

F.39 Status of N(10P) and what the root-number formula gives from a partial factorization (Remark). An ECM attempt on the 94-digit cofactor C_{94} of F.31 (40 curves with B1 = $5 \cdot 10^4$, 60 with B1 = 10^5, 100 with B1 = $2.5 \cdot 10^5$, sympy implementation, about 10 minutes) found no factor, which makes a prime factor of at most about 25 to 30 digits

unlikely but proves nothing; the quadratic sieve or the number field sieve is needed, and the factorization is not yet known. *What is already determined.* For a squarefree N with a known part and a composite cofactor C, the exact formula W = f·(−1)^s gives (−1)^{s_C} = (−1)^{ω(C)}·(ab/C) for the number s_C of split primes of C, where (ab/C) is the Jacobi symbol; hence W = ε·δ, and the only missing information is the parity of ω(C). For N(10P): ε_{10} = −1, ω(known part) = 3, so if C_{94} is a product of two primes then ω = 5, δ_{10} = −1 and W(E_{t(10P)}) = +1; if it is a product of three primes then ω = 6, δ_{10} = +1 and W = −1. In the first case Conjecture D (F.35) holds at n = 5 (the period-4 law is already refuted by F.38) and, with the non-torsion point (2, y) on E_t, F.33 gives an unconditional member with rank ⩾ 2 or Sha[2^∞] infinite. *Use of the formula.* It reduces the computation of W(E_t) to the factorization of F(U, V) (no conductor or L-function computation) and was confirmed on members up to 163 digits (F.31, F.38) and by an independent analytic computation (F.28); it does not make the factorization cheaper, and it does not help to prove H.

F.40 Problem H in terms of the root-number sums T_r(X) (Reformulation; with independent verification of F.28). *Exact formula.* Let N_n be the part of |F_n| prime to 6 (N_n = |F_n| along nP), ω the number of distinct prime factors, e_p the exponent of p in N_n, and η_n = Π_{p | N_n, e_p even} (ab/p). Then for n ⩾ 2, W_n = ε_{n mod 6} ·(−1)^{ω(N_n)}·η_n (the same as W = ε·δ·κ of F.24, because (−1)^{ω}η = δκ with δ = (−1)^{Ω} and κ = Π_{e_p even}(−(ab/p))); along nP + T replace ε_{n mod 6} by ε′_n (period 3). For squarefree N_n, η_n = 1, and the 19 factored members of F.31 and F.38 have all exponents equal to 1 (the product of the listed distinct primes equals N). *Sums.* For r mod 6 put M_r(X) = #{n ⩽ X : n ≡ r (mod 6)}, S_r(X) = Σ (−1)^{ω(N_n)} and T_r(X) = Σ W_n over the same indices; then T_r = ε_r·Σ (−1)^{ω(N_n)}η_n, equal to ε_r·S_r on squarefree indices, and #{n ⩽ X, n ≡ r, W_n = +1} = (M_r(X) + T_r(X))/2. Problem H is equivalent to sup_X Σ_r (M_r(X) + T_r(X)) = ∞ (a restatement). *Sufficient conditions.* |T_r(X)| ⩽ (1 − c)M_r(X) for all large X and some c > 0 and one residue class r (both signs of W with positive lower density) implies H; so does limsup (M_r + T_r) = ∞ for one r, which is H restricted to a class. Full cancellation T_r = o(X) is not needed. The relevant auxiliary quantity is a_n = (−1)^{ω(N_n)}η_n, not the unweighted parity S_r: bounding S_r says nothing about H unless η is controlled, and the pattern that must be excluded is a_n = −ε_n for all large n. *The squareful correction.* The set of n with F_n not squarefree has positive lower density: for each prime p of good reduction the set {n : p^2 | F_n} is periodic with period p·r_p, so it is a nonempty union of residue classes as soon as one n is known (p = 23, n = 32), hence of positive density. What is not established is that η_n = −1 on a set of positive density: at a prime p with p^2 ‖ N_n the factor (ab/p) depends only on the root of f modulo p to which t_n reduces, but other primes with even exponent can interfere, and the squarefreeness of the remaining part of N_n, needed to isolate p, is not known on a set of positive density. *Independent verification of F.28.* The sign computed analytically from the functional equation (method validated on 11a, 14a, 389a, 37a) agrees with the

exact formula for t = ±1/2, ±2, ±4, ±11/2, 8 of 8 cases with the unique conductors 907200, 410400, 5243616 and 17006400 respectively, with signs +1 and −1 both occurring (t = −4: +1, where the naive formula gives −1; t = −11/2: +1, a known rank-2 fibre).

F.41 Doubling and tripling laws for a_n, and the general squareclass independence (Proposition; proofs by divisors and exact checks). Put a_n = (−1)^{ω(N_n)}η_n, so that W_n = ε_na_n. *Consequences of the proved ε-laws.* From the table of ε modulo 6: ε_nε_{2n} = (−1)^n, ε_{3n} = (−1)^n, hence ε_nε_{2n}ε_{3n} = +1 for all n (checked for n < 300). Therefore if H fails, so that a_n = −ε_n for all n ⩾ n_0, then a_na_{2n} = (−1)^n and a_na_{2n}a_{3n} = −1 for all large n, and a_{3n} = −(−1)^n. Conversely, any proved law a_na_{2n} = χ(n) with χ(n) ≠ (−1)^n for infinitely many n, or any proved law a_na_{2n}a_{3n} = +1 for infinitely many n, would prove H; these are restatements of the negation of H, not independent routes. *Data.* For n = 6 and n = 12 all exponents are 1, so a_n = δ_n there and the corrected law a_{2n}a_n = +1 is still refuted (a_{12}a_6 = −1). The values of a_na_{2n} for n = 1, 2, 3, 4, 6 are +, +, +, +, − (the negation of H would give −, +, −, +, + for n = 1, 2, 3, 4, 6; they differ exactly at n = 1, 3, 6, where W = +1 occurs); the values of a_na_{2n}a_{3n} for n = 1, …, 4 are +, +, −, −. *Proposition (no squareclass identity among multiples and translates).* Let φ(Q) = h(x(Q)). For distinct pairs (m_i, T_i) with m_i ⩾ 1 integers and T_i torsion points, the functions φ(m_iQ + T_i) are linearly independent in $\bar{\mathbb{Q}}$(J)^×/ $\bar{\mathbb{Q}}$^×·($\bar{\mathbb{Q}}$(J)^×)² over $\mathbb{F}_2$; in particular no product of a finite set of values of φ along distinct multiples and translates of Q is a constant times a square, and no relation a_na_{2n} = χ(n), a_na_{2n}a_{3n} = ρ(n) can be derived from an identity of functions. *Proof.* With α a root of h and R_α a point with x(R_α) = α one has div(x − α) = (R_α) + (−R_α) − 2(O), so div φ(m·+T) ≡ Σ_α Σ_{mS+T = ±R_α} (S) modulo 2 (the poles are even). The R_α are not torsion (the orders of the reductions of the points (r, y) with h(r) = 0 in J($\mathbb{F}$_p) reach 1330 for p < 1500, and torsion orders over a fixed field are bounded). If S lies in the sets of two distinct pairs, eliminating S gives (m_jε$_1$ − m_iε$_2$)R_α = m_jT_i − m_iT_j for signs ε$_1$, ε$_2$, hence (non-torsion) m_i = m_j, and then T_i = T_j. So the supports are pairwise disjoint, each point occurs with odd multiplicity, and a nonempty sum of such divisors is not even. □ *What remains.* A relation would have to be arithmetic (it would use the primes dividing the values), not an identity of functions; this is the statement of F.34 in its general form, and it is exactly the case of the heuristic that predicts independent signs.

F.42 The even/odd asymmetry of divisibility along nP is a one-prime effect (Observation; exact computation). For a prime p ⩾ 5 let m_p = ord(P mod p). Then p | F_n exactly when nP is an affine point whose x-coordinate is a root of h, so the set of such n is a union of residue classes modulo m_p; let d_e(p) and d_o(p) be the densities of these n among the even and the odd integers (equal when m_p is odd). Put R(B) = Σ_{p ⩽ B} d_e(p) / Σ_{p ⩽ B} d_o(p). Exact computation (rational arithmetic) gives R(10000) = 1.67364670646, and for B = 100, 300, 1000, 3000, 10⁴, 2·10⁴, 4·10⁴: R = 4.90, 3.77, 2.58, 1.82, 1.67, 1.56, 1.53, i.e. decreasing. The difference Σ(d_e − d_o) stays bounded, 0.400, 0.404, 0.453, 0.410, 0.410, 0.391, 0.406, while the sums themselves grow like ln ln B (Σ d_e = 0.50,

0.74, 1.02, 1.17 and Σ d_o = 0.10, 0.29, 0.61, 0.76 at B = 100, 10^3, 10^4, $4\cdot10^4$). Almost all of the difference comes from the single prime p = 23 (m_23 = 10, d_e = 0.4, d_o = 0; p = 23 is a prime dividing disc(h) = $2^6 3^{18} 23^2$ and divides F_2, F_8, F_{12} and F_{32}): it contributes +0.4000 to the difference 0.4103 at B = 10^4; the contributions of all other primes sum to about +0.01 and change sign from range to range (+0.004, +0.049, −0.043, +0.0005, −0.019, +0.015 for the ranges [100, 300), [300, 1000), [1000, 3000), [3000, 10^4), [10^4, $2\cdot10^4$), [$2\cdot10^4$, $4\cdot10^4$)). Without p = 23 the ratio is 1.017 at B = 10^4 and 1.009 at B = $4\cdot10^4$. Hence the finite-prime ratio is not 1 for every computable bound, but this is a statement about p = 23 (and a few small primes), not a persistent asymmetry of the family; heuristically the signed contributions of the primes p > 100 have summable variance (of size $1/m_p^2$ each), the difference converges, and R(B) → 1 like 1 + 0.4/(ln ln B + c), which is extremely slow. This does not touch H: the contribution of the primes p ⩽ B to (−1)^ω is a periodic function of n (period dividing lcm of the p·m_p), which can be absorbed into the known periodic part, and the unknown is the tail, which this computation does not see; a difference in the mean of ω between even and odd n does not give a difference in the parity of ω, since the Poisson parity bias $e^{-2\mu}$ is negligible (F.40-type model).

F.E Second sections on the constant-x covers (F.43–F.46)

F.43 A second geometric section on the twist of a constant-x cover (Case analysis; estimates; the point left undecided here is settled in F.44–F.46). For c = p/q the surface of the twist E^{d_c}, $d_c = (1 + t^2)N_c(t)$, is again an elliptic K3 surface with fibres $2\cdot III + 2\cdot I_0^* + 6\cdot I_1$; its trivial lattice has rank 2 + 2·1 + 2·4 = 12, so by Shioda–Tate rank $E^{d_c}(\bar{\mathbb{Q}}(t)) = \rho - 12$, and rank $E(\mathbb{Q}(C_c))$ = rank $E(\mathbb{Q}(t))$ + rank $E^{d_c}(\mathbb{Q}(t))$ = rank $E^{d_c}(\mathbb{Q}(t))$ since the generic rank of E over $\mathbb{Q}(t)$ is 0. If ρ = 13 the geometric rank is 1 (this is proved for c = 2, §7). If ρ = 14 the geometric rank is 2, and there are two cases. (A) The second section is defined over $\mathbb{Q}(t)$: then rank $E(\mathbb{Q}(C_c)) = 2$ and, by Silverman's specialization theorem and the infinitude of $C_c(\mathbb{Q})$, infinitely many t have rank $E_t(\mathbb{Q}) \geqslant 2$, unconditionally. (B) The second section is defined only over an extension: the rank over $\mathbb{Q}(C_c)$ stays 1, and rank ⩾ 2 would occur only at t where the section becomes rational, which needs a further cover. *Evidence (Observed).* The Nagao-type estimate of rank $E^{d_c}(\mathbb{Q}(t))$, $\Sigma_p (-A_p/p) \log p / \Sigma_p \log p$ over the good primes 5 ⩽ p < 700 (A_p the sum over $t \in \mathbb{F}_p$ of the traces of the fibres; bad fibres contribute O(1)), is 0.80 for c = 2 (geometric rank 1, proved), and 1.04 for c = 13/9, 0.89 for c = −11/5, 0.96 for c = 14/45, 1.17 for c = 1 and 0.95 for c = 4: all consistent with rank 1 and none with rank 2, so case (A) is not supported for any of these covers. For case (B) with a quadratic field of definition $\mathbb{Q}(\sqrt{m})$ one would have rank $E^{d_c m}(\mathbb{Q}(t)) \geqslant 1$, detected by the same sum with the character (m/p) inserted; for all squarefree m with |m| ⩽ 30, m ≠ 1, the estimate is at most 0.20 for c = 13/9 (m = −26; median 0.00) and at most 0.13 for the control c = 2, so no such section appears. What is not decided: whether $\rho(S^{d_c}_{\bar{\mathbb{Q}}})$ is 13 or 14 for c ≠ 2 (this needs the Artin–Tate square-class argument at two primes, as in §7 for c = 2, not done), and sections

defined over fields of degree > 2 or with larger m. The search for a second rational section on the twist is the same search as the estimate above and the bounded searches of §8: it has found none.

F.44 The cover c = 13/9: ρ = 13 and geometric rank 1 [the p = 19 data below used the wrong sign; the corrected values are in F.45 (class 15 or 69), and the conclusion is unchanged] (Theorem, conditional on the Tate conjecture for K3 surfaces over $\mathbb{F}_p$ and the square order of Br, both known for these surfaces in characteristic p ⩾ 5; computations done with a C program validated on the data of §7). Let S^d be the K3 surface of the twist E^{d} with d = (1 + t²)N_c(t), c = 13/9, with the minimal model $Y^2 = X^3 + 4(1 - t^2)(1 + t^2)N_c(t)^2X - 16t(1 + t^2)^2N_c(t)^3$, $\Delta = 2^{12}(1 + t^2)^3N_c^6 \cdot f(t)$ up to sign, fibres $2\cdot\mathrm{III} + 2\cdot I_0^* + 6\cdot I_1$, smooth at ∞, and trivial lattice of rank 12. The good primes (those not in {2, 3, 5, 7, 13, 17, 23, 29, 31, 53, 307, 3989, 885403, 456084899}, the primes dividing the resultants and the leading coefficient of N_c) include 11, 19, 37, 41. For each such p put $N_k = -\Sigma_{t \in \mathbb{P}^1(\mathbb{F}_{p^k})} \Sigma_x \chi(x^3 + a_4(t)x + a_6(t))$ (valid for every fibre of this minimal model); then L(T) = exp(Σ N_kT^k/k) has degree 10 and the eigenvalues α with α·(p²/α) pairing. *Validation.* The program reproduces N_4 = 2462, −11826, 66252 at p = 7, 11, 13 and N_5 = 38 at p = 7 for c = 2 (the values of §7), and the discriminant classes 2, 2, 22, 1 at p = 7, 11, 17, 19 (Λ_p = 512/49, 1152/121, 1408/289, 484/361). *Results for c = 13/9.* p = 11 ($N_1 \ldots N_4$ = −18, −82, −5106, −31650): ρ(S^d_{$\bar{\mathbb{F}}_{11}$}) = 16, no information. p = 19 (N = −26, −714, 2602, −32554, N_4 checked by brute force and by baby-step giant-step): the eigenvalues ±19 occur once, the other eight lie on |α| = 19, ρ = 14 and Λ_{19} = 2628/361, absolute square class **73**. p = 41 (N = −40, −3896, 10046, −3727844, reproduced with two different defining polynomials of $\mathbb{F}_{41^4}$ and after a repair of the point counter): ρ = 14, all eight remaining roots on |α| = 41, Λ_{41} = 6768/1681, absolute class **47**. p = 37 (N = 6, −4254, 153174, −6068482 with the repaired counter): ρ = 16 (all roots on the circle), no information; the first run at p = 37 gave a non-Weil polynomial and exposed an error of the point counter for fibres whose points have small order, now repaired (such points are skipped). The class does not depend on the choice of q = p^{2m}: it is 73 for m = 1, 2, 3, 6 at p = 19 and 47 for m = 1, 2, 3, 6 at p = 41. *Conventions.* $\Lambda_p = \Pi(1 - \alpha^2/p^2)$ over the eight remaining eigenvalues is a product over conjugate pairs and is positive, so these are classes of |disc NS|; the signed determinant of a rank-14 lattice of signature (1, 13) is negative, so the signed classes are −73 and −47, and the comparison is the same. *Theorem.* ρ(S^d_{$\bar{\mathbb{Q}}$}) = 13 and rank E^d($\bar{\mathbb{Q}}$(t)) = 1; hence rank E($\bar{\mathbb{Q}}$(C_{13/9})) = 1 and Case 2 of F.43 (A and B) does not occur for c = 13/9. *Proof.* ρ(S^d_{$\bar{\mathbb{Q}}$}) ⩾ 13 by the section x = c, and ⩽ ρ(S^d_{$\bar{\mathbb{F}}_{19}$}) = 14. If it were 14, specialization (injective on NS and compatible with the intersection form) would identify NS(S^d_{$\bar{\mathbb{Q}}$}) with a sublattice of finite index in both NS(S^d_{$\bar{\mathbb{F}}_{19}$}) and NS(S^d_{$\bar{\mathbb{F}}_{41}$}), so the two discriminants would have the same square class; by the Artin–Tate formula (Tate conjecture, Br of square order, q = p^{2m} a square) these are 73 and 47 modulo squares, a contradiction. □ The same computation, two primes with ρ = 14 and different classes, applies to the other covers C_c.

F.45 The covers c = −11/5, 14/45, 1, 4: ρ = 13; the sign of the functional equation; corrections to F.44 and §7 (Observed/Theorem as in F.44; the point counts were done with the validated C program). *The sign.* The counts $N_1, \ldots, N_4$ do not determine the L-function: besides the case ε = −1 (R(x) divisible by $x^2 - p^2$, $e_5 = 0$) there are in general further Weil-valid polynomials with ε = +1 whose eigenvalues p·ζ are other roots of unity, and the choice changes the discriminant class. The sign ε of the functional equation is the product of the local root numbers over the bad places of $\mathbb{F}_p(t)$: the places over ±i (type III) and over the roots of N_c (type I_0^*) contribute +1 (they come in conjugate pairs or have residue field $\mathbb{F}_{p^2}$), and each place over an irreducible factor g of f (type I_1) contributes −1 if the node is split, i.e. if $3x_0 = 18t(1 + t^2)N_c(t)/(1 - t^2)$ is a square in the residue field. Hence ε = (−1)^r·χ_p(Res(f, 18t(1 + t²)N_c)/Res(f, 1 − t²)), with r the number of irreducible factors of f modulo p. *Validation of the rule.* The count N_5 was computed for c = 2: at p = 7, $N_5 = 38$ selects ε = −1 (the rule gives −1); at p = 19, $N_5 = 4354984$ selects the ε = +1 polynomial ($e_5 = -4170272$) and rejects the ε = −1 polynomial (which predicts −16496376), as the rule predicts. In four further cases (c = 2, p = 17; c = 1, p = 17; c = 14/45, p = 29; c = 13/9, p = 37) only one sign gives a Weil-valid polynomial, again the sign predicted by the rule. *Correction to §7 and F.44.* (i) For c = 2 the class at p = 19 is 11, not 1 (the value 1 came from the polynomial with ε = −1, which is wrong at p = 19); the classes of c = 2 at p = 7, 11, 17, 19 are 2, 2, 22, 11, so the conclusion ρ(S^d_{$\bar{\mathbb{Q}}$}) = 13 of §7 stands (p = 7 against p = 17 already gives it). (ii) For c = 13/9 the sign at p = 19 is +1, not −1: the class at p = 19 is 15 or 69 (two ε = +1 polynomials remain, $e_5 = 521284$ or −651605, both with ρ = 14; N_5 at p = 19 would choose between them) and not 73; the class at p = 41 (ε = −1, unique polynomial) is 47; since 47 differs from 15 and from 69, the conclusion ρ = 13 and rank E($\bar{\mathbb{Q}}$(C_{13/9})) = 1 of F.44 stands. (The earlier pair of values 73 and 15 corresponded to the two signs.) *Results.* With the predicted sign, the polynomial is unique, or all candidates have the same ρ and the classes below, in these cases (class of |disc NS| when ρ($\mathbb{F}_p$) = 14; "16" = ρ($\mathbb{F}_p$) = 16, no information): c = 1: p = 11: 16; 13: 1; 17: 2; 19: 16; 29: 16; 31: 13; 37: 26. c = 4: p = 11: 1; 13: 13; 17: 16; 29: 22; (p = 31: three candidates, classes 91, 97, 65). c = −11/5: p = 19: 16; 29: 229; 37: 16; (p = 31: three candidates, classes 91, 97, 65, all ≠ 229). c = 14/45: p = 11: 11; 29: 13; 31: 13; 37: 39. In every case two primes with ρ($\mathbb{F}_p$) = 14 and different square classes exist, so ρ(S^{d_c}_{$\bar{\mathbb{Q}}$}) = 13 and rank E^{d_c}($\bar{\mathbb{Q}}$(t)) = 1, hence rank E($\bar{\mathbb{Q}}$(C_c)) = 1, for c = 1, 4, −11/5, 14/45 (as for 2 and 13/9): no second section exists over $\bar{\mathbb{Q}}$ for these covers, under the Tate conjecture for K3 surfaces over $\mathbb{F}_p$ (a theorem for p ⩾ 5), the square order of Br and the sign rule above. For c = −11/5 the argument also covers the ambiguous prime 31, because all three candidates have ρ = 14 and a class different from 229. Primes with bad reduction of the configuration were excluded (for example 5, 7, 11, 13, 17 for c = −11/5; 5, 7, 13, 17, 19 for c = 14/45; 5, 7, 47, 79 for c = 1; 5, 7, 19, 59 for c = 4, in addition to 2, 3, 23).

F.46 Is ρ = 13 for every c? (Remark and scan.) *Symmetry.* N_{−c}(t) = −N_c(−t) and E_{−t} is the quadratic twist of E_t by −1, so S^{d_{−c}} and S^{d_c} are isomorphic over $\bar{\mathbb{Q}}$ and c > 0 suffices. *What is easy.* The family c ↦ S^{d_c} is a family of K3 surfaces whose Néron–

Severi group contains the trivial lattice and the section x = c, of rank 13. Since ρ(S^{d_c}_{$\bar{\mathbb{Q}}$}) = 13 for c = 2, 13/9, −11/5, 14/45, 1, 4 (F.44, F.45), the generic Picard number of the family is 13 and ρ(S^{d_c}) = 13 for all c outside a countable set. *What is not easy.* $\mathbb{Q}$ is countable, so "outside a countable set" says nothing about rational c; and for algebraic c the claim "always ρ = 13" is expected to fail: in a non-isotrivial one-parameter family of lattice-polarized K3 surfaces the Noether–Lefschetz points (where a further class appears) form a countable, dense set of algebraic parameters. The expected situation is that of rational CM points on a modular curve (dense CM points, but only finitely many rational ones): rational c with ρ = 14 are expected to be rare, possibly absent, and a uniform proof for all rational c is not apparent (a proof by two primes cannot be uniform, because the class at p is a function of c mod p, and by the Chinese remainder theorem every pair of residues occurs). *Scan.* For the 24 rational c = a/b (0 < c, a ⩽ 13, b ⩽ 4) not treated before and sorted by height, with point counts at the good primes among 11, 13, 17, 19 and the sign rule of F.45: ρ = 13 is proved for c = 1/4, 1/3, 2/3, 3/4, 5/4, 7/4, 9/4, 5/2, 3, 10/3, 5, 11/2 (two primes with ρ($\bar{F}_p$) = 14 and disjoint square classes); for c = 1/2, 4/3, 3/2, 5/3, 7/3, 8/3, 11/4, 13/4, 7/2, 11/3, 13/3, 9/2 fewer than two of these primes have ρ($\bar{F}_p$) = 14, so nothing can be said without further primes; no c was flagged (all informative classes overlapping, which would indicate ρ = 14). Together with F.44 and F.45 this gives ρ = 13 for 18 rational values of c and no counterexample. *Meaning.* An "always" statement for rational c would say that no cover C_c carries a second geometric section; it would not bear on H, since rank ⩾ 2 fibres on these covers would then arise only through the parity mechanism, which is the situation assumed in the conditional theorem of §6.

Appendix G. Code and validation values for the Picard-number computations

This appendix contains the two programs behind F.43–F.46 and the values against which they were validated. The program fibrecount.c counts points on the fibres of the twist surface over finite fields; lfunc.py turns the counts into the Picard number of the geometric fibre and the discriminant class; the sign of the functional equation is computed by the rule of F.45. The first version of the counter had a bug for fibres whose points have small order (F.44); it is repaired in the code below, which reproduces the earlier values.

Method.

1. For a prime p of good reduction and k = 1, …, 4 compute $N_k = -\Sigma$ over t in $P^1(F_{p^k})$ of $\Sigma_x \chi(x^3 + a_4(t)x + a_6(t))$ with fibrecount.c (compile with gcc -O2 -o fibrecount fibrecount.c; run ./fibrecount p k n2 n1 n0, where $N_c(t) = n2\,t^2 + n1\,t + n0 = (c^3 - 4c)t^2 - 16t + (c^3 + 4c)$ modulo p).

2. Newton's identities give the elementary symmetric functions $e_1, \ldots, e_4$ of the ten Frobenius eigenvalues.
3. The functional equation $\alpha \leftrightarrow p^2/\alpha$ and the sign ε of F.45 determine the characteristic polynomial except for e_5 when $\varepsilon = +1$; e_5 is fixed by requiring an even number (at least 2) of eigenvalues $p\cdot\zeta$ with ζ a root of unity, and all eigenvalues on $|\alpha| = p$.
4. ρ of the geometric fibre is 12 plus the number of eigenvalues $p\cdot\zeta$.
5. If $\rho = 14$, $\Lambda_p = \Pi(1 - \alpha^2/p^2)$ over the other eight eigenvalues; its squarefree class is the class of $|\mathrm{disc}\ NS|$. Two primes with $\rho = 14$ and different classes give $\rho(S_{\bar{\mathbb{Q}}}) = 13$ (F.44).

Validation values.

Cover	p	$N_1 \ldots N_4$	Sign	ρ of the fibre	Class
c = 2	7	−2, −74, −218, 2462 ($N_5 = 38$)	−1	14	2
c = 2	11	12, −166, −1506, −11826	−1	14	2
c = 2	17	−12, −382, −810, −270194	−1	14	22
c = 2	19	−36, −574, 5328, −268106 ($N_5 = 4354984$)	+1	14	11
c = 13/9	19	−26, −714, 2602, −32554	+1	14	15 or 69
c = 13/9	41	−40, −3896, 10046, −3727844	−1	14	47
c = 1	13	−10, −256, −1312, −17540	+1	14	1

fibrecount.c

```
// fibrecount.c : N_k = sum over t in P^1(F_q) of a_t,  q = p^k,  a_t = -
sum_x chi(x^3 + a4(t) x + a6(t)).
// Model of the twist K3 surface (minimal at every place):
//   Y^2 = X^3 + 4(1-t^2)(1+t^2)N(t)^2 X - 16 t (1+t^2)^2 N(t)^3,   N(t) = n2
t^2 + n1 t + n0.
// Usage: ./fibrecount p k n2 n1 n0     (n_i integers mod p; p >= 5 prime; k
<= 5)
// Fields: log/antilog tables from a primitive polynomial, Zech logarithms
for addition.
// Fibres over F_q whose t has Frobenius orbit of full size k >= 4 are
counted by baby-step giant-step.
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
typedef long long ll;
int p, k; ll q, Q1;
int *expt, *logt, *Z;

static int mulx(int code, int *poly) {            // multiply by x modulo x^k
= -poly(x); digits in base p
```

```
    int d[8], c = code;
    for (int i = 0; i < k; i++) { d[i] = c % p; c /= p; }
    int top = d[k-1];
    for (int i = k-1; i > 0; i--) d[i] = d[i-1];
    d[0] = 0;
    for (int i = 0; i < k; i++) d[i] = ((d[i] - top*poly[i]) % p + p) % p;
    int r = 0;
    for (int i = k-1; i >= 0; i--) r = r*p + d[i];
    return r;
}
static int addcode(int a, int b) {
    int r = 0, m = 1;
    for (int i = 0; i < k; i++) { r += (((a % p) + (b % p)) % p) * m; a /= p;
b /= p; m *= p; }
    return r;
}
static void build(void) {
    q = 1; for (int i = 0; i < k; i++) q *= p;
    Q1 = q - 1;
    expt = malloc(sizeof(int)*q); logt = malloc(sizeof(int)*q); Z =
malloc(sizeof(int)*q);
    int poly[8];
    for (ll cand = 0; cand < q; cand++) {          // first primitive
polynomial x^k + ... + poly[0]
        ll c = cand;
        for (int i = 0; i < k; i++) { poly[i] = c % p; c /= p; }
        if (poly[0] == 0 && k > 1) continue;
        int cur = 1, ok = 1;
        memset(logt, -1, sizeof(int)*q);
        for (ll i = 0; i < Q1; i++) {
            if (logt[cur] != -1) { ok = 0; break; }
            expt[i] = cur; logt[cur] = i; cur = mulx(cur, poly);
        }
        if (ok && cur == 1) break;
    }
    for (ll n = 0; n < Q1; n++) { int s = addcode(1, expt[n]); Z[n] = (s ==
0) ? -1 : logt[s]; }
}
// arithmetic on logarithms; -1 stands for 0
static inline ll ladd(ll a, ll b) { if (a < 0) return b; if (b < 0) return a;
ll d = b - a; if (d < 0) d += Q1; int z = Z[d]; if (z < 0) return -1; ll r =
a + z; if (r >= Q1) r -= Q1; return r; }
static inline ll lmul(ll a, ll b) { if (a < 0 || b < 0) return -1; ll r = a +
b; if (r >= Q1) r -= Q1; return r; }
static inline ll lneg(ll a) { if (a < 0) return -1; ll r = a + Q1/2; if (r >=
Q1) r -= Q1; return r; }
```

```
static inline ll lsub(ll a, ll b) { return ladd(a, lneg(b)); }
static inline ll linv(ll a) { return (Q1 - a) % Q1; }
static ll lconst(int v) { v %= p; if (v < 0) v += p; return v == 0 ? -1 :
logt[v]; }
static ll lpow(ll a, int e) { return a < 0 ? -1 : (a*e) % Q1; }

static ll legsum(ll A, ll B) {                    // sum over x of chi(x^3 +
A x + B), A and B given as logs
    ll s = 0;
    if (B >= 0) s += (B % 2 == 0) ? 1 : -1;       // x = 0
    for (ll L = 0; L < Q1; L++) {
        ll f = ladd(ladd((3*L) % Q1, (A < 0) ? -1 : (A + L) % Q1), B);
        if (f >= 0) s += (f & 1) ? -1 : 1;
    }
    return s;
}
// ---- point counting on y^2 = x^3 + A x + B over F_q by baby-step giant-
step ----
typedef struct { ll x, y; int inf; } Pt;
static ll LA, LB, L2, L3;
static Pt padd(Pt P, Pt Q) {
    if (P.inf) return Q;
    if (Q.inf) return P;
    ll lam;
    if (P.x == Q.x) {
        if (P.y == lneg(Q.y) || (P.y < 0 && Q.y < 0)) { Pt O = {0, 0, 1};
return O; }
        lam = lmul(ladd(lmul(L3, lmul(P.x, P.x)), LA), linv(lmul(L2, P.y)));
    } else lam = lmul(lsub(Q.y, P.y), linv(lsub(Q.x, P.x)));
    ll x3 = lsub(lsub(lmul(lam, lam), P.x), Q.x);
    Pt R = {x3, lsub(lmul(lam, lsub(P.x, x3)), P.y), 0};
    return R;
}
static Pt pmul(Pt P, ll n) {
    Pt R = {0, 0, 1};
    if (n < 0) { n = -n; if (!P.inf) P.y = lneg(P.y); }
    while (n) { if (n & 1) R = padd(R, P); P = padd(P, P); n >>= 1; }
    return R;
}
static unsigned long long rng = 88172645463325252ULL;
static unsigned long long rnd(void) { rng ^= rng << 13; rng ^= rng >> 7; rng
^= rng << 17; return rng; }
static int randpt(Pt *P) {
    for (int tries = 0; tries < 200; tries++) {
        ll L = rnd() % Q1;
        ll f = ladd(ladd((3*L) % Q1, (LA < 0) ? -1 : (LA + L) % Q1), LB);
```

```
        if (f >= 0 && f % 2 == 0) { P->x = L; P->y = f/2; P->inf = 0; return
1; }
    }
    return 0;
}
#define MAXC 600
static int candidates(Pt P, ll W, ll *cand) {      // all M in [q+1-W, q+1+W]
with M*P = O
    ll m = 1; while (m*m < W) m++; m += 1;
    Pt baby[1024]; baby[0].inf = 1; baby[0].x = 0; baby[0].y = 0;
    Pt cur = P;
    for (ll j = 1; j < m; j++) { baby[j] = cur; cur = padd(cur, P); }
    Pt mP = cur, T = pmul(P, q + 1);
    if (!T.inf) T.y = lneg(T.y);                   // T = -(q+1)P
    ll amax = W/m + 2; int nc = 0;
    Pt S = padd(T, pmul(mP, amax));                // S_a = -(q+1)P - a mP,
started at a = -amax
    Pt negm = mP; if (!negm.inf) negm.y = lneg(negm.y);
    for (ll a = -amax; a <= amax; a++) {
        for (ll b = 0; b < m; b++) {
            int eq = (S.inf && baby[b].inf) || (!S.inf && !baby[b].inf && S.x
== baby[b].x && S.y == baby[b].y);
            if (eq) { ll kk = a*m + b; if (kk >= -W && kk <= W && nc < MAXC)
cand[nc++] = q + 1 + kk; }
        }
        S = padd(S, negm);
    }
    return nc;
}
static ll count_points(ll A, ll B) {               // #E(F_q), or -1 on
failure
    LA = A; LB = B;
    ll W = 2; while (W*W <= q) W++; W *= 2;
    ll cand[MAXC], cur[MAXC]; int nc = -1, informative = 0;
    for (int it = 0; it < 60; it++) {
        Pt P; if (!randpt(&P)) return -1;
        int n = candidates(P, W, cand);
        if (n >= MAXC - 1 || n > 24) continue;     // a point of small order
says too little: skip it
        informative++;
        if (nc < 0) { nc = n; memcpy(cur, cand, sizeof(ll)*n); }
        else { int m2 = 0; for (int i = 0; i < nc; i++) for (int j = 0; j <
n; j++) if (cur[i] == cand[j]) { cur[m2++] = cur[i]; break; } nc = m2; }
        if (nc == 0) return -1;
        if (nc == 1 && informative >= 4) return cur[0];
    }
```

```
    return -1;
}
int main(int argc, char **argv) {
    p = atoi(argv[1]); k = atoi(argv[2]);
    int n2 = atoi(argv[3]), n1 = atoi(argv[4]), n0 = atoi(argv[5]);
    build();
    L2 = lconst(2); L3 = lconst(3);
    ll cn2 = lconst(n2), cn1 = lconst(n1), cn0 = lconst(n0), total = 0;
    ll Ainf = lneg(lmul(lconst(4), lmul(cn2, cn2)));          // t =
infinity: a4 = -4 n2^2, a6 = 0
    if (Ainf >= 0) total -= legsum(Ainf, -1);
    ll A0 = lmul(lconst(4), lmul(cn0, cn0));                  // t = 0: a4 =
4 N(0)^2, a6 = 0
    if (A0 >= 0) total -= legsum(A0, -1);
    char *seen = calloc(Q1, 1);
    for (ll m = 0; m < Q1; m++) {                             // t = g^m, one
t per Frobenius orbit
        if (seen[m]) continue;
        int osz = 0; ll mm = m;
        do { seen[mm] = 1; osz++; mm = (mm*p) % Q1; } while (mm != m);
        ll t = m, t2 = lpow(t, 2);
        ll omt2 = ladd(0, lneg(t2)), opt2 = ladd(0, t2);
        ll N = ladd(ladd(lmul(cn2, t2), lmul(cn1, t)), cn0);
        ll A = lmul(lconst(4), lmul(lmul(omt2, opt2), lpow(N, 2)));
        ll B = lneg(lmul(lconst(16), lmul(lmul(t, lpow(opt2, 2)), lpow(N,
3))));
        if (A < 0 && B < 0) continue;                         // additive
fibre, a_t = 0
        ll disc = ladd((A < 0) ? -1 : lmul(lconst(4), lpow(A, 3)), (B < 0) ?
-1 : lmul(lconst(27), lpow(B, 2)));
        ll at;
        if (osz == k && k >= 4 && disc >= 0) {
            ll npts = count_points(A, B);
            at = (npts > 0) ? q + 1 - npts : -legsum(A, B);
        } else at = -legsum(A, B);
        total += osz * at;
    }
    printf("%d %d %lld", p, k, total); putchar(10);
    return 0;
}
```

lfunc.py

```
# lfunc.py : from the counts N_1..N_4 to the Picard number rho(S_{F_p-bar})
of the twist surface
#            and the square class of |disc NS| (Artin-Tate), for the cover
with x = c.
# Needs: python3, sympy, numpy.  Usage: see the table at the end of the file.
import math, cmath
import numpy as np
from fractions import Fraction as Fr
from sympy import symbols, Poly, Integer, factor_list, resultant, factorint

x, y, t = symbols('x y t')
f6 = t**6 - 30*t**4 - 24*t**2 - 1

def chi(a, p):
    a %= p
    return 0 if a == 0 else (1 if pow(a, (p-1)//2, p) == 1 else -1)

def sign_of_functional_equation(c, p):
    """epsilon = product of local root numbers: only the places over the
roots of f contribute
       (-1 if the node is split, i.e. if 3 x0 = 18 t (1+t^2) N(t) / (1-t^2)
is a square there)."""
    cc = (c.numerator * pow(c.denominator, -1, p)) % p
    N = ((cc**3 - 4*cc) % p) * t**2 + ((-16) % p) * t + ((cc**3 + 4*cc) % p)
    r = len(factor_list(f6, modulus=p)[1])
    fp = Poly(f6, t, modulus=p)
    num = Poly(18*t*(1 + t**2)*N, t, modulus=p)
    den = Poly(1 - t**2, t, modulus=p)
    pi = int(resultant(fp, num, t)) * pow(int(resultant(fp, den, t)), -1, p)
% p
    return (-1)**r * chi(pi, p)

def elementary(Ns):
    """e_1..e_4 of the eigenvalues from the power sums p_k = -N_k (Newton's
identities)."""
    P = [None] + [-n for n in Ns]
    e = [Fr(1)]
    for k in range(1, len(Ns) + 1):
        e.append(sum((-1)**(i-1) * e[k-i] * P[i] for i in range(1, k+1)) / k)
    return e

def char_poly(e, p, eps, e5):
    """R(x) = prod (x - alpha_i): functional equation e_j = eps p^(2(j-5))
e_(10-j) for j > 5."""
    ee = [Fr(v) for v in e[:5]] + [Fr(e5)]
    for j in range(6, 11):
```

```
        ee.append(eps * Fr(p)**(2*(j-5)) * ee[10-j])
    return ee, Poly(sum((-1)**k * Integer(int(ee[k])) * x**(10-k) for k in
range(11)), x)

def on_circle(R, p):
    r = np.roots([float(c) for c in R.all_coeffs()])
    return bool(np.allclose(np.abs(r), p, atol=3e-4*p))

def eigenvalues_p_times_root_of_unity(R, p):
    """number of eigenvalues of the form p*zeta (zeta a root of unity);
rho(F_p-bar) = 12 + this number."""
    total = 0
    for g, mult in factor_list(R.as_expr())[1]:
        gp = Poly(g, x); k = gp.degree()
        gy = Poly(sum(Integer(int(c)) * (p*y)**(k-j) for j, c in
enumerate(gp.all_coeffs())), y)
        if all(int(c) % p**k == 0 for c in gy.all_coeffs()):
            cy = Poly(gy.as_expr() / p**k, y)
            if any(Poly(y**m - 1, y).rem(cy).is_zero for m in range(1, 61)):
                total += k * mult
    return total

def squarefree_class(fr):
    v = fr.numerator * fr.denominator
    s = -1 if v < 0 else 1
    for pr, e in factorint(abs(v)).items():
        if e % 2: s *= pr
    return s

def discriminant_class(R, p):
    """Lambda_p = prod over the 8 eigenvalues that are not p*zeta of (1 -
alpha^2/p^2); returns its squarefree class."""
    rest = 1
    for g, mult in factor_list(R.as_expr())[1]:
        gp = Poly(g, x); k = gp.degree()
        gy = Poly(sum(Integer(int(c)) * (p*y)**(k-j) for j, c in
enumerate(gp.all_coeffs())), y)
        if not all(int(c) % p**k == 0 for c in gy.all_coeffs()):
            rest *= gp.as_expr()**mult
    rest = Poly(rest, x)
    return squarefree_class(Fr(int(resultant(rest.as_expr(), x**2 - p*p, x)))
/ Fr(p)**16) if rest.degree() == 8 else None

def analyse(c, p, Ns):
    """list of (e5, rho, class) for the L-functions that are Weil-valid, have
the predicted sign,
```

```
       and have an even number >= 2 of eigenvalues p*zeta."""
    eps = sign_of_functional_equation(c, p)
    e = elementary(Ns[:4]); out = []
    if eps == -1: trial = [0]
    else:
        trial = set()
        for m in (1, 2, 3, 4, 5, 6, 8, 10, 12):
            for j in range(m):
                if math.gcd(j, m) != 1: continue
                z = cmath.exp(2j*math.pi*j/m)
                A = sum((-1)**k * complex(char_poly(e, p, 1, 0)[0][k]) *
(p*z)**(10-k) for k in range(11))
                v = A / (p*z)**5
                if abs(v.imag) < 1e-6*(abs(v.real) + 1) and abs(v.real -
round(v.real)) < 1e-3: trial.add(int(round(v.real)))
    for e5 in sorted(trial):
        ee, R = char_poly(e, p, eps, e5)
        if not on_circle(R, p): continue
        n = eigenvalues_p_times_root_of_unity(R, p)
        if n >= 2 and n % 2 == 0:
            out.append((e5, 12 + n, discriminant_class(R, p) if n == 2 else
None))
    return eps, out

if __name__ == '__main__':
    tests = [(Fr(2), 7, [-2, -74, -218, 2462]), (Fr(2), 17, [-12, -382, -810,
-270194]), (Fr(2), 19, [-36, -574, 5328, -268106]),
             (Fr(13, 9), 19, [-26, -714, 2602, -32554]), (Fr(13, 9), 41,
[-40, -3896, 10046, -3727844]),
             (Fr(1), 13, [-10, -256, -1312, -17540])]
    for c, p, Ns in tests: print('c =', c, ' p =', p, ' (sign, [(e5, rho,
class)]) =', analyse(c, p, Ns))
```

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